

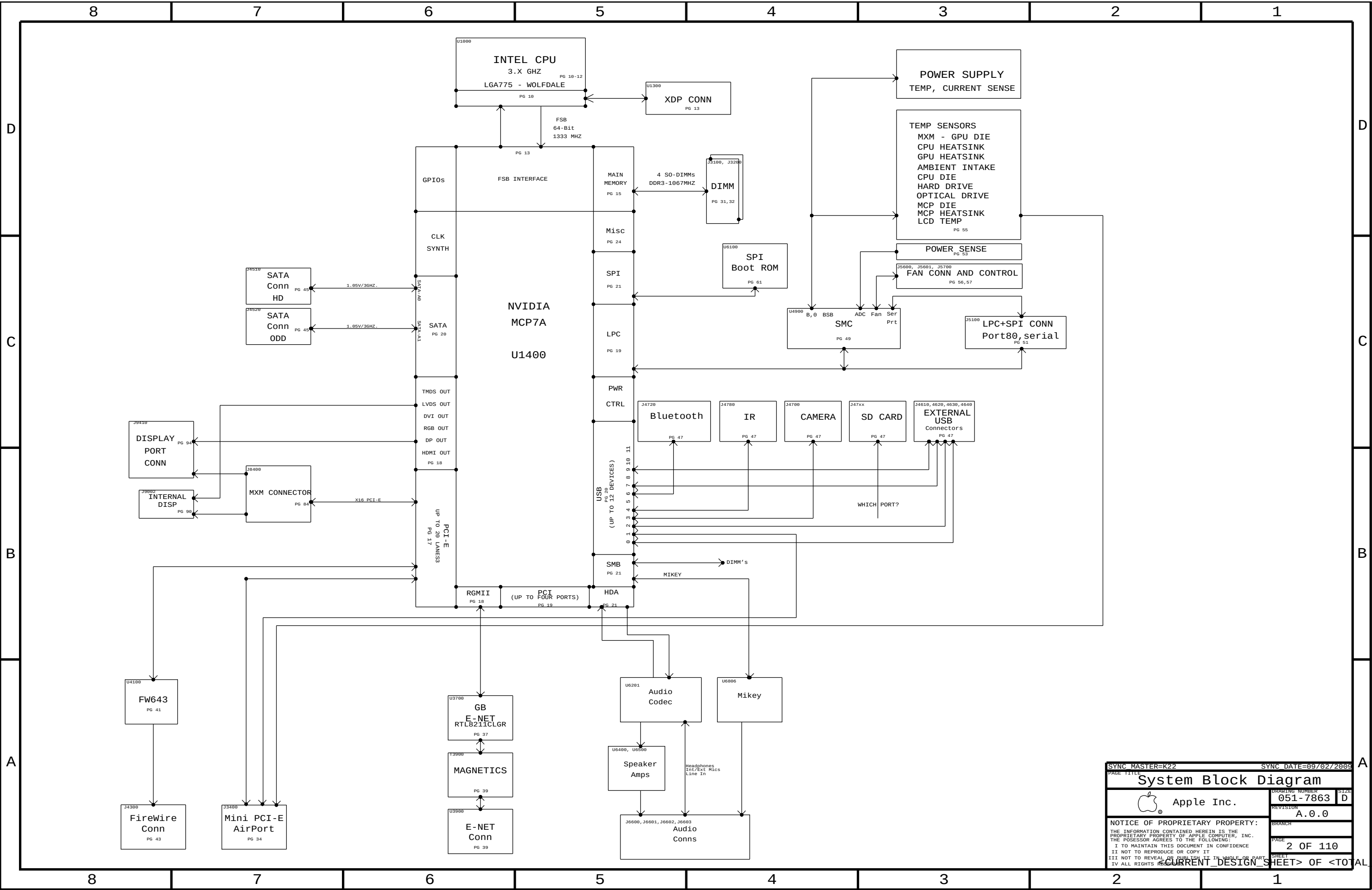
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

K23

REV	ECN	DESCRIPTION OF REVISION	CK APPD / DATE
A	0000774489	PRODUCTION RELEASED	2009-08-20

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5	Power Conn / Alias	MASTER	N/A
6	Holes	MASTER	N/A
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26	MEMORY CAPS	MASTER	N/A
27	DDR3 SO-DIMMs 0 & 2	K22	07/06/2009
28	DDR3 SO-DIMM CONNECTOR B	K22	09/02/2009
29	DDR3 SUPPORT AND BITSWAPS	MASTER	N/A
30	PCI-E Wireless Connector	K22	05/28/2009
31	Ethernet PHY (RTL8211CL)	K22	09/02/2009
32	Ethernet Support	K22	09/02/2009
33	ETHERNET CONNECTOR	MASTER	N/A
34	FireWire LLC/PHY (XIO2213B)	K22	09/02/2009
35	FW: 1394B MISC	K22	09/02/2009
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60	VREG: PPVCORE_S0_CPU	K22	09/02/2009
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71	Display: Int DP Connector	MARKVIDEO	03/12/2009
72	Display: BiDiVi Mux1	MARKVIDEO	03/12/2009
73	BIDIVI DP MUX2	MASTER	N/A
74	Display: Ext DP Connector	MARKVIDEO	03/12/2009
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81	FireWire Constraints	K22	09/02/2009
82	SMC Constraints	K22	09/02/2009
83	Graphics Constraints	MASTER	N/A
84	K22/K23 SPECIFIC CONSTRAINTS	K22	09/02/2009
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System Block Diagram

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C

B

A

TOP	SIGNAL
2	GROUND
3	SIGNAL
4	POWER
5	POWER
6	SIGNAL
7	GROUND
BOTTOM	SIGNAL

4

PARTS

(338S0489 - BLNK)

4

A

D

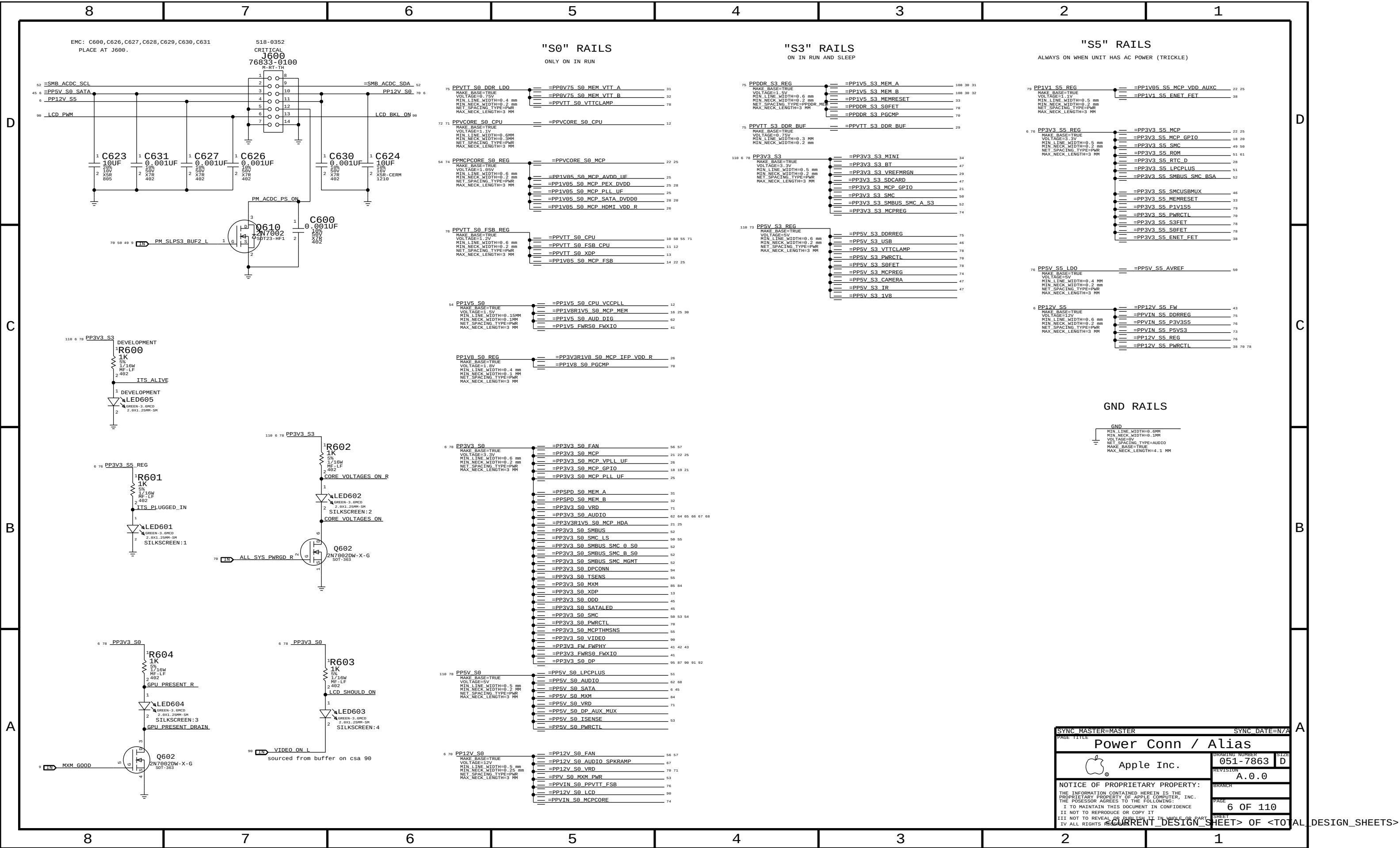
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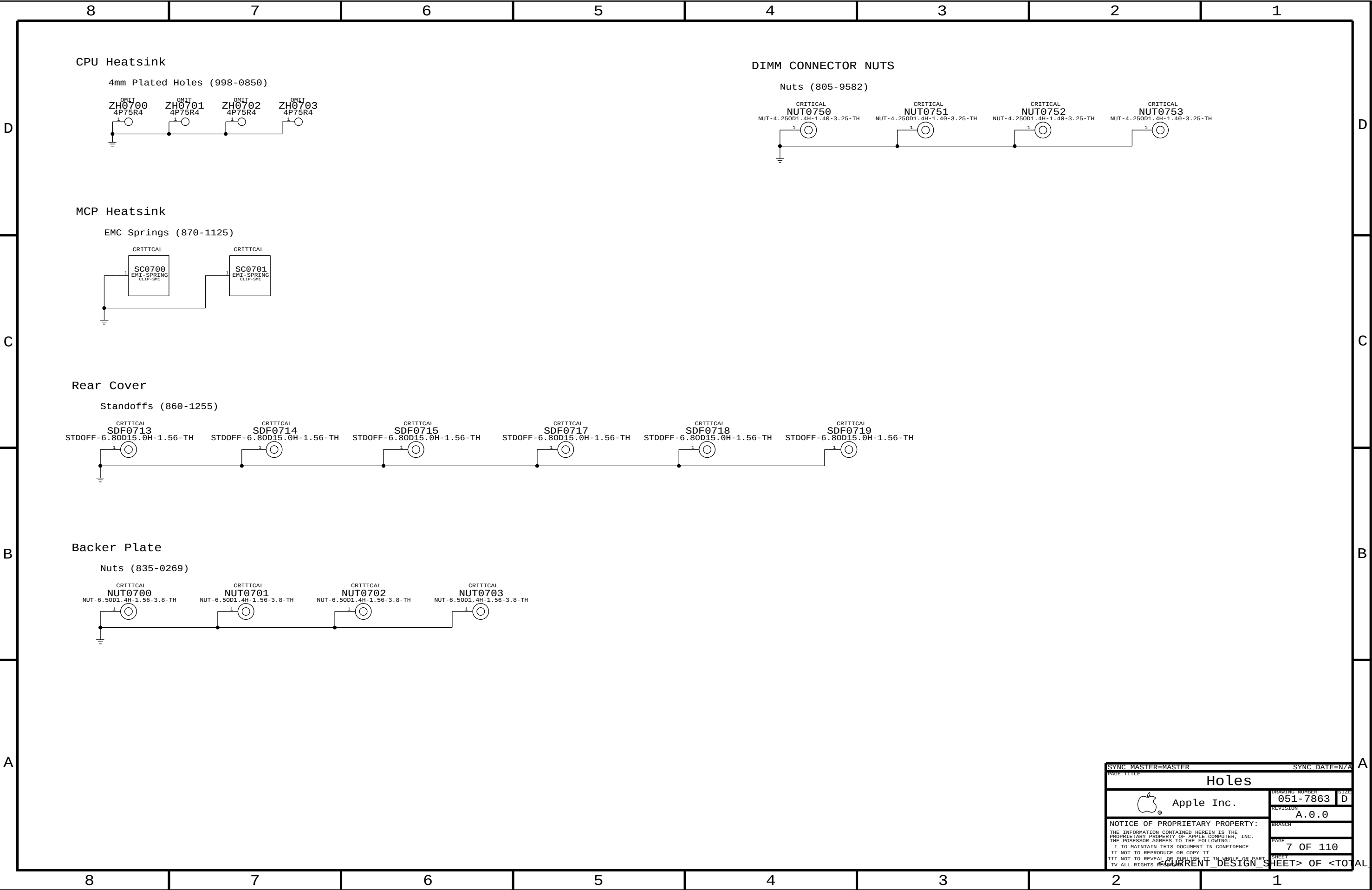
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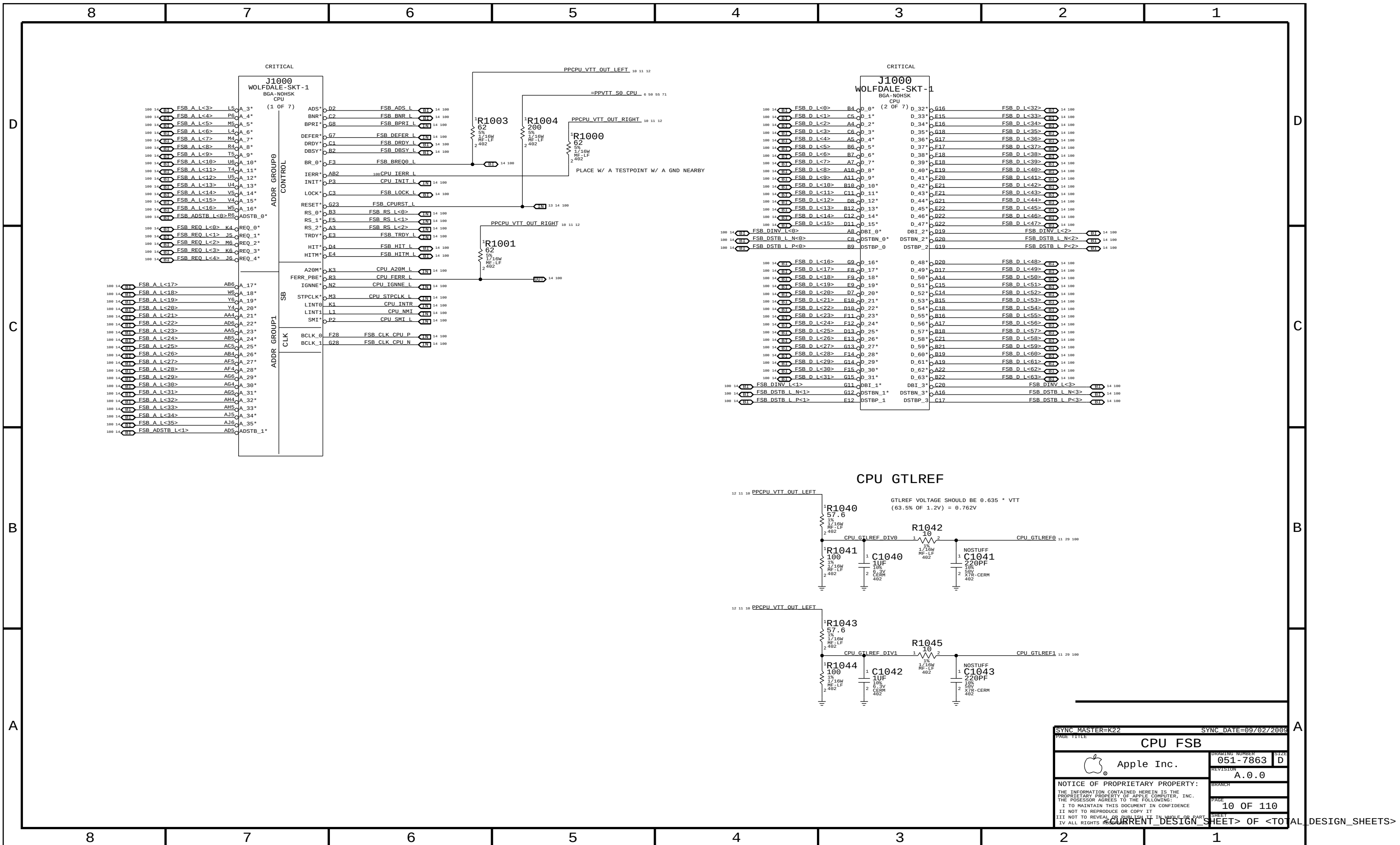


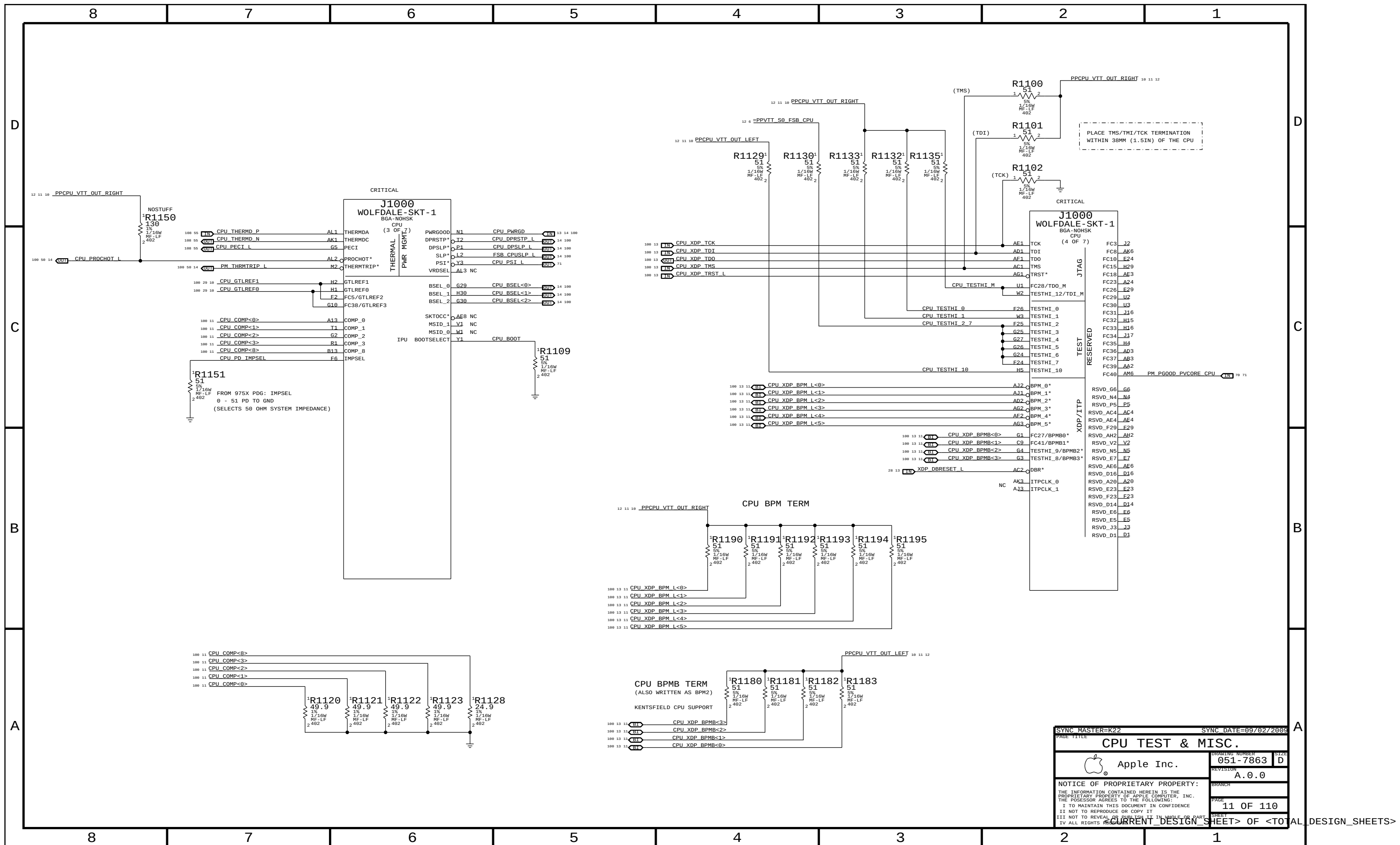
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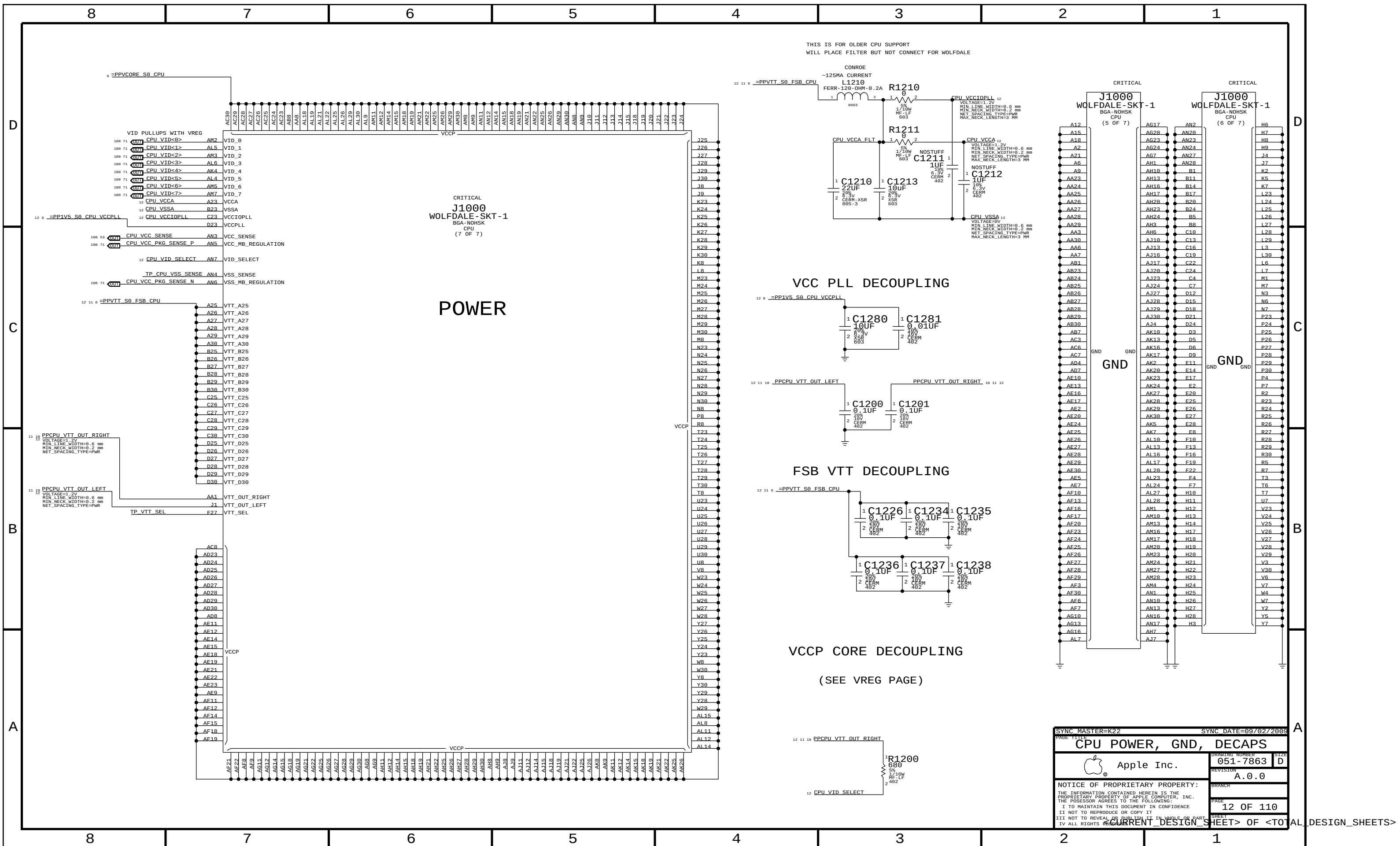


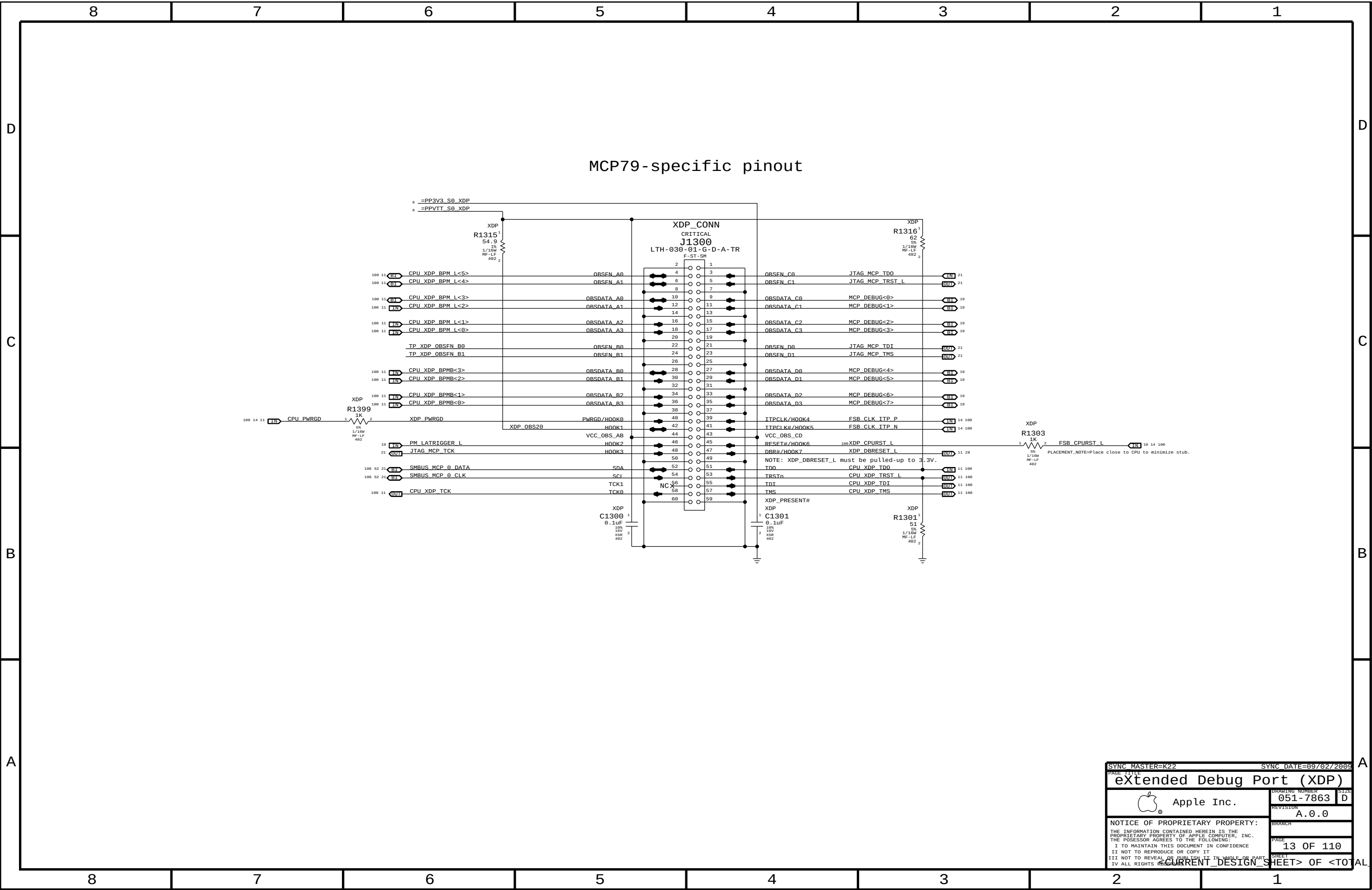
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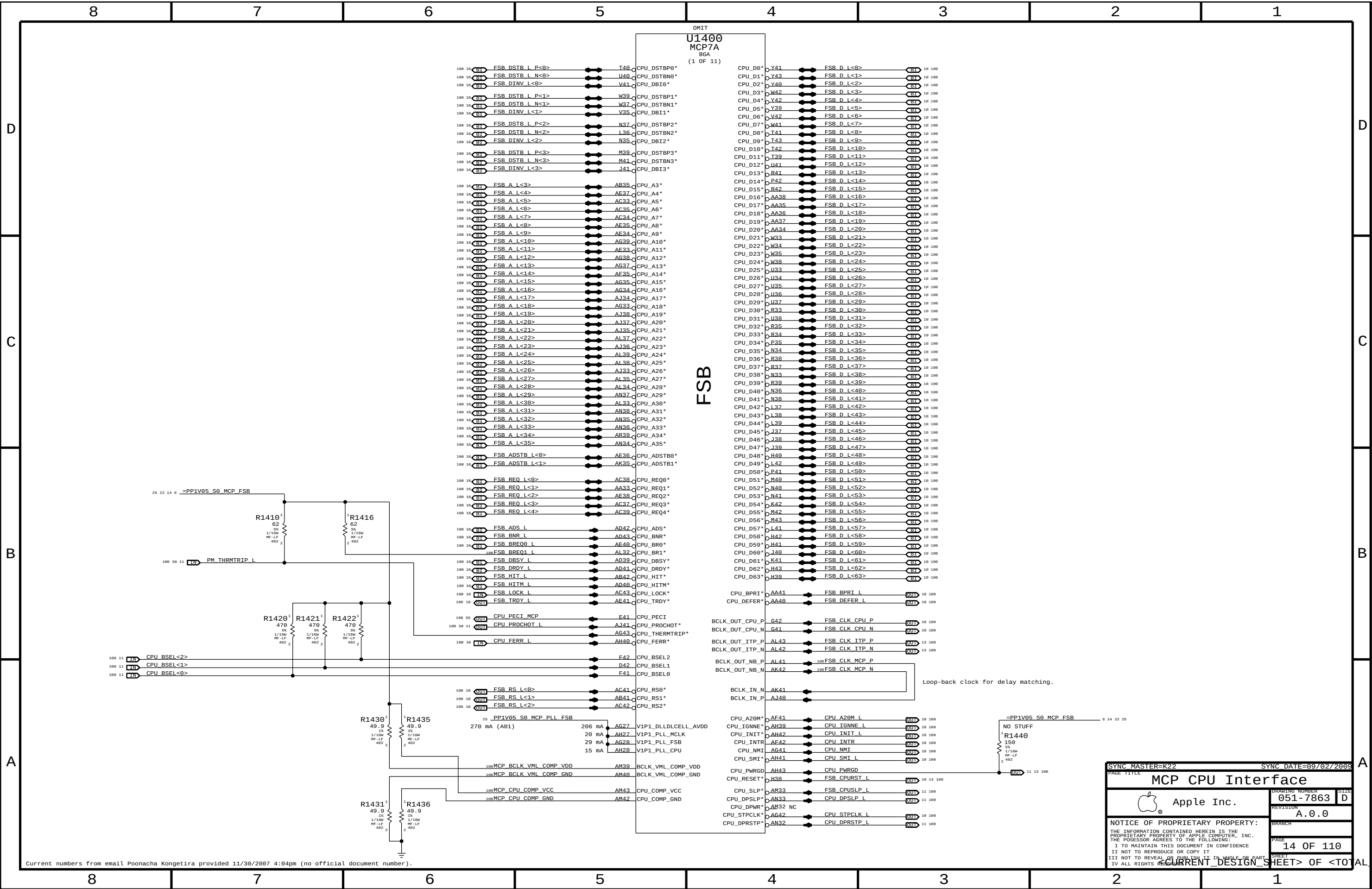
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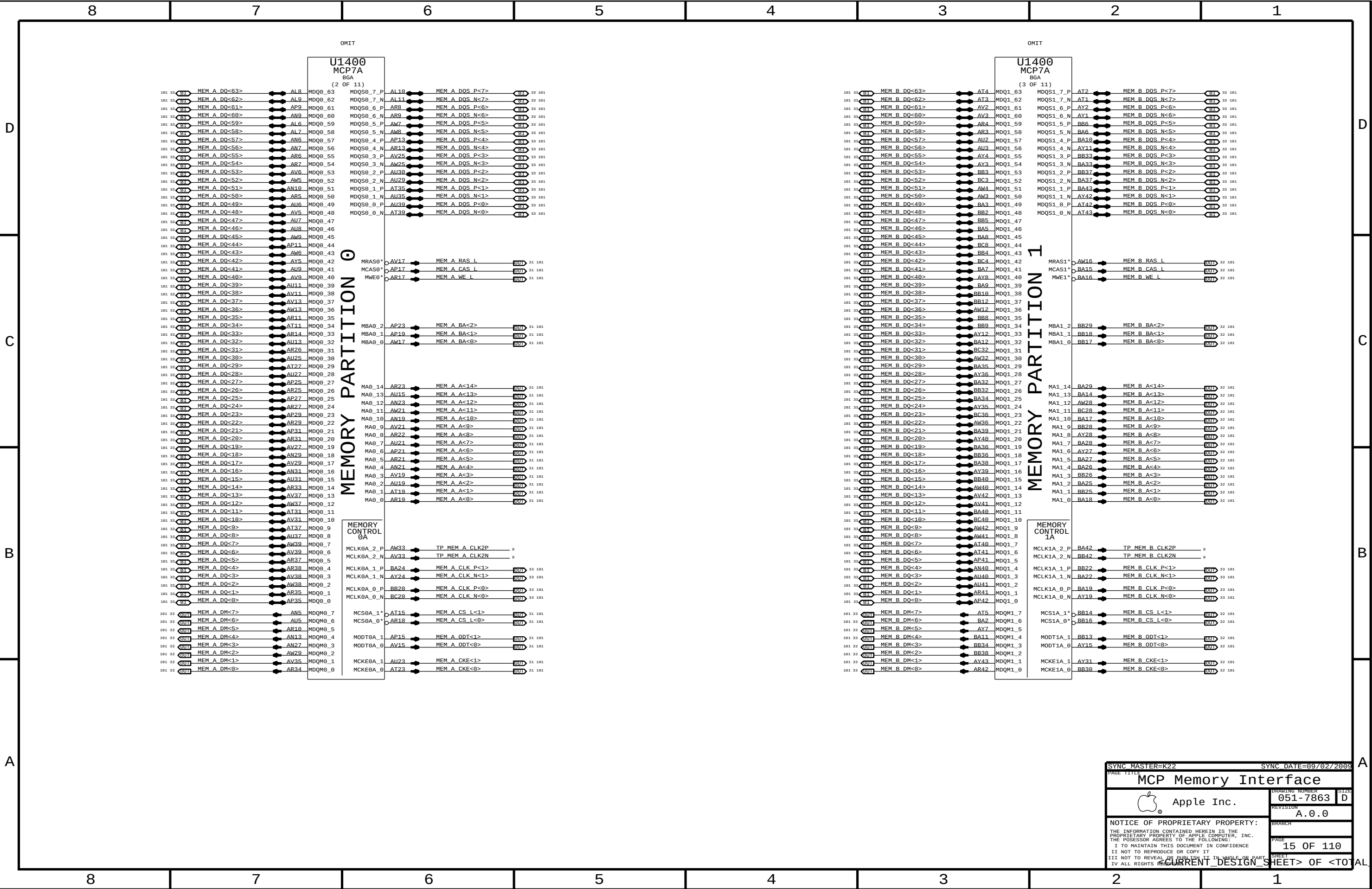


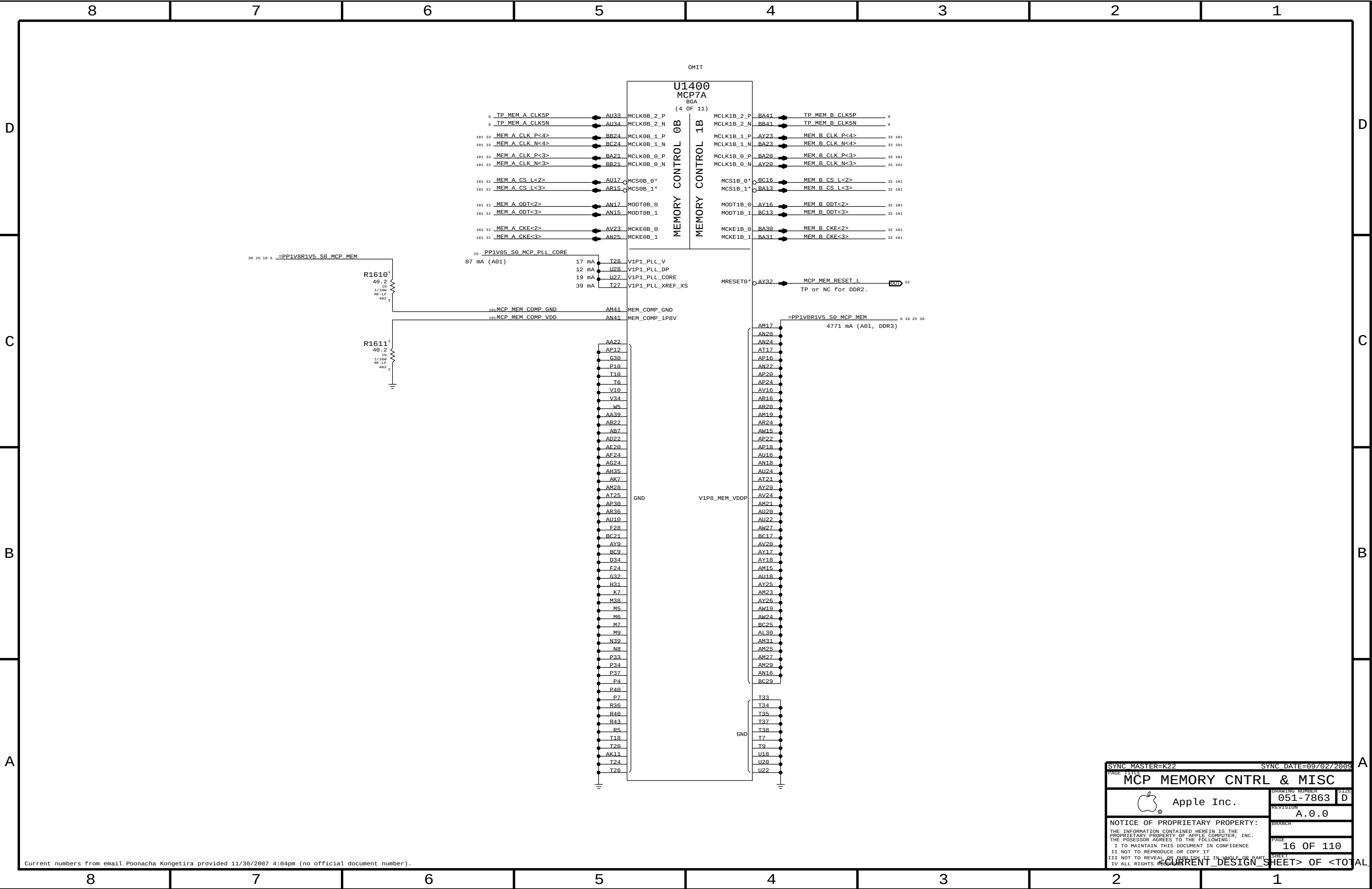




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PAGE TITLE MCP MEMORY CNTRL & MISC

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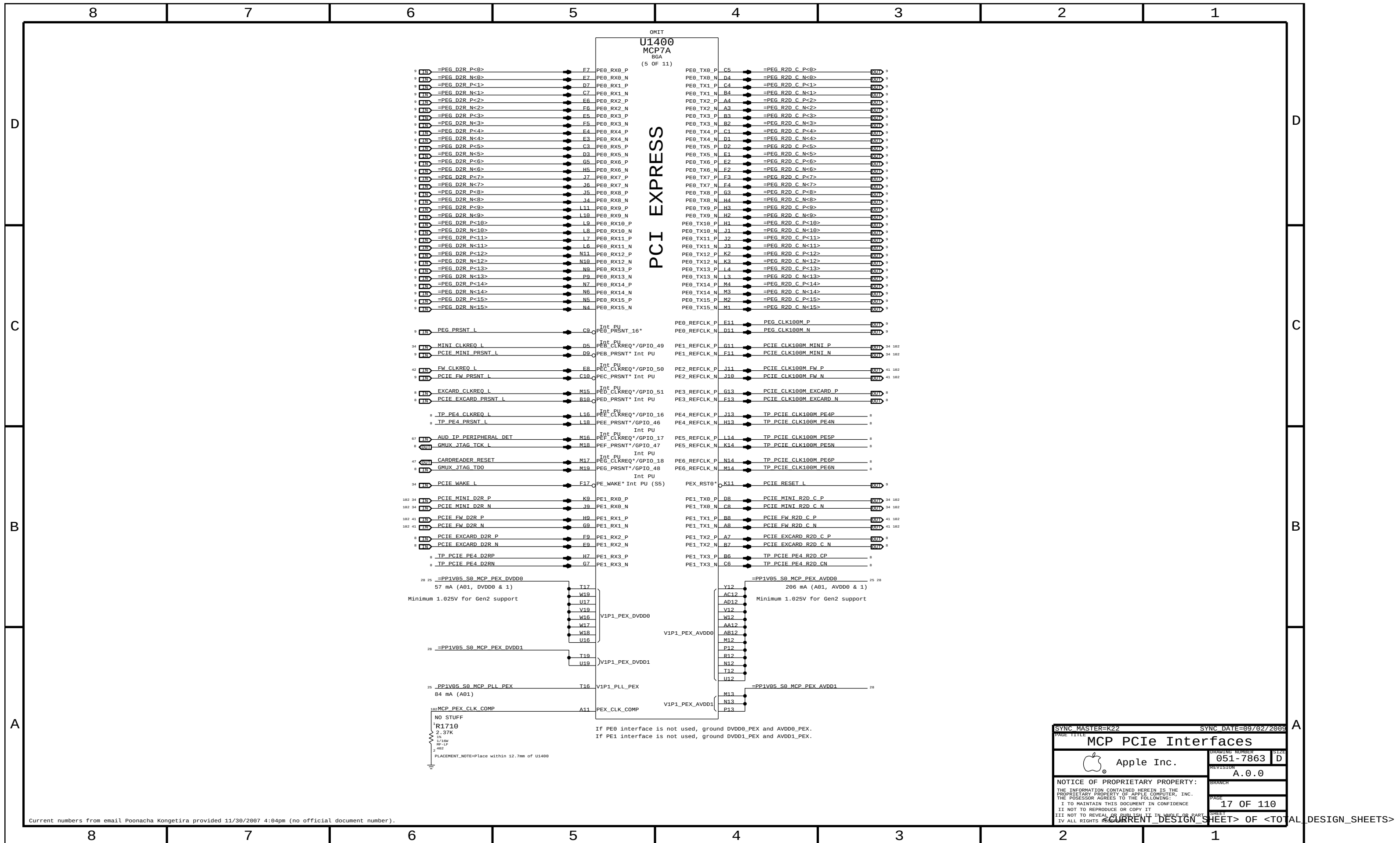
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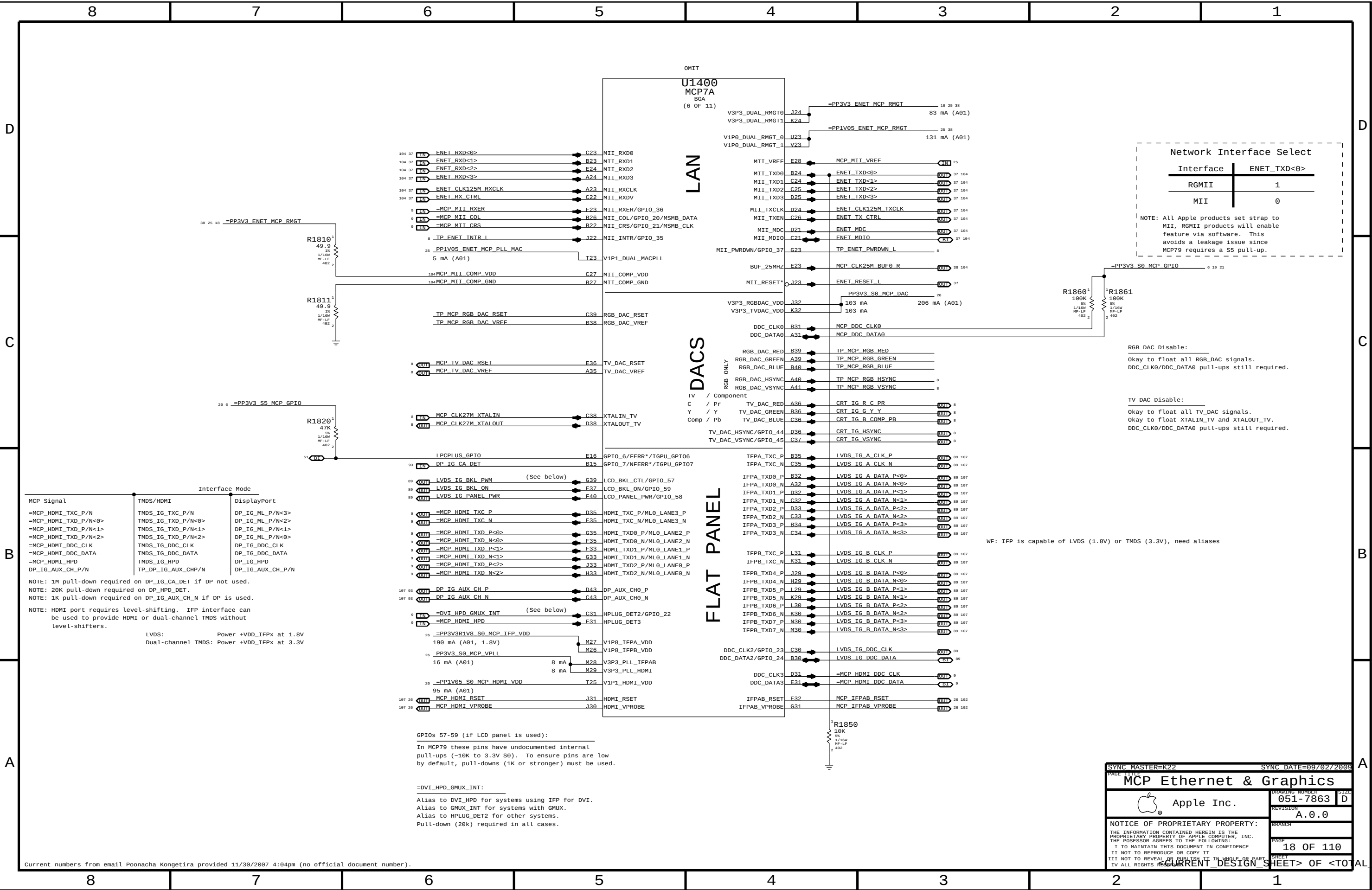
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Network Interface Select	
Interface	ENET_TXD<0>
RGMII	1
MII	0

NOTE: All Apple products set strap to MII, RGMII products will enable feature via software. This avoids a leakage issue since MCP79 requires a S5 pull-up.

RGB DAC Disable:
Okay to float all RGB_DAC signals.
DDC_CLK0/DDC_DATA0 pull-ups still required.

TV DAC Disable:
Okay to float all TV_DAC signals.
Okay to float XTALIN_TV and XTALOUT_TV.
DDC_CLK0/DDC_DATA0 pull-ups still required.

WF: IFP is capable of LVDS (1.8V) or TMDS (3.3V), need aliases

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MCP Ethernet & Graphics

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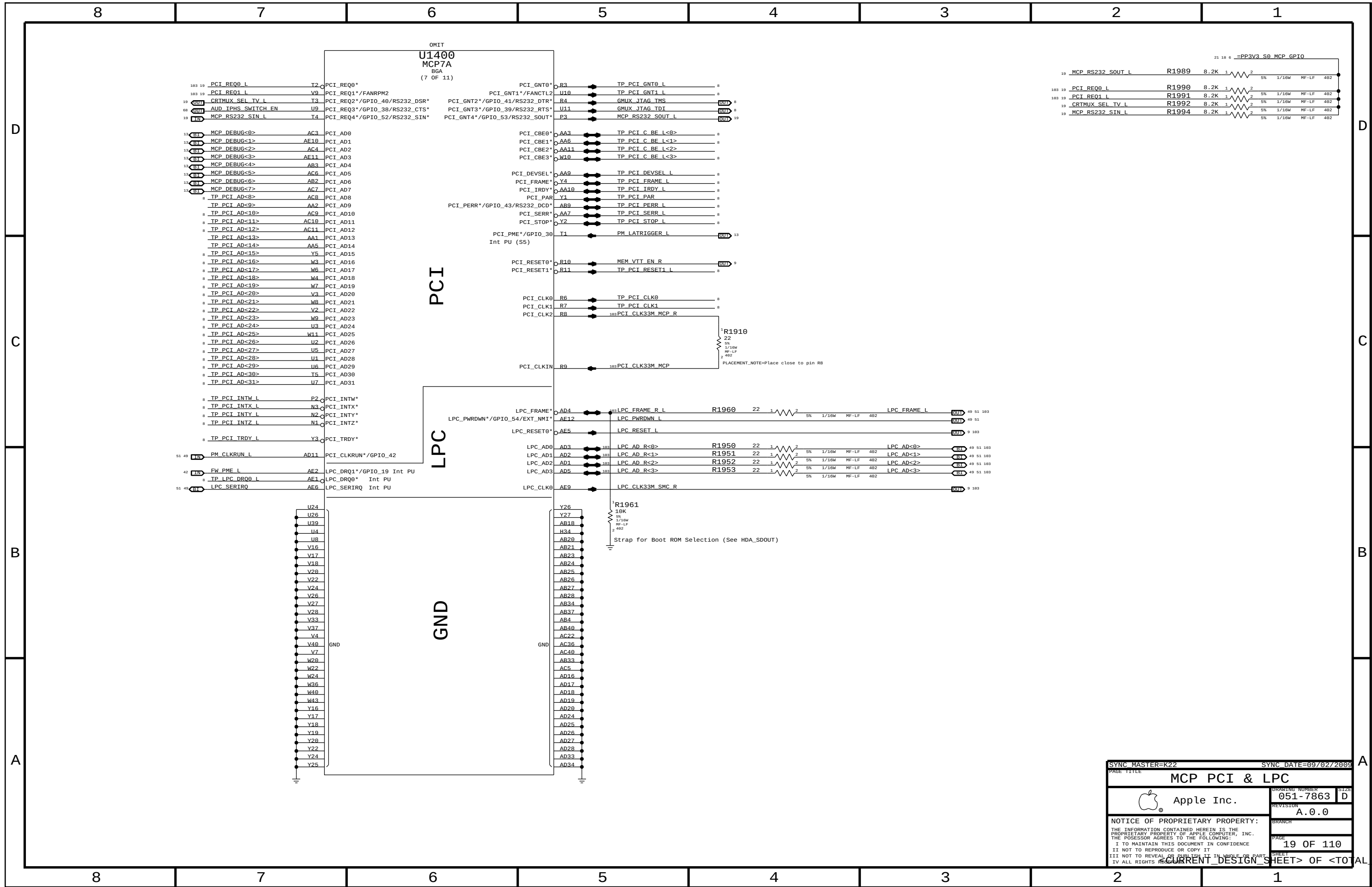
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GPIOs 57-59 (if LCD panel is used):
In MCP79 these pins have undocumented internal pull-ups (~10K to 3.3V S0). To ensure pins are low by default, pull-downs (1K or stronger) must be used.

=DVI_HPD_GMUX_INT:
Alias to DVI_HPD for systems using IFP for DVI.
Alias to GMUX_INT for systems with GMUX.
Alias to HPLUG_DET2 for other systems.
Pull-down (20K) required in all cases.



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PAGE TITLE

MCP PCI & LPC

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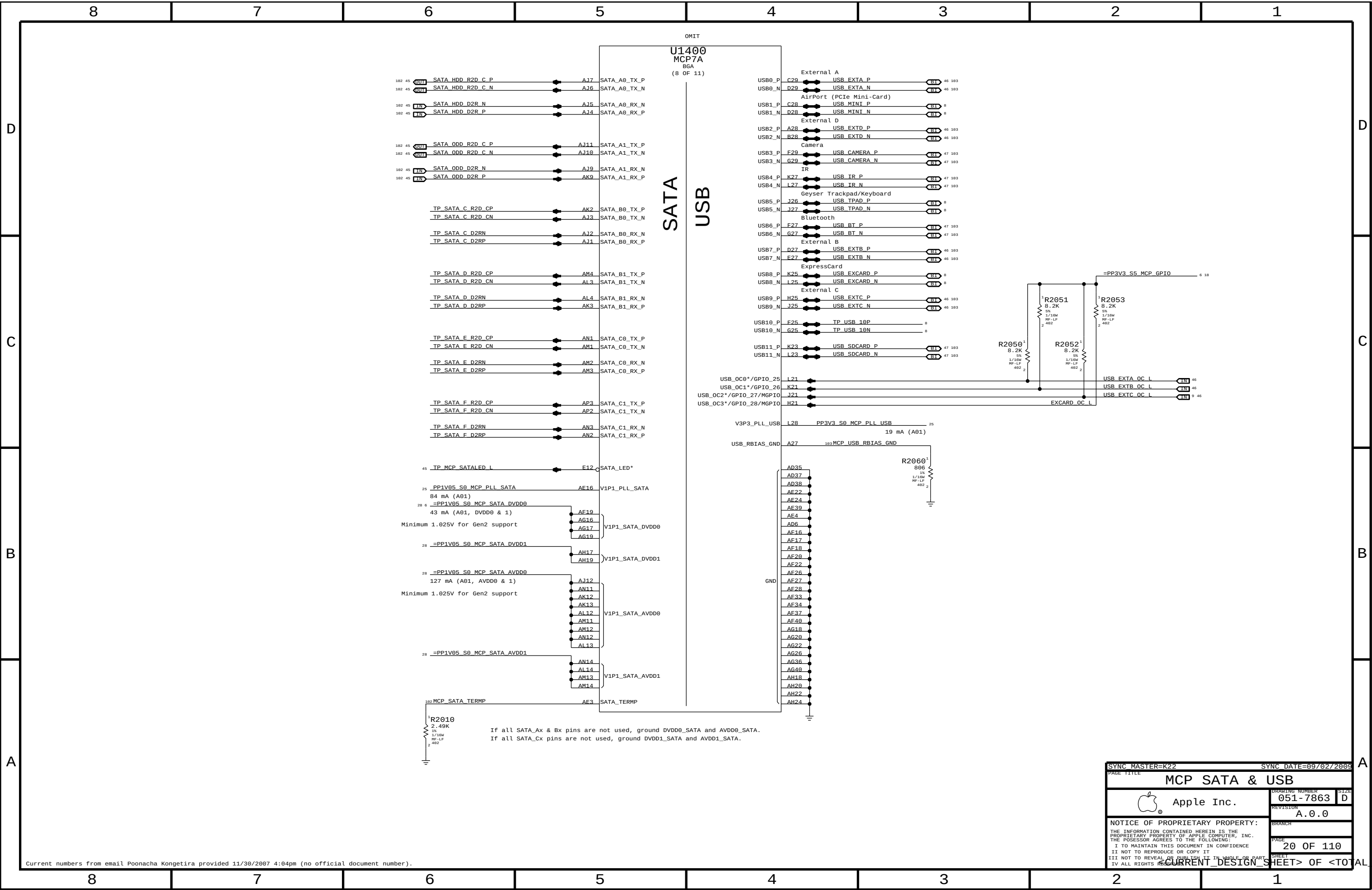
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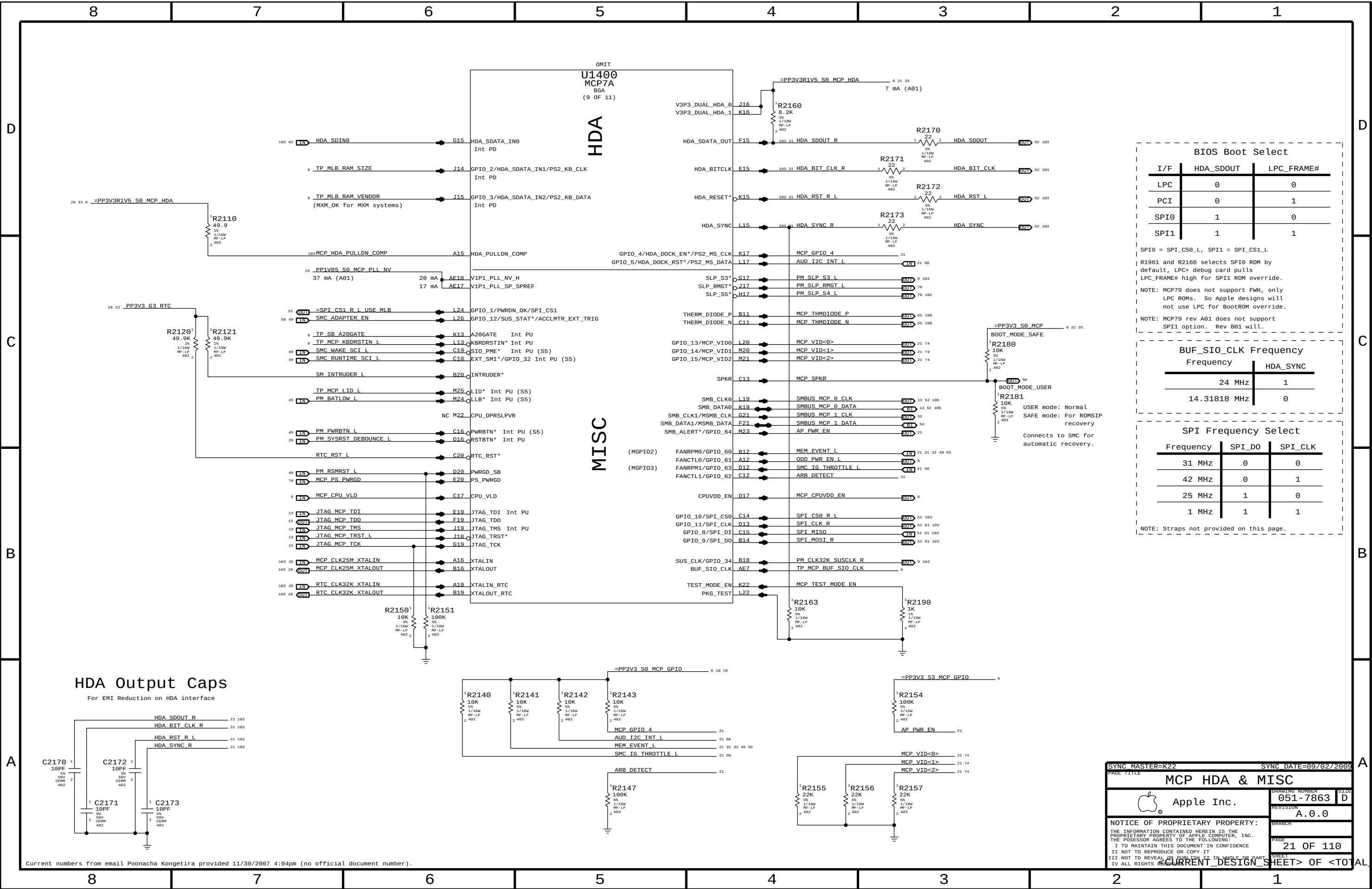
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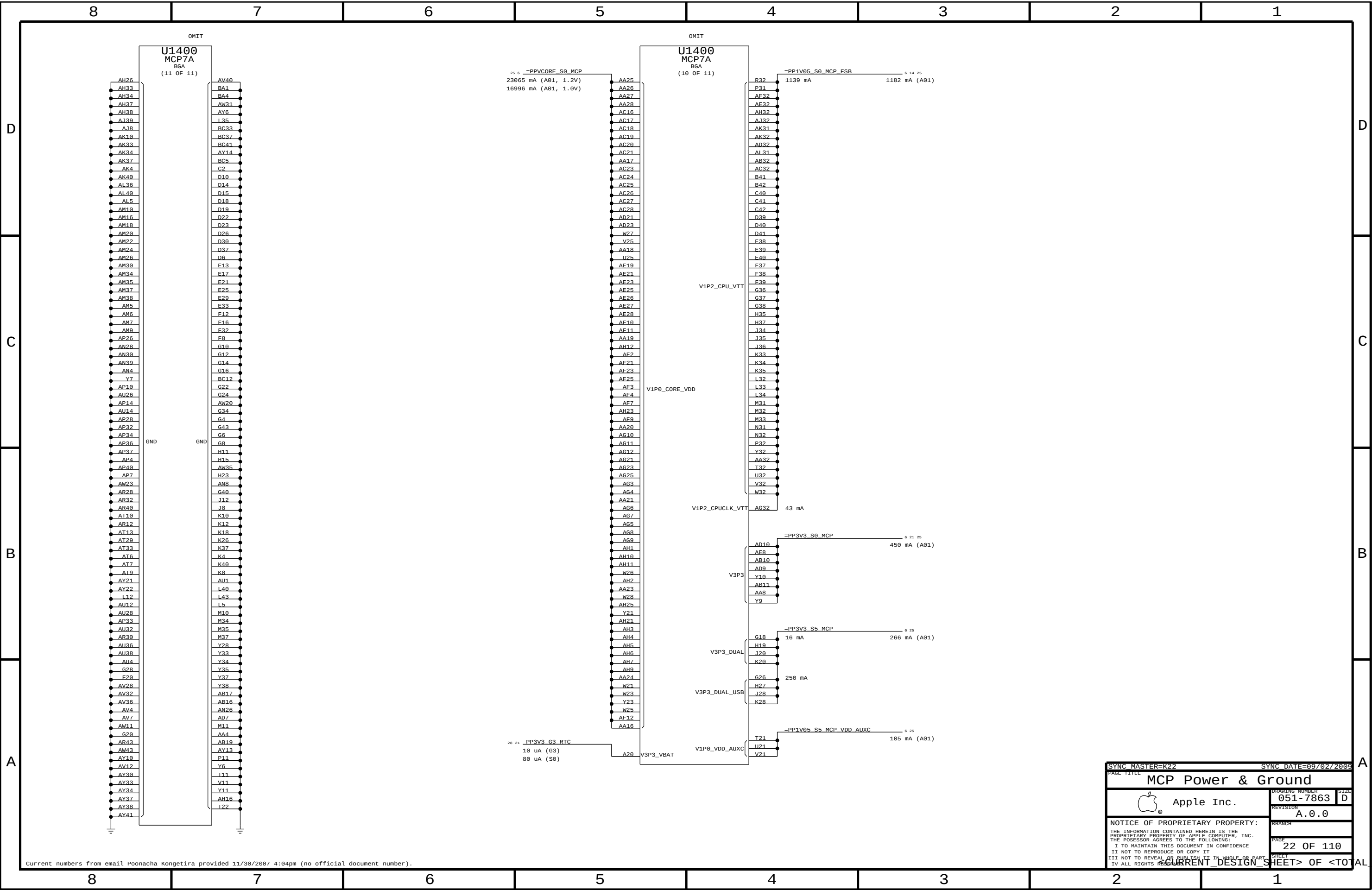
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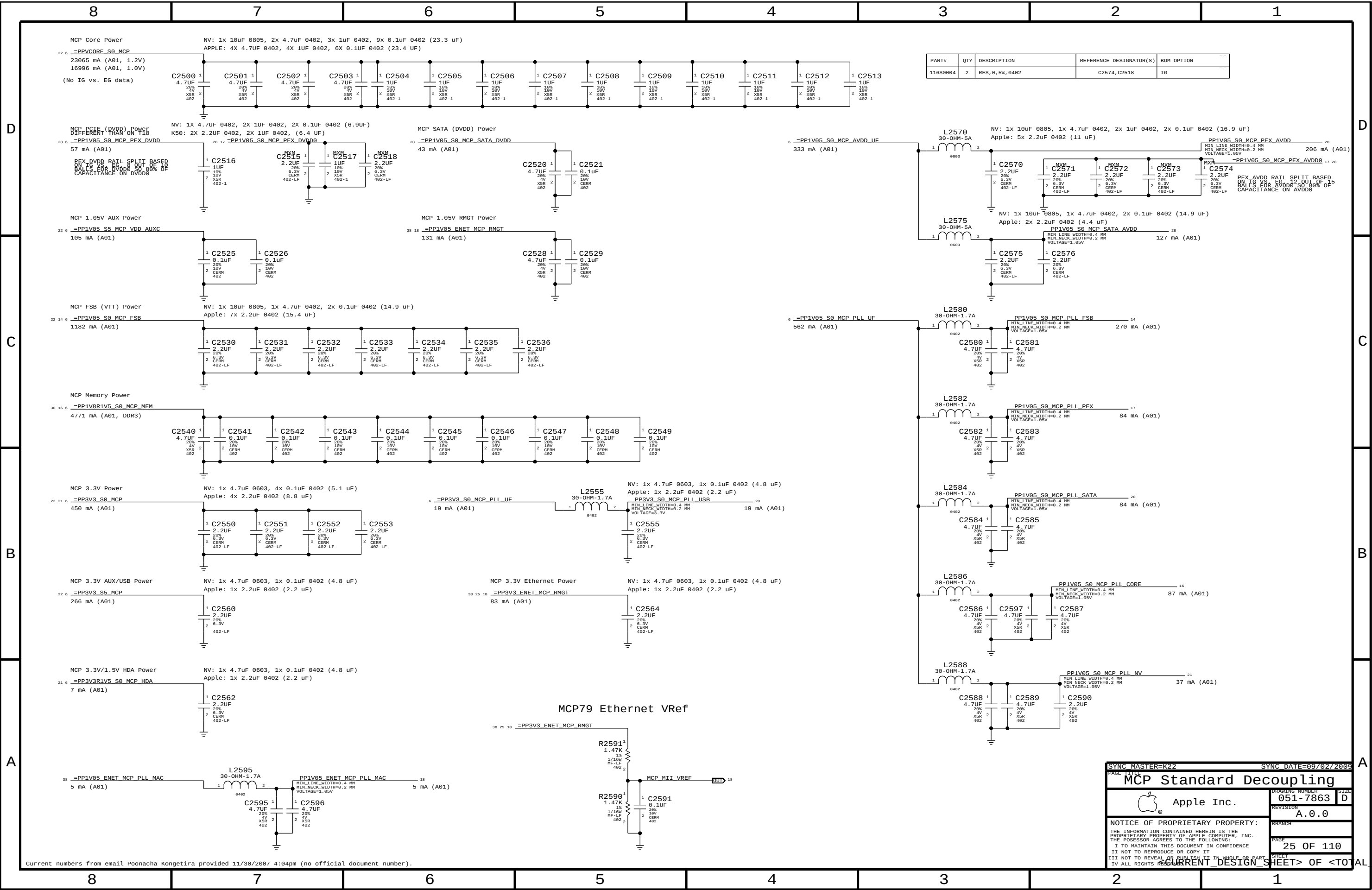
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MCP 1.05V AUX Power

=PP1V05_S5_MCP_VDD_AUXC

105 mA (A01)

C2525

0.1uF

20%
10V
CERM
402

1

2

C2526

0.1uF

20%
10V
CERM
402

1

2

MCP FSB (VTT) Power

NV: 1x 10uF 0805, 1x 4.7uF 0402, 2x 0.1uF 0402 (14.9 uF)
Apple: 7x 2.2uF 0402 (15.4 uF)

22 14 0

=PP1V05_S0_MCP_FSB

1182 mA (A01)

C2530

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2531

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2532

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2533

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2534

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2535

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2536

2.2uF

20%
6.3V
CERM
402-LF

1

2

MCP Memory Power

=PP1V8R1V5_S0_MCP_MEM

4771 mA (A01, DDR3)

C2540

4.7uF

20%
4V
XSR
402

1

2

C2541

0.1uF

20%
10V
CERM
402

1

2

C2542

0.1uF

20%
10V
CERM
402

1

2

C2543

0.1uF

20%
10V
CERM
402

1

2

C2544

0.1uF

20%
10V
CERM
402

1

2

C2545

0.1uF

20%
10V
CERM
402

1

2

C2546

0.1uF

20%
10V
CERM
402

1

2

C2547

0.1uF

20%
10V
CERM
402

1

2

C2548

0.1uF

20%
10V
CERM
402

1

2

C2549

0.1uF

20%
10V
CERM
402

1

2

MCP 3.3V Power

NV: 1x 4.7uF 0603, 4x 0.1uF 0402 (5.1 uF)
Apple: 4x 2.2uF 0402 (8.8 uF)

22 21 6

=PP3V3_S0_MCP

450 mA (A01)

C2550

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2551

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2552

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2553

2.2uF

20%
6.3V
CERM
402-LF

1

2

MCP 3.3V AUX/USB Power

NV: 1x 4.7uF 0603, 1x 0.1uF 0402 (4.8 uF)
Apple: 1x 2.2uF 0402 (2.2 uF)

22 6

=PP3V3_S5_MCP

266 mA (A01)

C2560

2.2uF

20%
6.3V
CERM
402-LF

1

2

MCP 3.3V/1.5V HDA Power

NV: 1x 4.7uF 0603, 1x 0.1uF 0402 (4.8 uF)
Apple: 1x 2.2uF 0402 (2.2 uF)

21 6

=PP3V3R1V5_S0_MCP_HDA

7 mA (A01)

C2562

2.2uF

20%
6.3V
CERM
402-LF

1

2

MCP79 Ethernet VRef

=PP3V3_ENET_MCP_RMG

83 mA (A01)

C2564

2.2uF

20%
6.3V
CERM
402-LF

1

2

MCP MII VREF

R2591 1.47K 1% 1/16W NF-LF 402 2
R2590 1.47K 1% 1/16W NF-LF 402 2
C2591 0.1uF 20% 10V CERM 402

MCP Core Power

NV: 1x 10uF 0805, 1x 4.7uF 0402, 2x 1uF 0402, 2x 0.1uF 0402 (16.9 uF)
Apple: 5x 2.2uF 0402 (11 uF)

6

=PP1V05_S0_MCP_AVDD_UF

333 mA (A01)

L2570

30-OHM-5A

0603

1

2

C2570

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2571

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2572

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2573

2.2uF

20%
6.3V
CERM
402-LF

1

2

C2574

2.2uF

20%
6.3V
CERM
402-LF

1

2

MCP 1.05V RMGT Power

=PP1V05_ENET_MCP_RMG

131 mA (A01)

C2528

4.7uF

20%
4V
XSR
402

1

2

C2529

0.1uF

20%
10V
CERM
402

1

2

MCP PLL FSB

L2580 30-OHM-1.7A 0402
C2580 4.7uF 20% 4V XSR 402
C2581 4.7uF 20% 4V XSR 402

270 mA (A01)

MCP PLL PEX

L2582 30-OHM-1.7A 0402
C2582 4.7uF 20% 4V XSR 402
C2583 4.7uF 20% 4V XSR 402

84 mA (A01)

MCP PLL SATA

L2584 30-OHM-1.7A 0402
C2584 4.7uF 20% 4V XSR 402
C2585 4.7uF 20% 4V XSR 402

84 mA (A01)

MCP PLL CORE

L2586 30-OHM-1.7A 0402
C2586 4.7uF 20% 4V XSR 402
C2597 4.7uF 20% 4V XSR 402
C2587 4.7uF 20% 4V XSR 402

87 mA (A01)

MCP PLL NV

L2588 30-OHM-1.7A 0402
C2588 4.7uF 20% 4V XSR 402
C2589 4.7uF 20% 4V XSR 402
C2590 2.2uF 20% 4V XSR 402

37 mA (A01)

11650004

2

RES, 0.5%, 0402

C2574, C2518

IG

8

7

6

5

4

3

2

1

Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

8

7

6

5

4

3

2

1

SYNC MASTER=K22

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MCP Standard Decoupling

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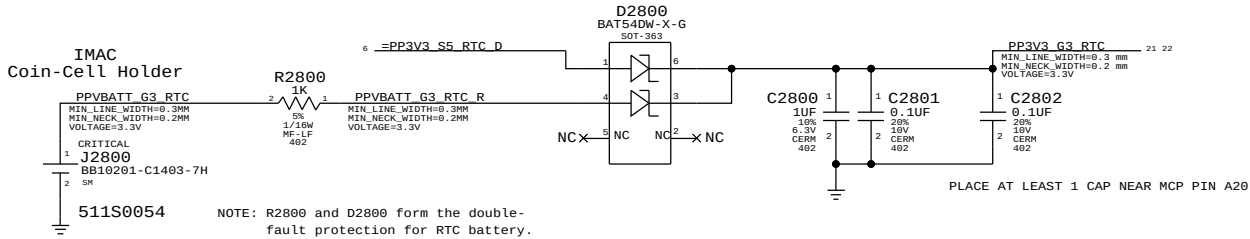
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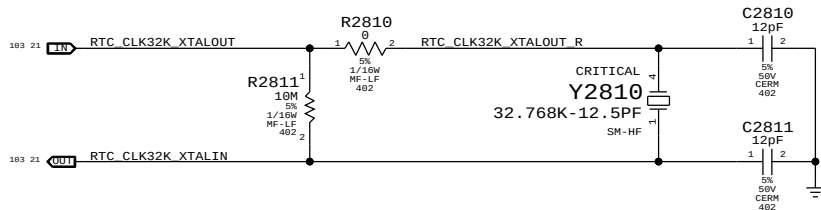
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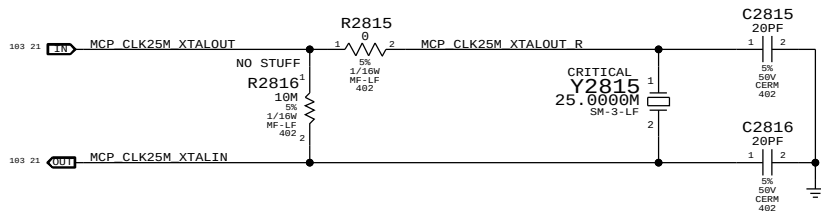
RTC Power Sources



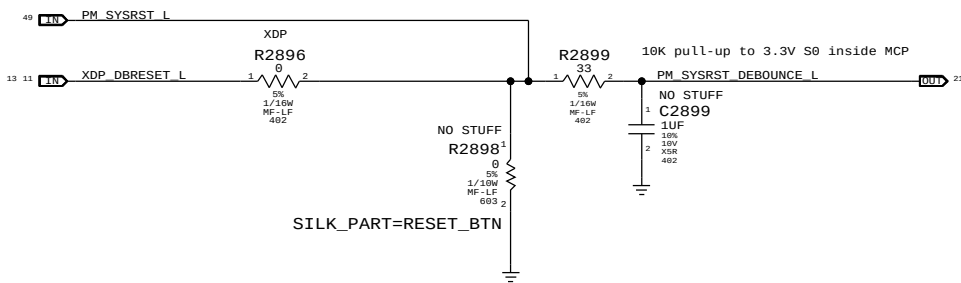
RTC Crystal



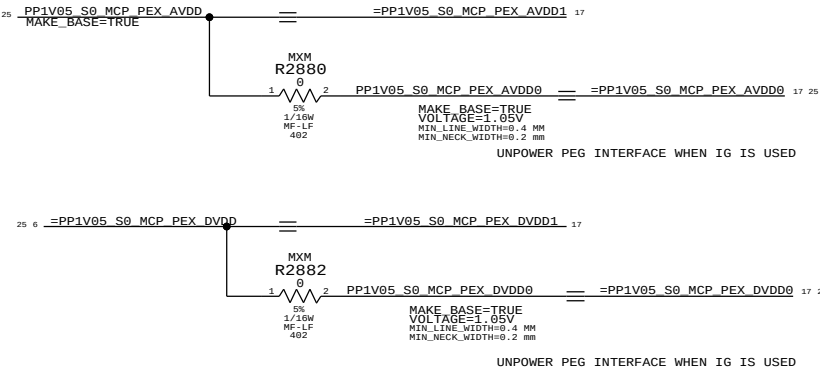
MCP 25MHz Crystal



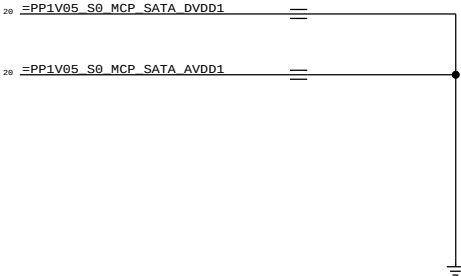
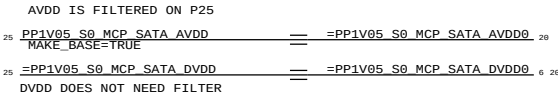
Reset Button



PEG POWER ALIAS/OPTION TO GND UNUSED POWER PIN



SATA ALIAS/GROUNDING UNUSED DVDD1 AND AVDD1



SYNC MASTER=K22		SYNC DATE=09/02/2009	
PAGE TITLE		SB Misc	
DRAWING NUMBER		051-7863	
REVISION		A.0.0	
BRANCH			
PAGE		28 OF 110	
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Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PP3V3_S5_VREFMRGN
- =PPVTT_S3_DDR_BUF

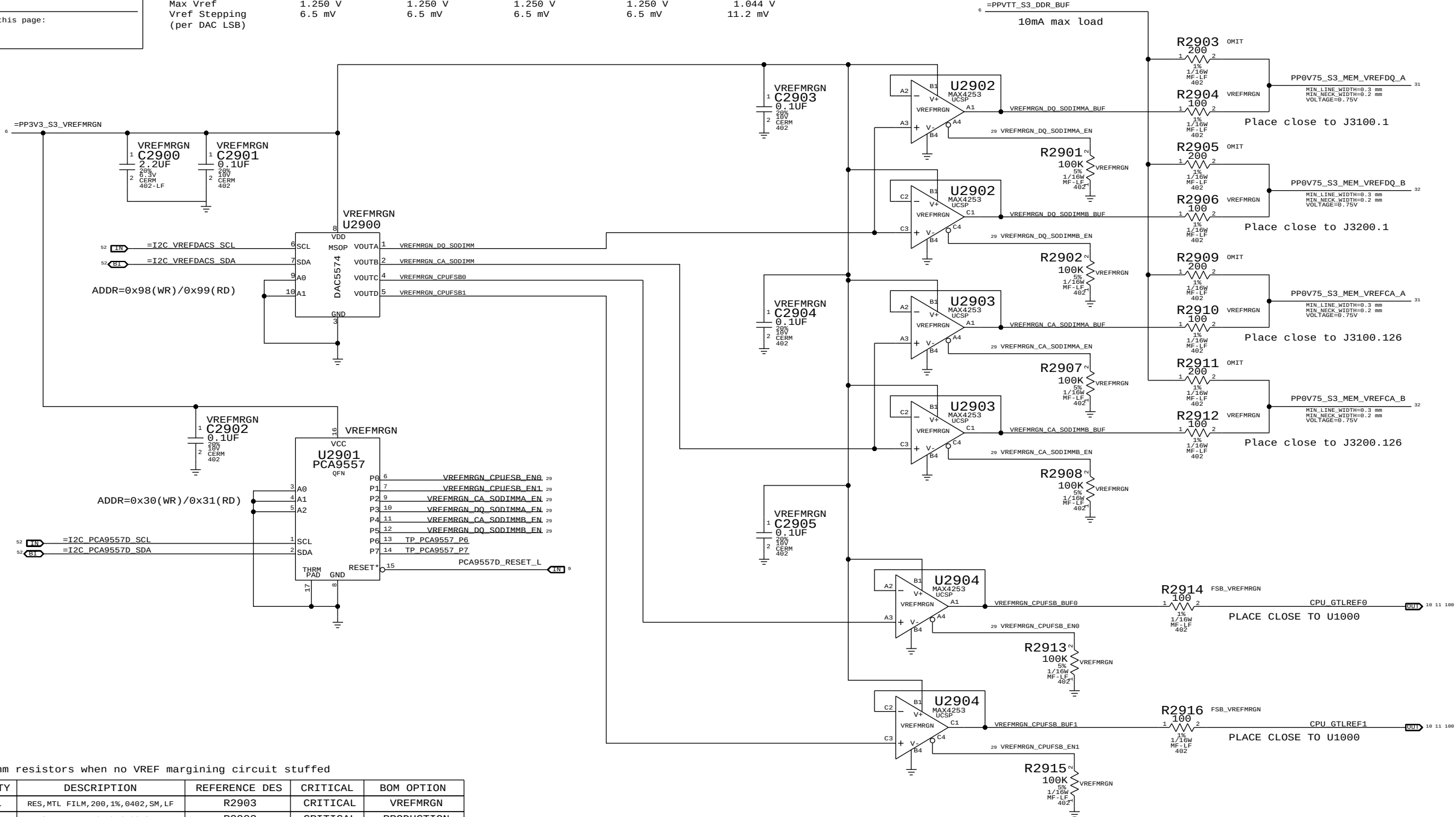
Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
VREFMRGN
PRODUCTION

DAC channel
Min DAC code
Max DAC code
Max sink I
Max source I
Nominal Vref
Min Vref
Max Vref
Vref Stepping
(per DAC LSB)

	MEM A VREF DQ	MEM A VREF CA	MEM B VREF DQ	MEM B VREF CA	CPU FSB VREF
A	0x00	0x00	0x00	0x00	0x00
B	0x87	0x87	0x87	0x87	0x55
Max sink I	-3.75 mA	-3.75 mA	-3.75 mA	-3.75 mA	-0.91 mA
Max source I	5 mA	5 mA	5 mA	5 mA	0.52 mA
Nominal Vref	0.75 V	0.75 V	0.75 V	0.75 V	0.70 V
Min Vref	0.375 V	0.375 V	0.375 V	0.375 V	0.091 V
Max Vref	1.250 V	1.250 V	1.250 V	1.250 V	1.044 V
Vref Stepping (per DAC LSB)	6.5 mV	6.5 mV	6.5 mV	6.5 mV	11.2 mV

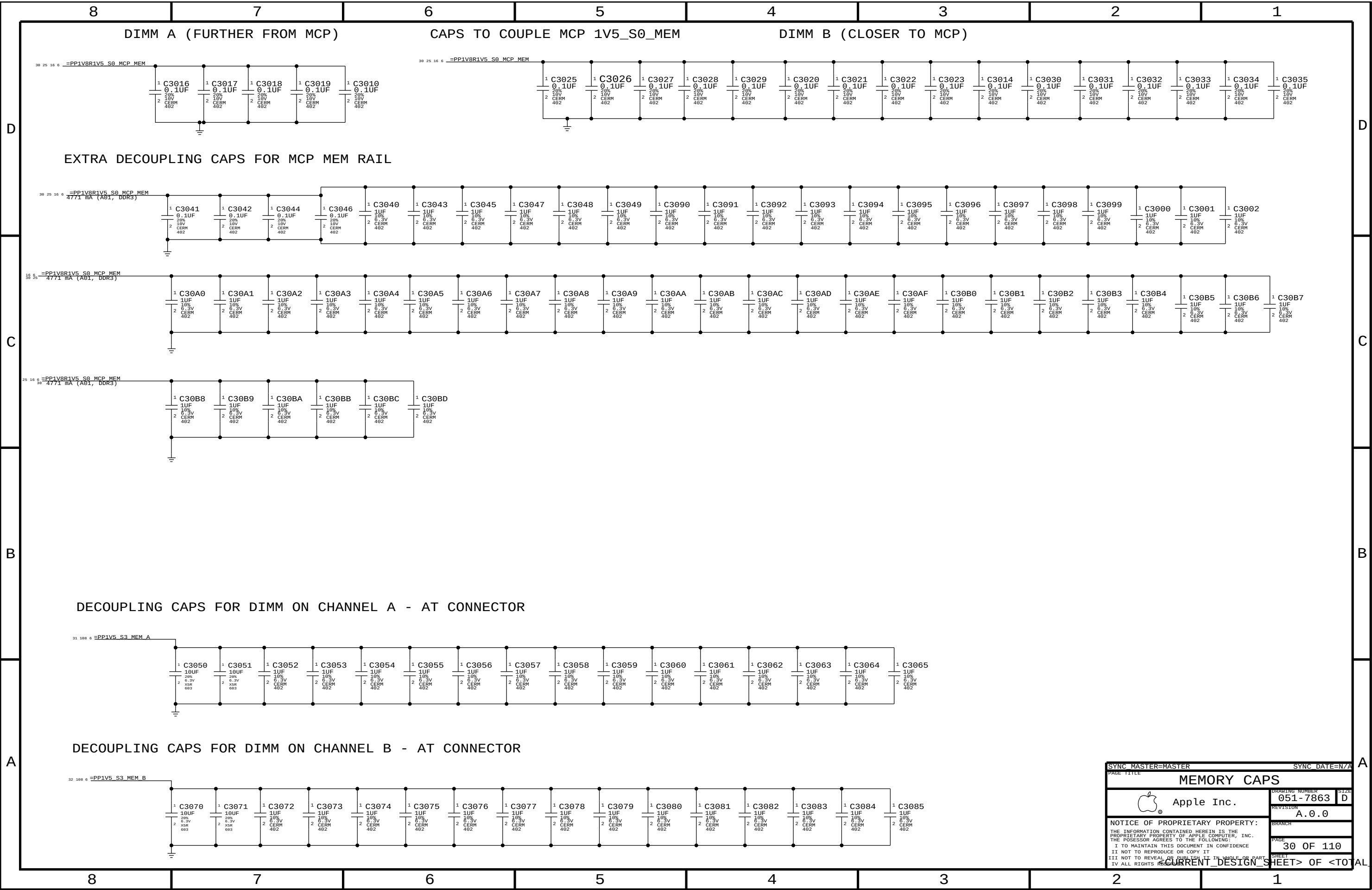
SO-DIMM A and SO-DIMM B Vref settings should be margined separately
(i.e. not simultaneously) due to current limitation of TPS51116 regulator.



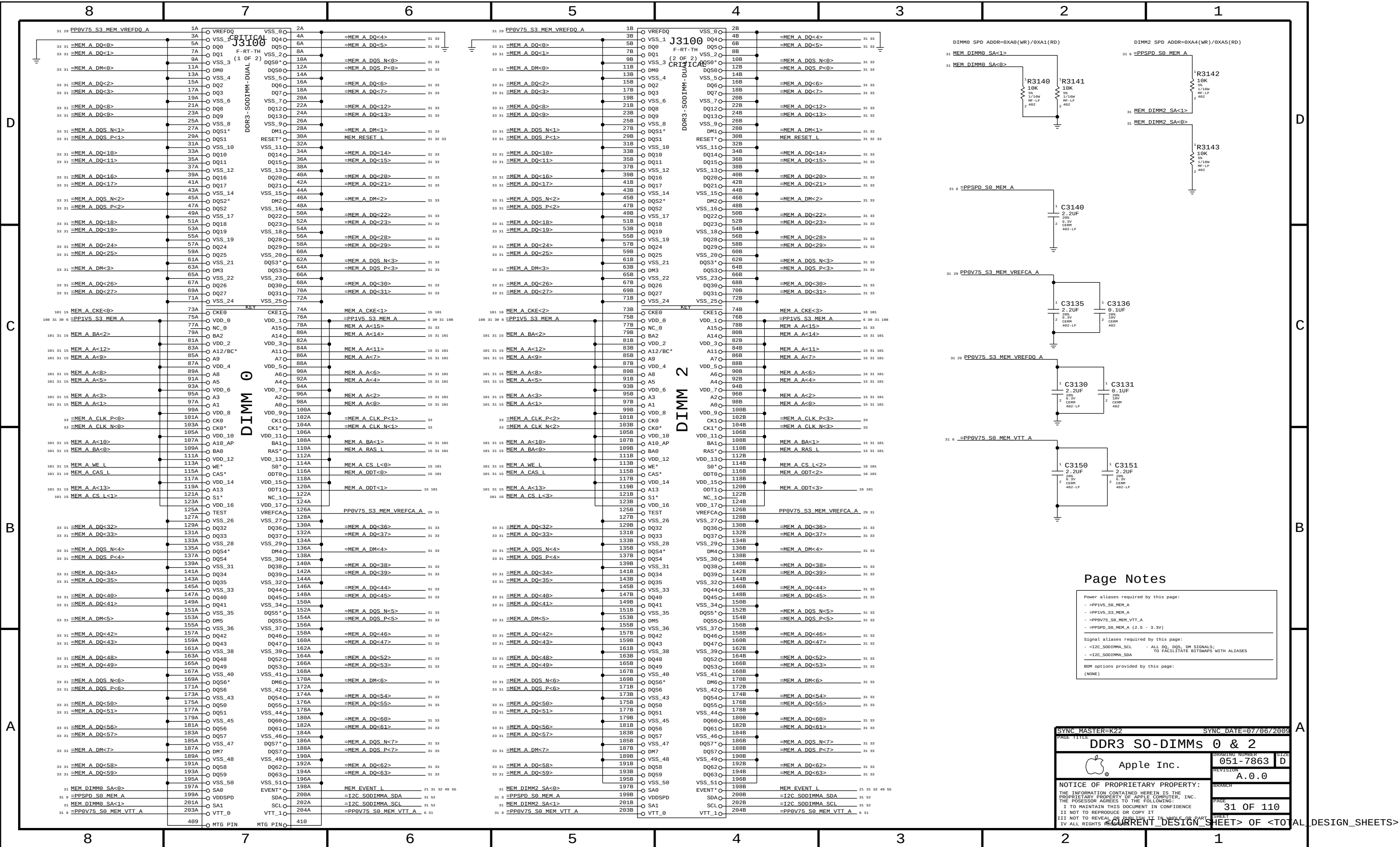
Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0149	1	RES,MTL FILM,200,1%,0402,SM,LF	R2903	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2903	CRITICAL	PRODUCTION
114S0149	1	RES,MTL FILM,200,1%,0402,SM,LF	R2905	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2905	CRITICAL	PRODUCTION
114S0149	1	RES,MTL FILM,200,1%,0402,SM,LF	R2909	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2909	CRITICAL	PRODUCTION
114S0149	1	RES,MTL FILM,200,1%,0402,SM,LF	R2911	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2911	CRITICAL	PRODUCTION

SYNC MASTER=K22		SYNC DATE=09/02/2009	
PAGE TITLE			
FSB/DDR3 Vref Margining			
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BRANCH			
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PAGE TITLE					
MEMORY CAPS					
	DRAWING NUMBER		SIZE		
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REVISION		A.0.0			
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

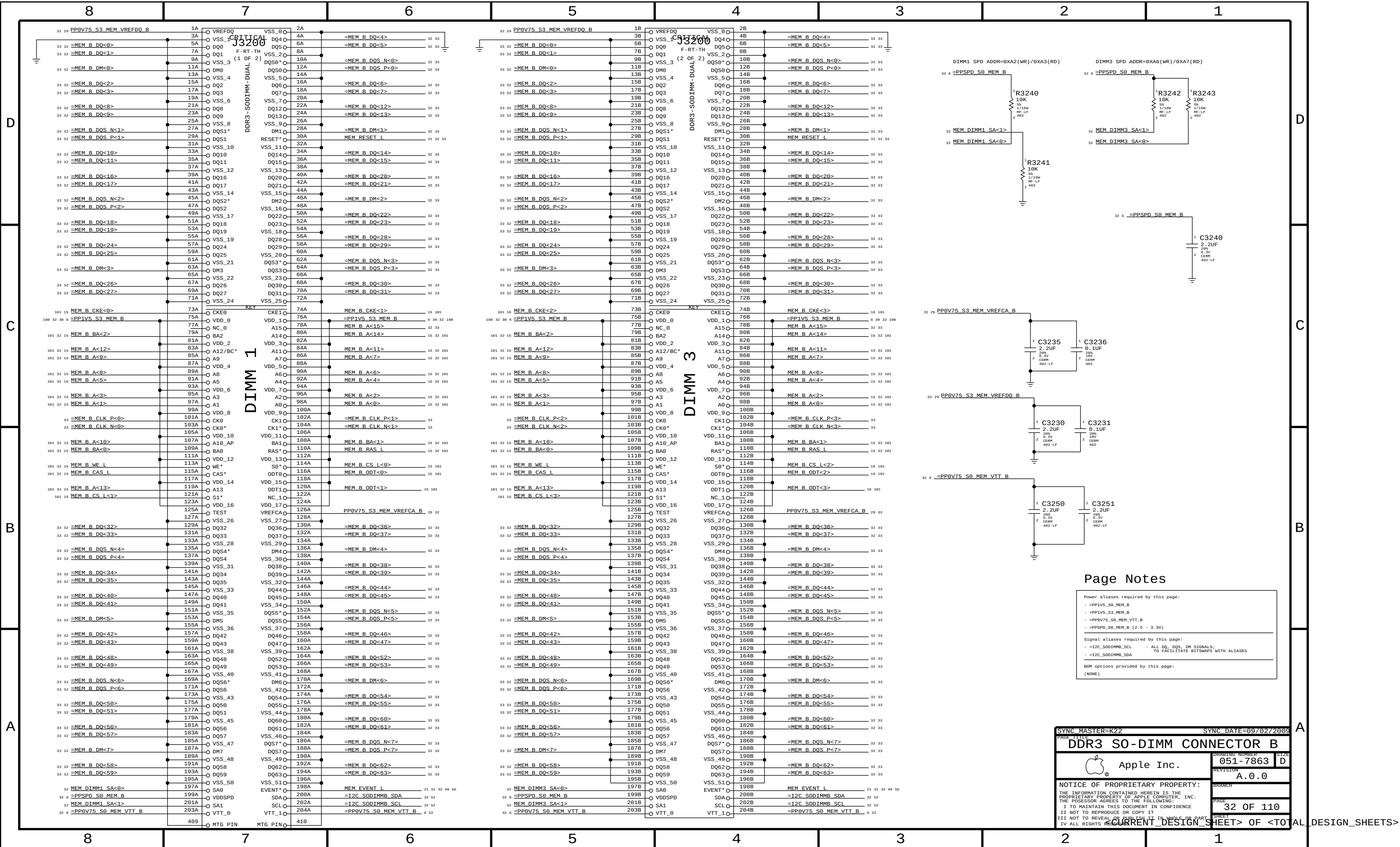
Signal aliases required by this page:

- =I2C_S0DIMMA_SCL - ALL DQ, DQS, DM SIGNALS: TO FACILITATE BITSLAPS WITH ALIASES
- =I2C_S0DIMMA_SDA

BOM options provided by this page:

(NONE)

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DDR3 SO-DIMMS 0 & 2		DRAWING NUMBER	
Apple Inc.		051-7863	
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0DIMMB_SCL - ALL DQ, DQS, DM SIGNALS: TO FACILITATE BITSLAPS WITH ALIASES
- =I2C_S0DIMMB_SDA

BOM options provided by this page:

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SYNC MASTER=K22

SYNC DATE=09/02/2009

DDR3 SO-DIMM CONNECTOR B

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SHEET

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D									D
C									C
B									B
A									A
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SYNC DATE=12/02/2008

PAGE TITLE

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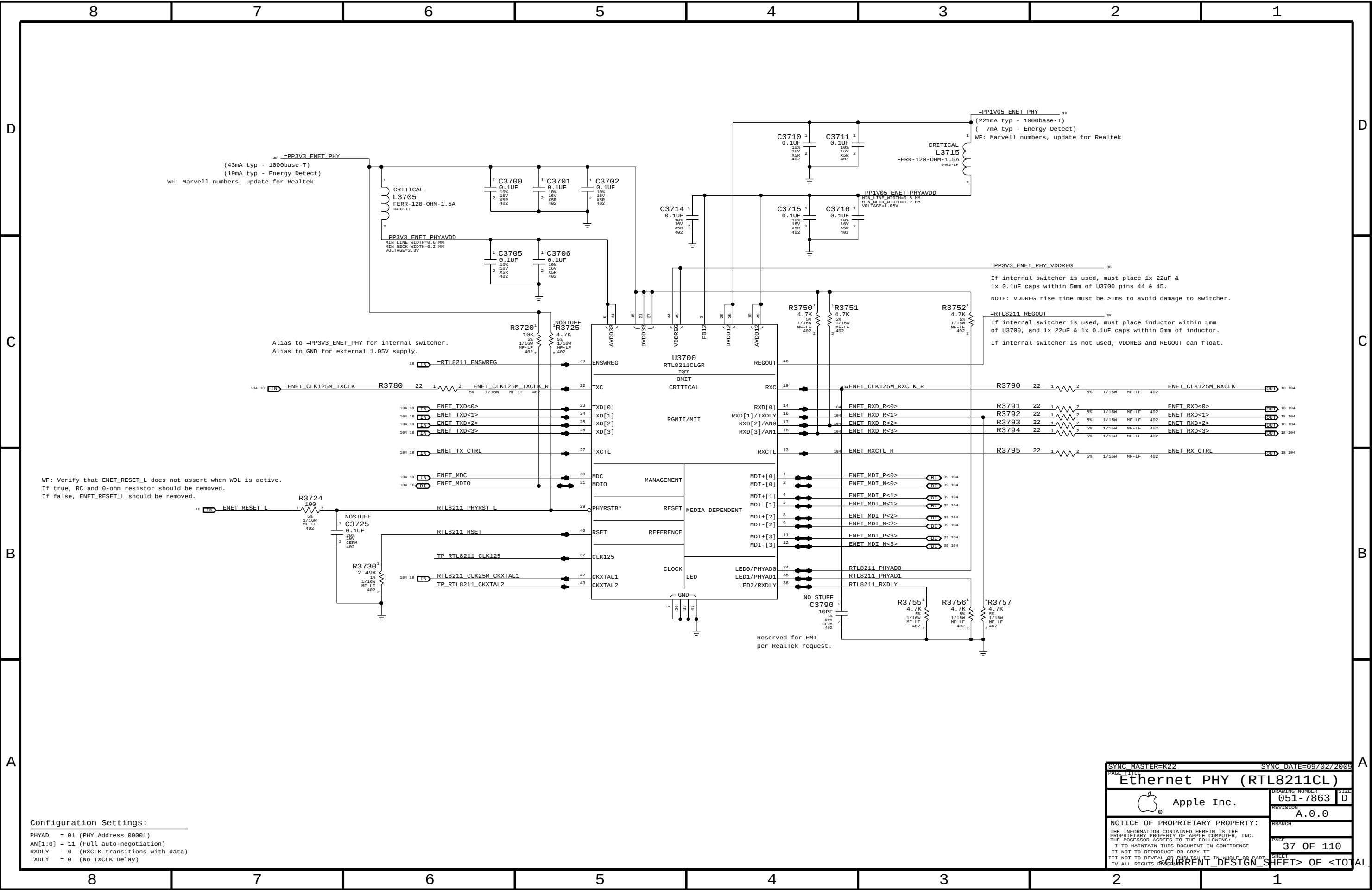
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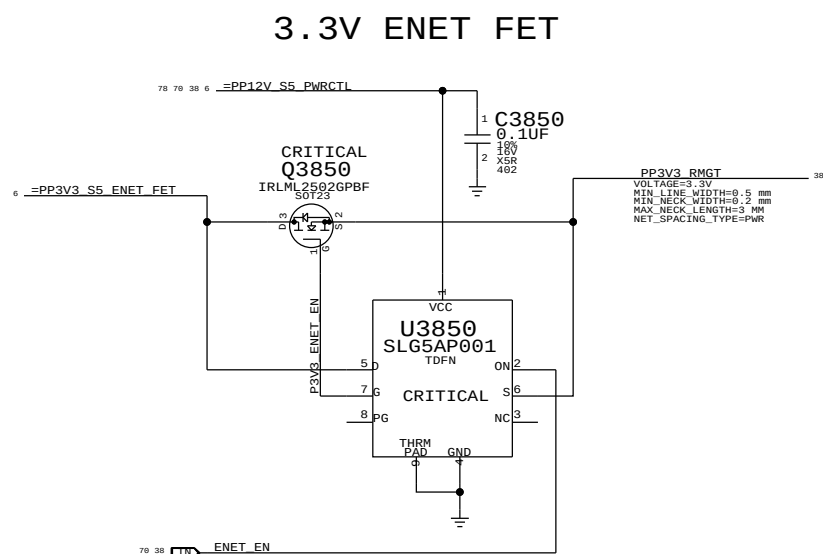
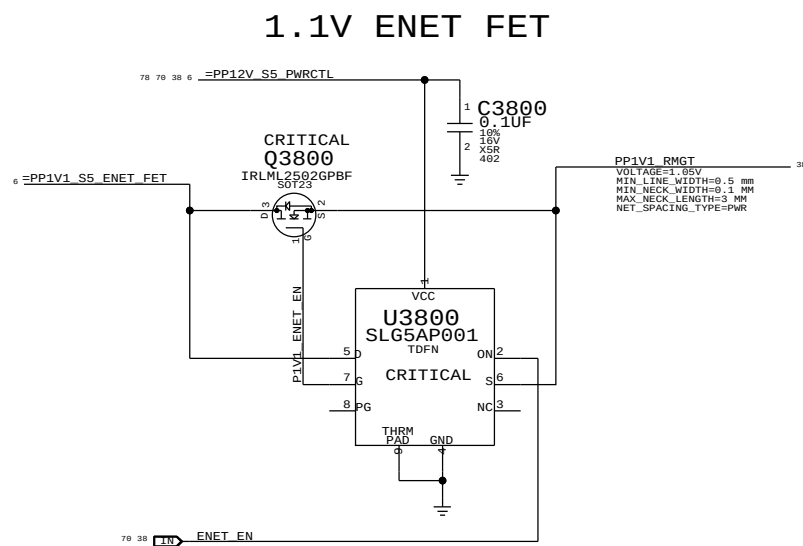
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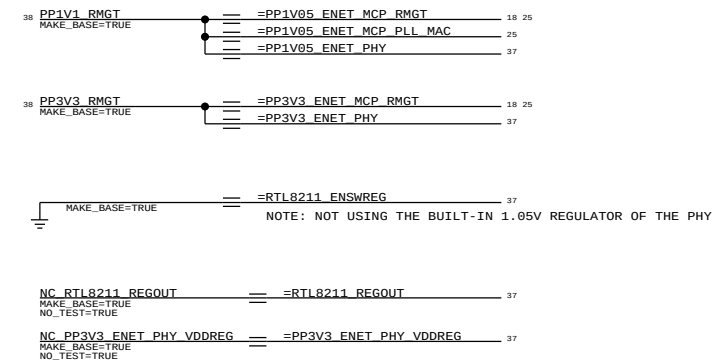
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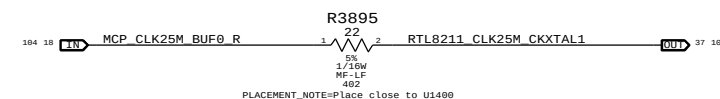


ENET ALIASES

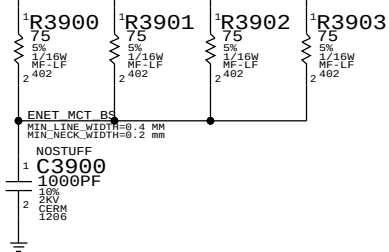
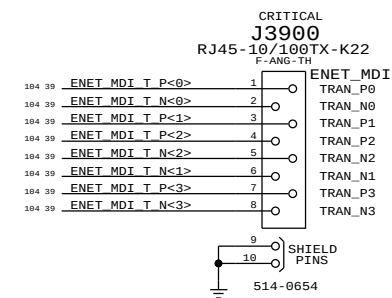


RTL8211 25MHz Clock

NOTE: MCP79 can provide 25MHz clock, but clock runs whenever RMGT rails are powered. Designs must ensure PHY is powered whenever RMGT rails are, or use separate crystal.



SYNC MASTER=K22		SYNC DATE=09/02/2009	
PAGE TITLE			
Ethernet Support			
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		38 OF 110	<ELECTRONIC DESIGN SHEET> OF <TOTAL DESIGNS SHEETS>



NOTE: BOB SMITH TERMINATION FOR EMC.

SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
ETHERNET CONNECTOR			
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SHEET 39 OF 110		SHEET 39 OF 110	

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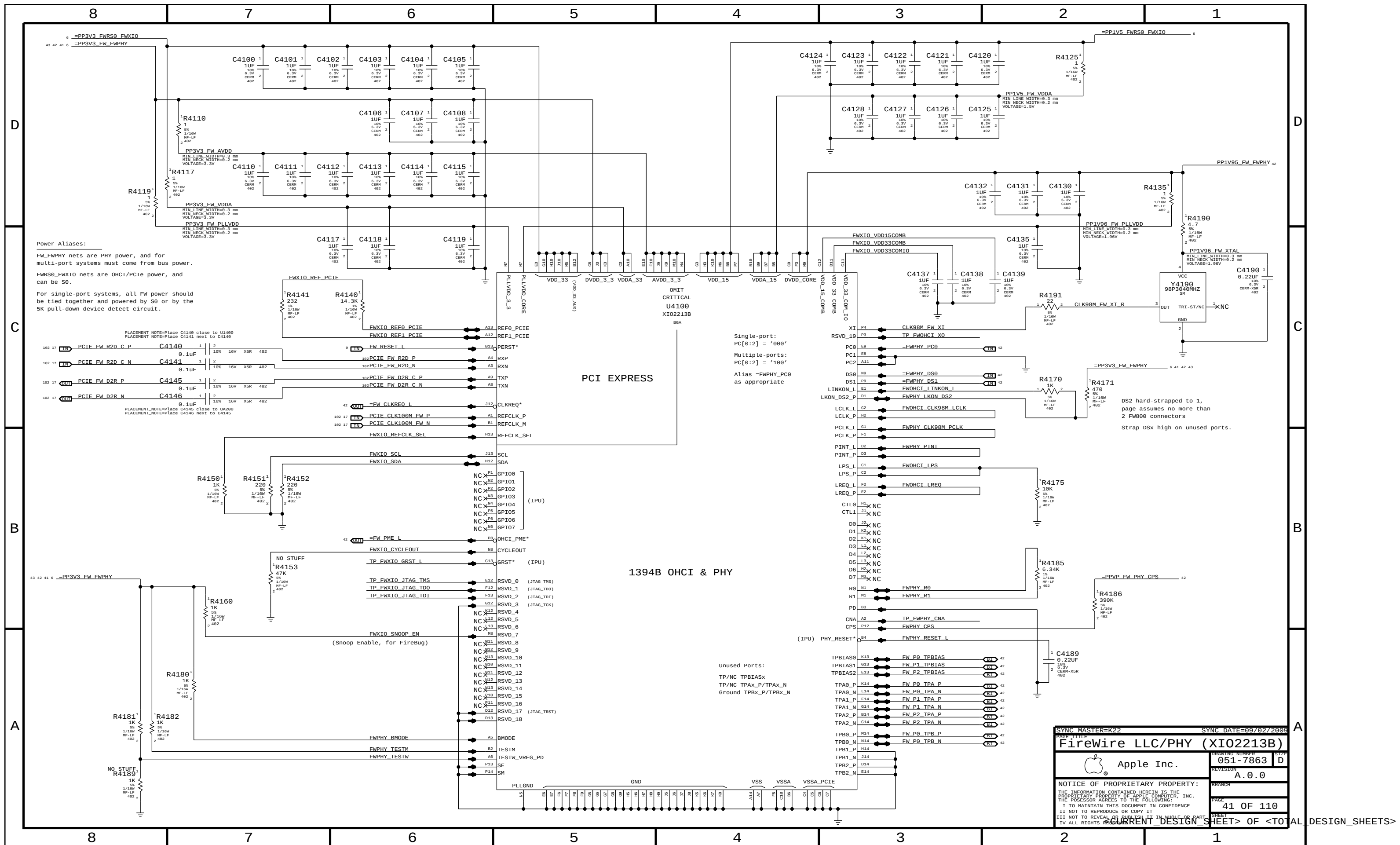
40 OF 110

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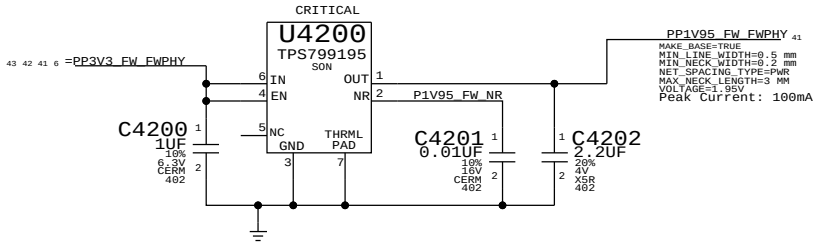
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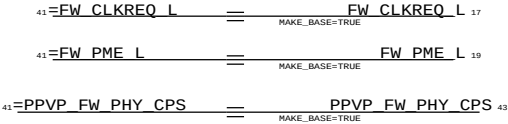
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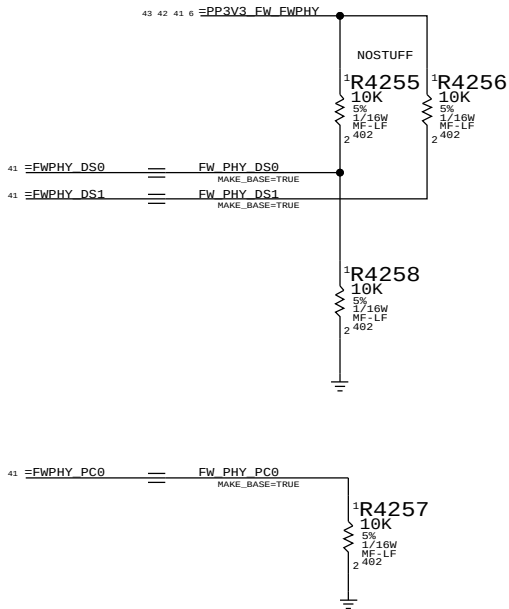
1394 PHY 1.95V SUPPLY



FireWire Aliases For Connectivity



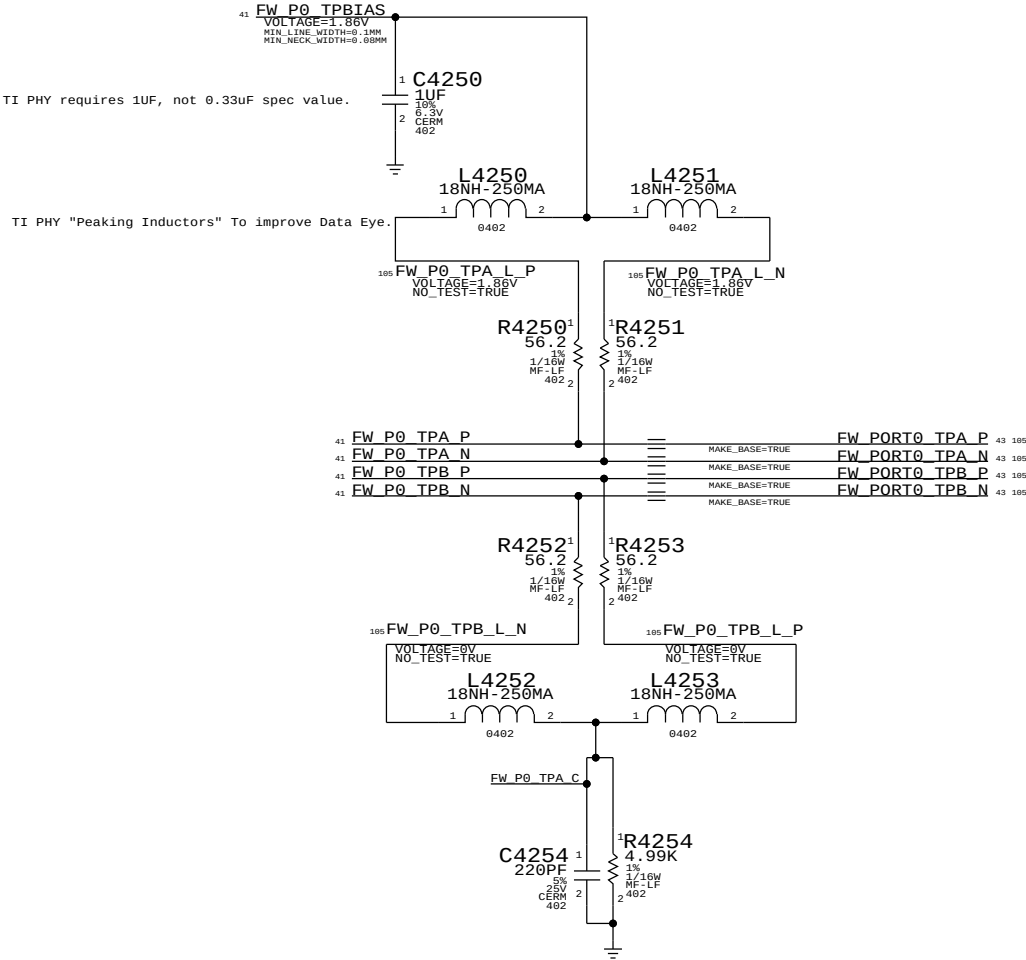
1394 PHY STRAPPING OPTIONS



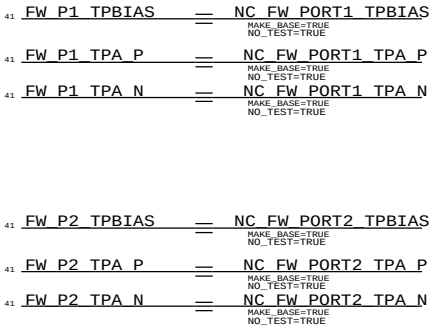
THERE ARE THREE FIREWIRE PORTS, BUT ONLY ONE IS USED.NO STUFF MEANS THAT IT IS IN BILINGUL MODE PULL-UPS ASSERT/ENABLE DATA STROBE ONLY MODE.

iMacs are now one port only and have Power Code "000"

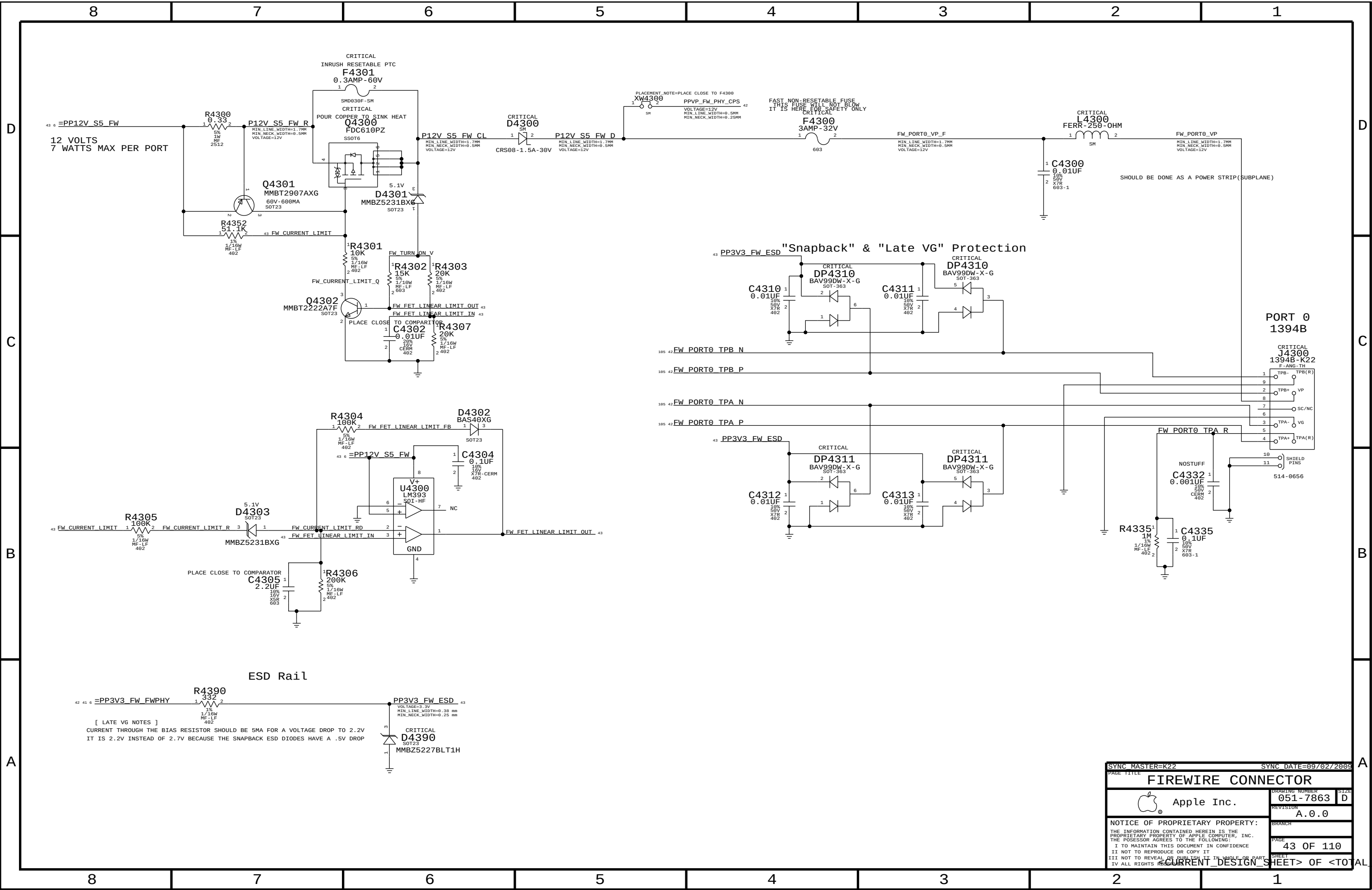
Termination
Place close to FireWire PHY



2ND & 3RD TPA/TPB PAIR UNUSED



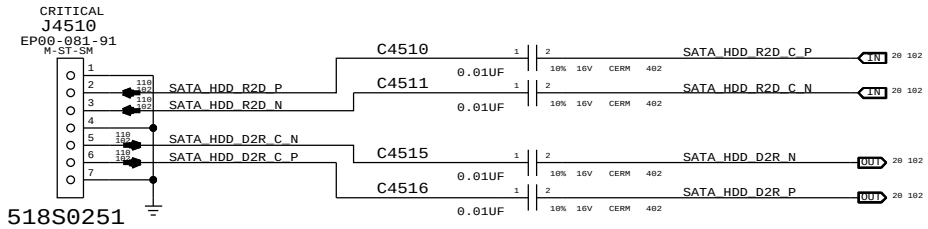
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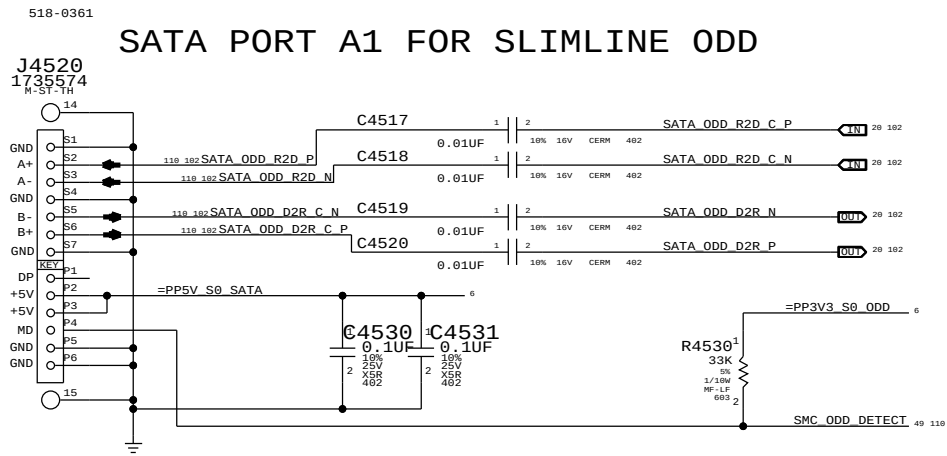
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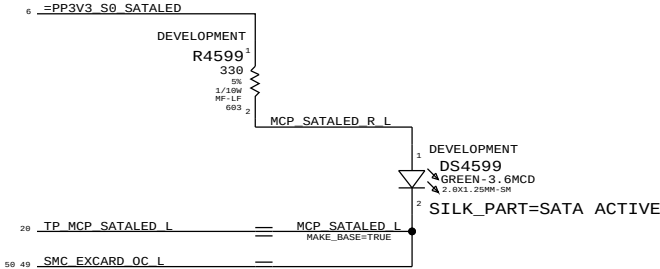
SATA PORT A0 FOR HDD

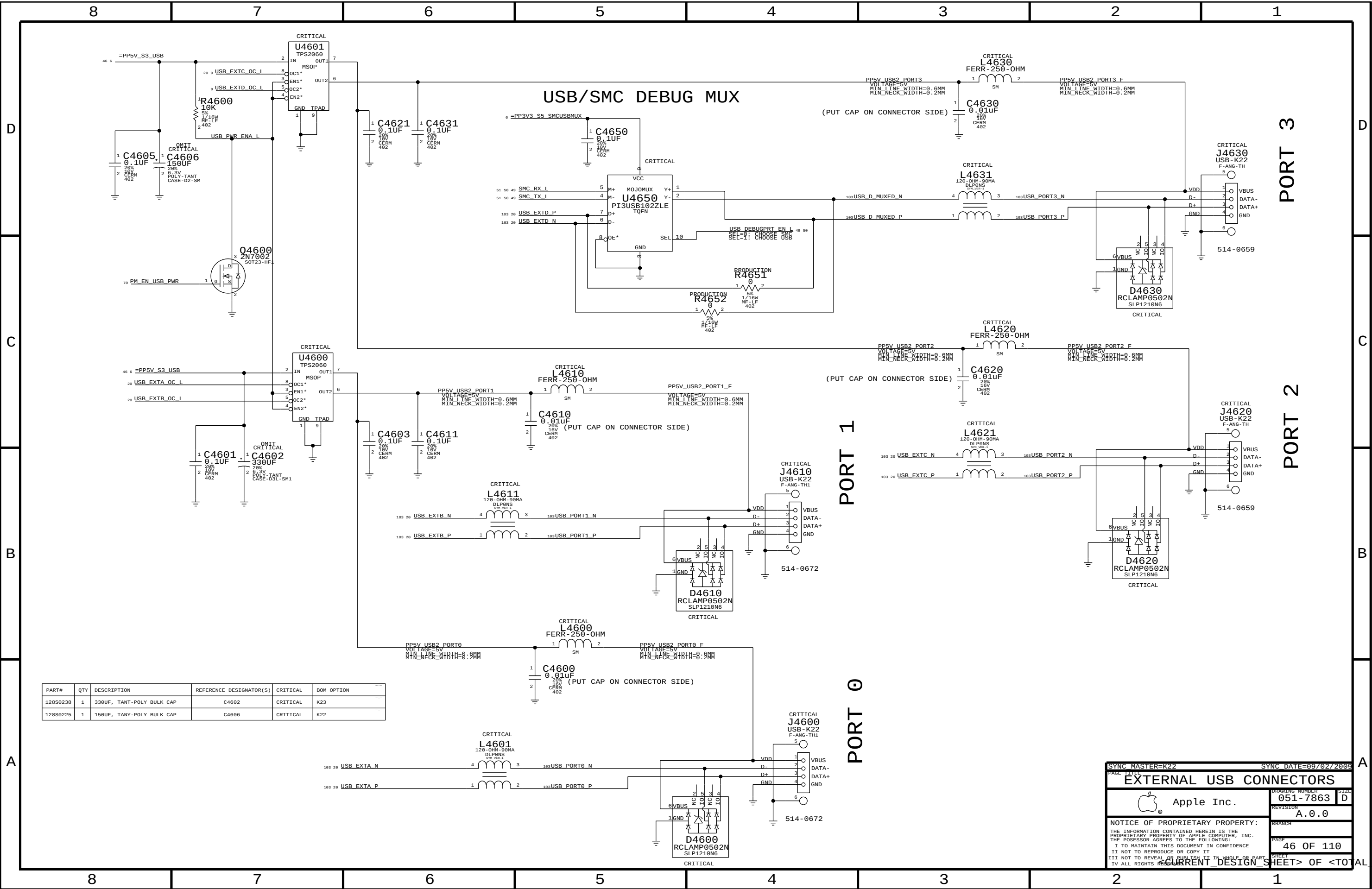


SATA PORT A1 FOR SLIMLINE ODD



SATA Activity LED





SYNC MASTER=K22

SYNC DATE=09/02/2009

EXTERNAL USB CONNECTORS

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<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>

CAMERA CONNECTOR & FILTER

LAYOUT NOTE:
PLACE C4700, C4701 & L4700
NEAR J4700 PINS 4 AND 5 IN THE
ORDER LISTED, AND NOT ON
BOTH SIDES OF THE PIN.

K37L (BLUETOOTH) CONNECTOR

IR RECEIVER CONNECTOR

SD Card Reader Board Connector

SYNC MASTER=K22		SYNC DATE=09/02/2009	
PAGE TITLE		Internal USB Connections	
DRAWING NUMBER		051-7863	SIZE D
REVISION		A.0.0	
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SHEET			
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CAMERA CONNECTOR & FILTER

LAYOUT NOTE:
PLACE C4700, C4701 & L4700
NEAR J4700 PINS 4 AND 5 IN THE
ORDER LISTED, AND NOT ON
BOTH SIDES OF THE PIN.

K37L (BLUETOOTH) CONNECTOR

IR RECEIVER CONNECTOR

SD Card Reader Board Connector

SYNC MASTER=K22		SYNC DATE=09/02/2009	
Internal USB Connections			
Apple Inc.		DRAWING NUMBER	051-7863
		REVISION	A.0.0
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CAMERA CONNECTOR & FILTER

LAYOUT NOTE:
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ORDER LISTED, AND NOT ON
BOTH SIDES OF THE PIN.

K37L (BLUETOOTH) CONNECTOR

IR RECEIVER CONNECTOR

SD Card Reader Board Connector

SYNC MASTER=K22		SYNC DATE=09/02/2009	
PAGE TITLE		Internal USB Connections	
DRAWING NUMBER		051-7863	SIZE D
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CAMERA CONNECTOR & FILTER

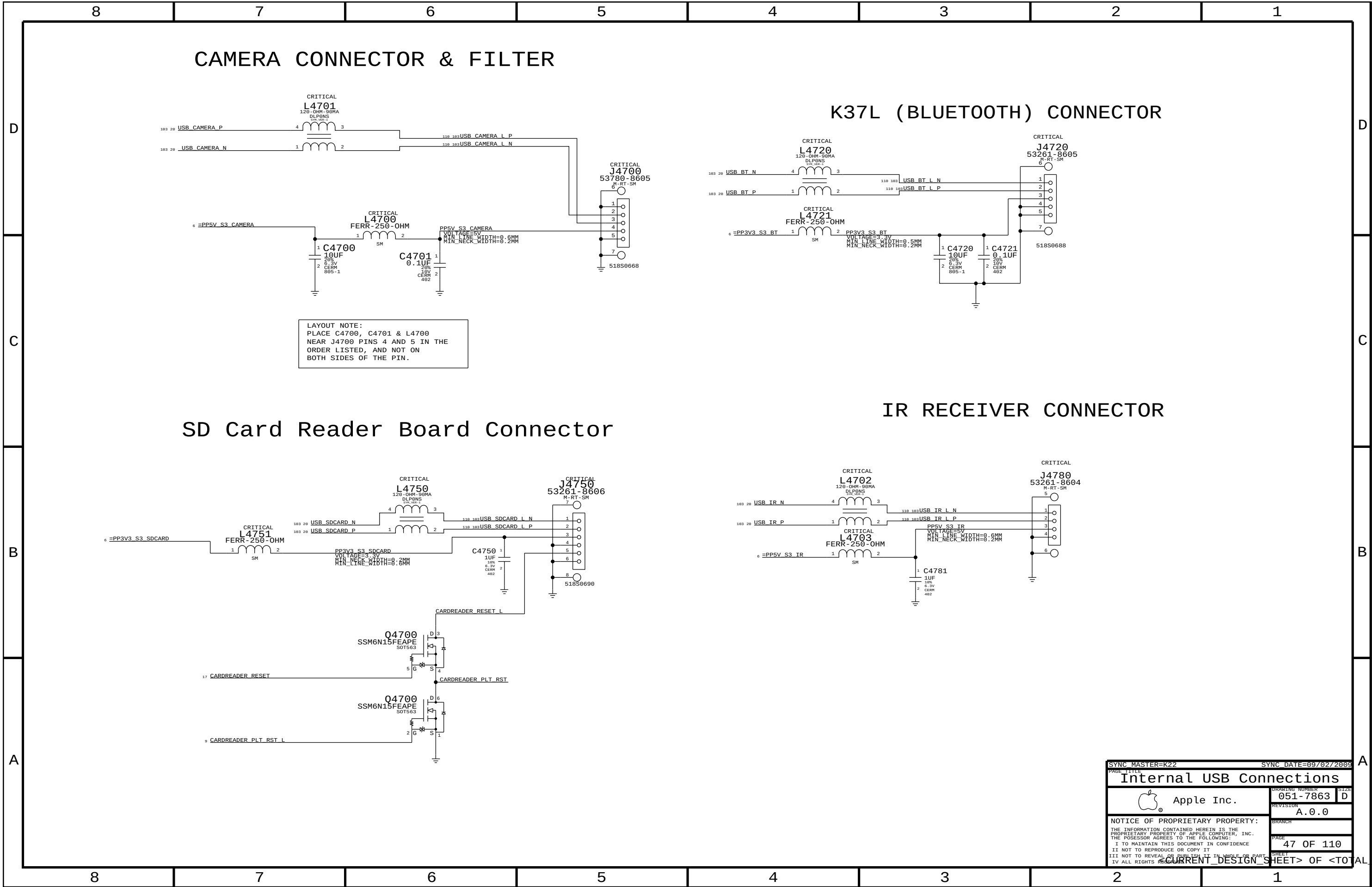
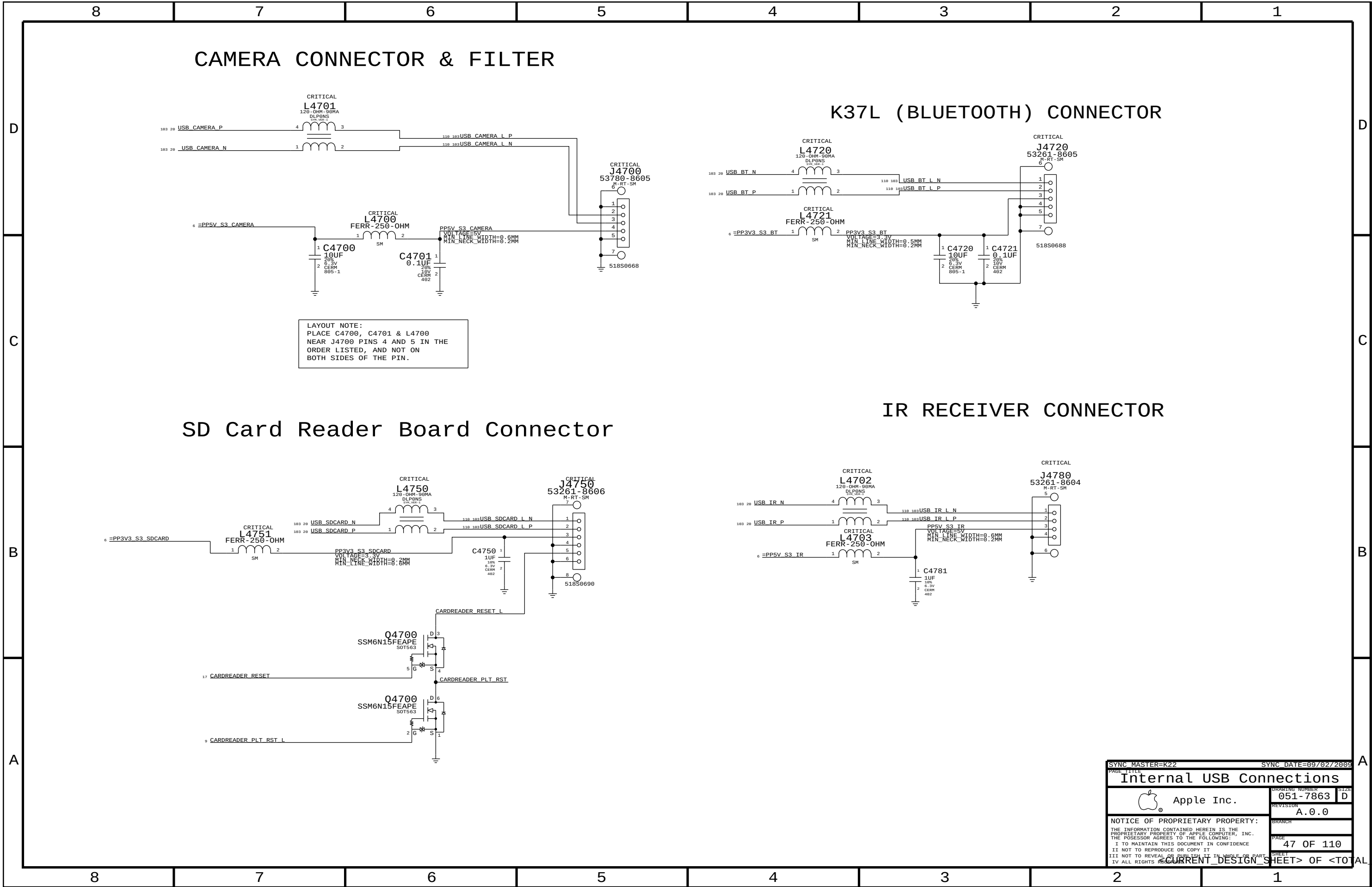
LAYOUT NOTE:
PLACE C4700, C4701 & L4700
NEAR J4700 PINS 4 AND 5 IN THE
ORDER LISTED, AND NOT ON
BOTH SIDES OF THE PIN.

K37L (BLUETOOTH) CONNECTOR

IR RECEIVER CONNECTOR

SD Card Reader Board Connector

SYNC MASTER=K22		SYNC DATE=09/02/2009	
PAGE TITLE		Internal USB Connections	
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CAMERA CONNECTOR & FILTER

LAYOUT NOTE:
PLACE C4700, C4701 & L4700
NEAR J4700 PINS 4 AND 5 IN THE
ORDER LISTED, AND NOT ON
BOTH SIDES OF THE PIN.

K37L (BLUETOOTH) CONNECTOR

SD Card Reader Board Connector

IR RECEIVER CONNECTOR

SYNC MASTER=K22		SYNC DATE=09/02/2009	
PAGE TITLE		Internal USB Connections	
DRAWING NUMBER		051-7863	SIZE D
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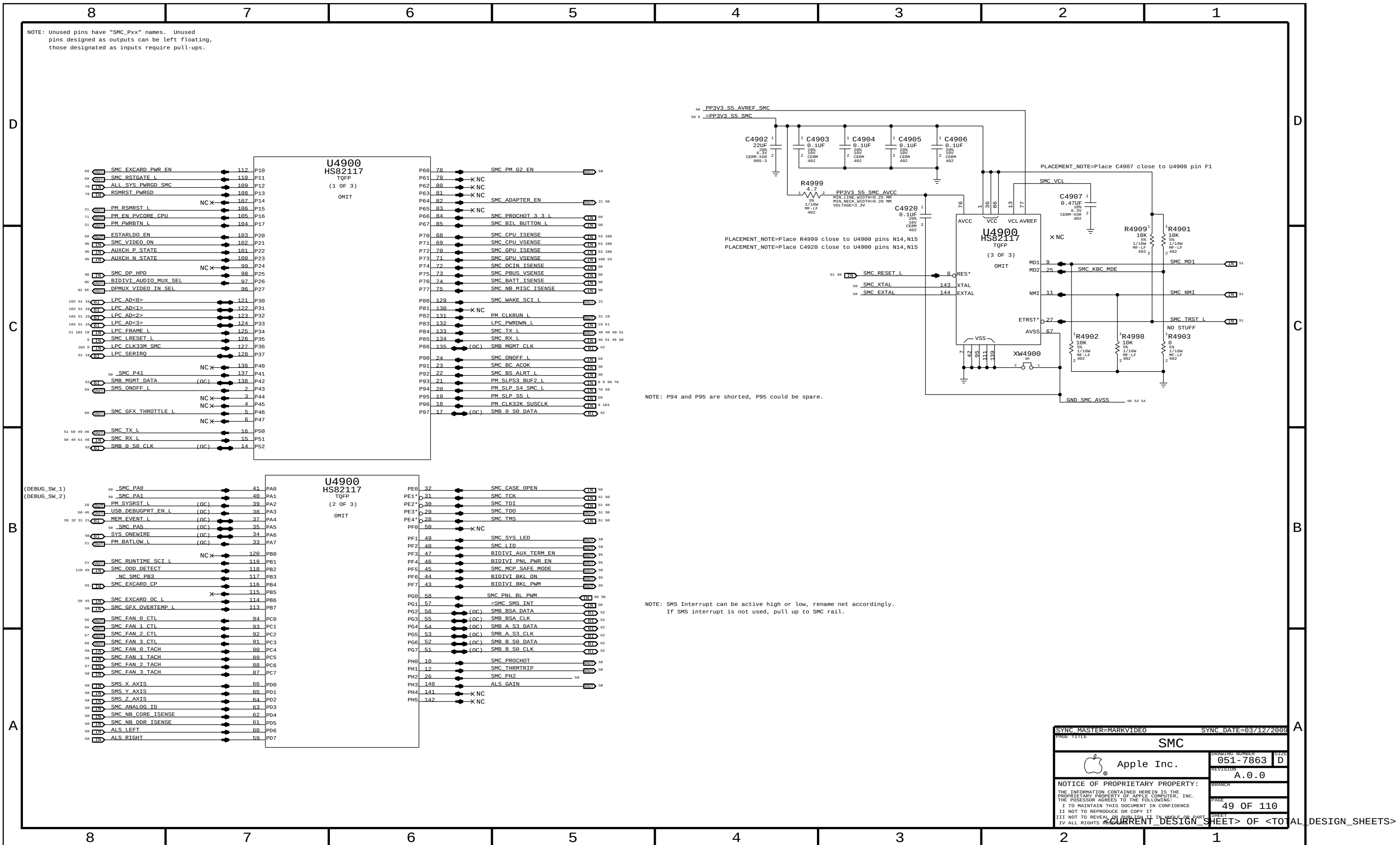
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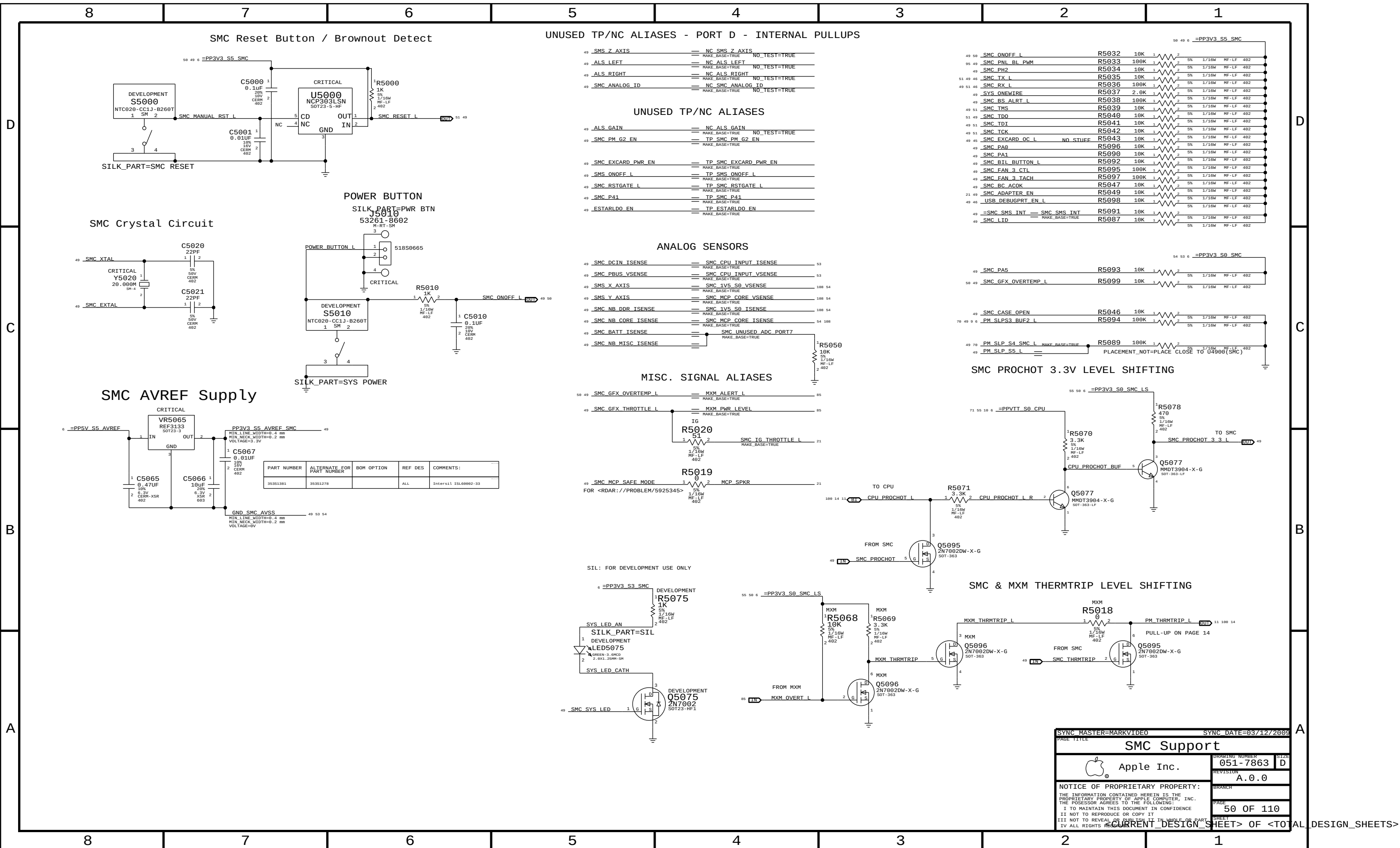
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LPC+SPI Connector

FRANK CONNECTOR
CRITICAL
LPCPLUS
J5100
55909-0374
M-ST-SM

=PP3V3 S5 LPCPLUS
=PP5V S0 LPCPLUS

LPC AD<0>
LPC AD<1>

SPI ALT MOSI
SPI ALT MISO
LPC FRAME L
PM CLKRUN L
SMC TMS
DEBUG RESET L
SMC TDQ
SMC TRST L
SMC MD1
SMC TX L

LPC CLK33M LPCPLUS
LPC AD<2>
LPC AD<3>

SPIROM USE MLB
SPI ALT CLK
SPI ALT CS L
LPC SERIRQ
LPC PWRDWN L
SMC TDI
SMC TCK
SMC RESET L
SMC NMI
SMC RX L
LPCPLUS GPIO

516S0573

Alternate SPI ROM Support

=PP3V3 S5 LPCPLUS
=PP3V3 S5 ROM

R5144 20K 1/16W MF-LF 402

C5144 0.1UF 20V CERM 402

SPI MLB CS L

SPI ALT CS L

Pull-up on debug card

LPCPLUS

U5100 NC7SB3157PAG 196 SC78 SEL 6 VER 1 CRITICAL PRODUCTION

R5146 1/16W MF-LF 402

PLACEMENT_NOTE=PLACE NEXT TO U5100

SPIROM USE MLB

=SPE CS1 R L USE MLB

R5145 100K 5% 1/16W MF-LF 402

PLACEMENT_NOTE=Place near U1400

SPI Bus Series Resistance Option

LPCPLUS

R5156 1/16W MF-LF 402

SPI ALT CLK

PLACEMENT_NOTE=Place next to R6150

SPI CLK R

LPCPLUS

R5157 1/16W MF-LF 402

SPI ALT MOSI

PLACEMENT_NOTE=Place next to R6152

SPI MOSI R

LPCPLUS

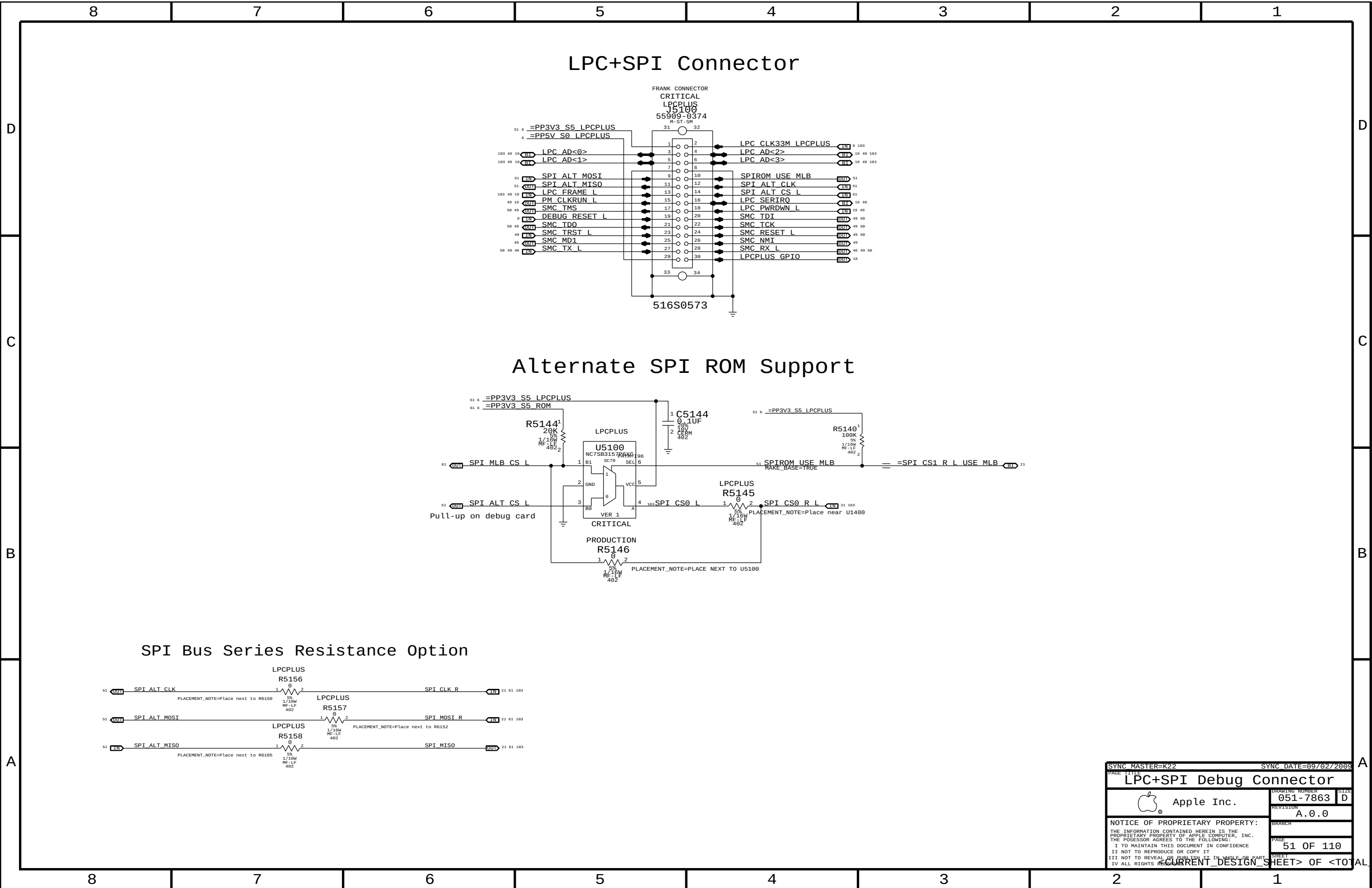
R5158 1/16W MF-LF 402

SPI ALT MISO

PLACEMENT_NOTE=Place next to R6105

SPI MISO

PAGE TITLE		SYNC MASTER=K22		SYNC DATE=09/02/2003	
LPC+SPI Debug Connector					
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LPC+SPI Connector

FRANK CONNECTOR
CRITICAL
LPCPLUS
J5100
55909-0374
M-ST-SM

51 6 =PP3V3 S5 LPCPLUS
6 =PP5V S0 LPCPLUS

103 49 19 B1 LPC AD<0>
103 49 19 B2 LPC AD<1>

51 51 B1 SPI ALT MOSI
51 51 B2 SPI ALT MISO
103 49 19 B1 LPC FRAME L
103 49 19 B2 PM CLKRUN L
50 49 50 B1 SMC TMS
50 49 50 B2 DEBUG RESET L
50 49 50 B1 SMC TD0
50 49 50 B2 SMC TRST L
49 50 49 B1 SMC MD1
49 50 49 B2 SMC TX L

1 2 LPC CLK33M LPCPLUS
3 4 LPC AD<2>
5 6 LPC AD<3>

9 10 SPIROM USE MLB
11 12 SPI ALT CLK
13 14 SPI ALT CS L
15 16 LPC SERIRQ
17 18 LPC PWRDWN L
19 20 SMC TDI
21 22 SMC TCK
23 24 SMC RESET L
25 26 SMC NMI
27 28 SMC RX L
29 30 LPCPLUS GPIO

516S0573

Alternate SPI ROM Support

51 6 =PP3V3 S5 LPCPLUS
61 6 =PP3V3 S5 ROM

R5144 20K
1/16W MF-LF 402

51 51 B1 SPI MLB CS L
51 51 B2 SPI ALT CS L

1 2 C5144 0.1UF
2 2 TERM 402

51 51 B1 SPIROM USE MLB
51 51 B2 MAKE_BASE=TRUE

51 51 B1 SPI CS0 L
51 51 B2 SPI CS0 R L

R5145 100K
1/16W MF-LF 402

51 51 B1 SPI CS0 L
51 51 B2 SPI CS0 R L

R5146 100K
1/16W MF-LF 402

51 51 B1 SPI CS0 L
51 51 B2 SPI CS0 R L

SPI Bus Series Resistance Option

LPCPLUS
R5156
5% 1/16W MF-LF 402

51 51 B1 SPI ALT CLK
51 51 B2 SPI CLK R

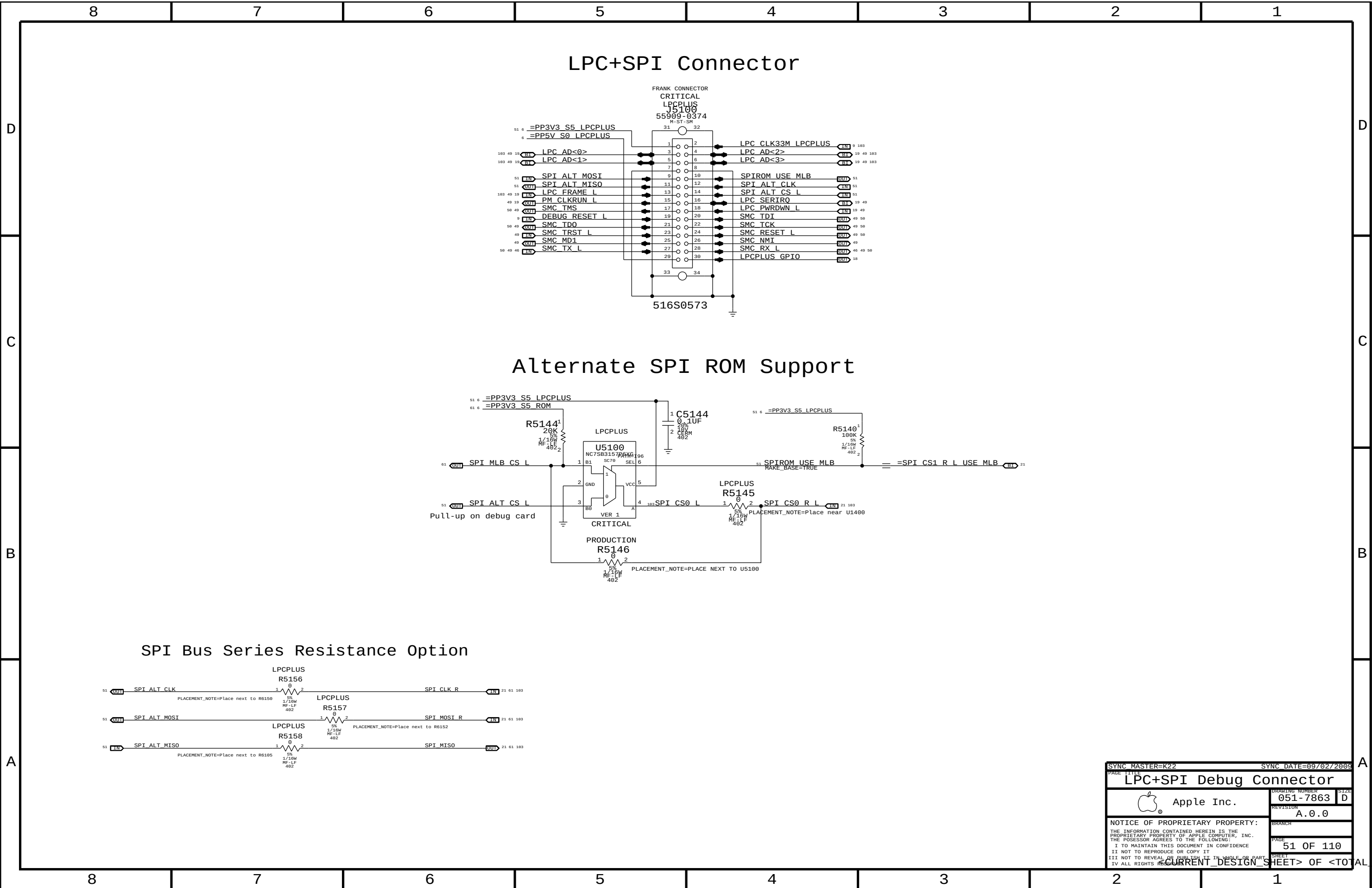
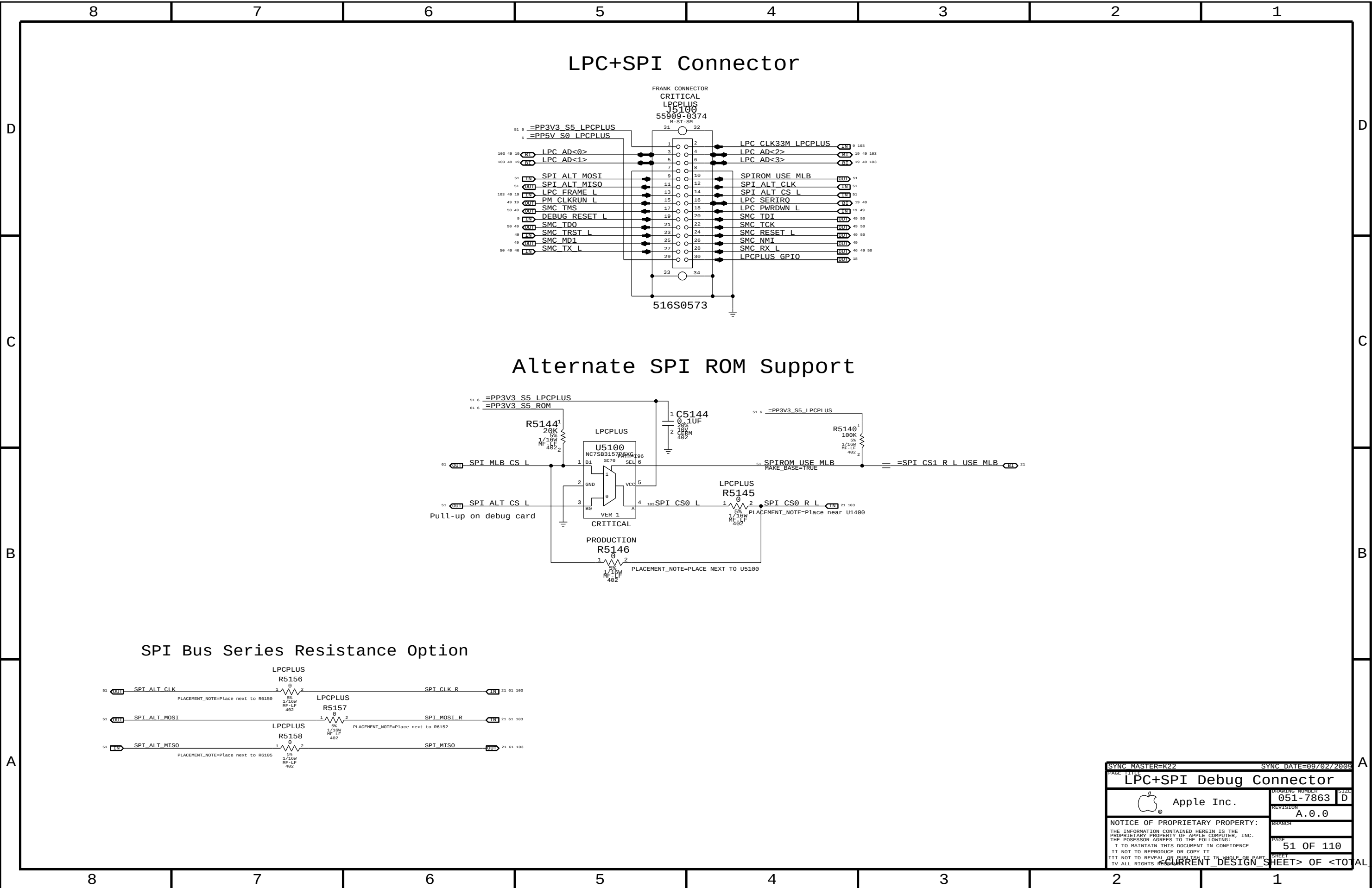
LPCPLUS
R5157
5% 1/16W MF-LF 402

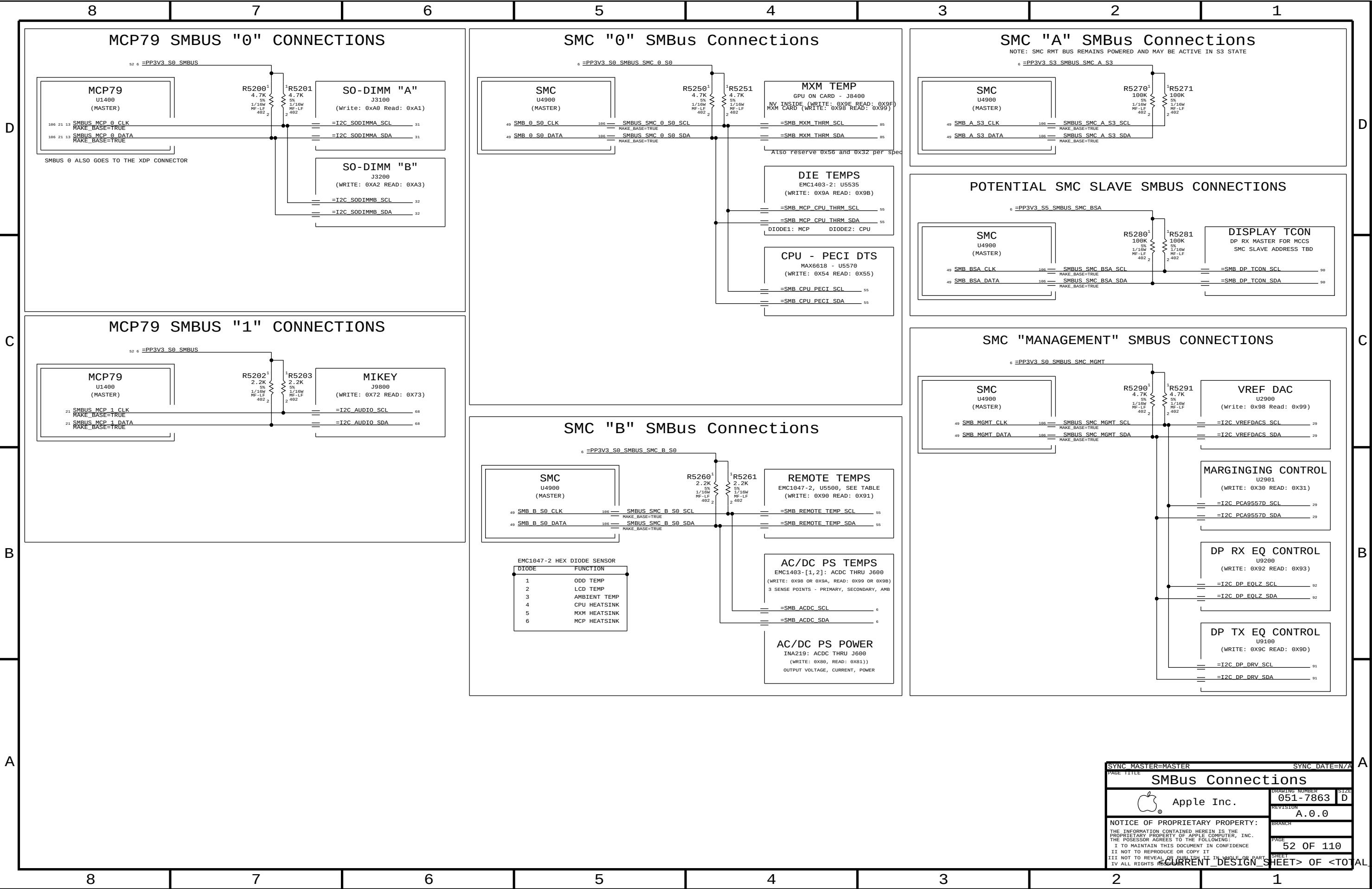
51 51 B1 SPI ALT MOSI
51 51 B2 SPI MOSI R

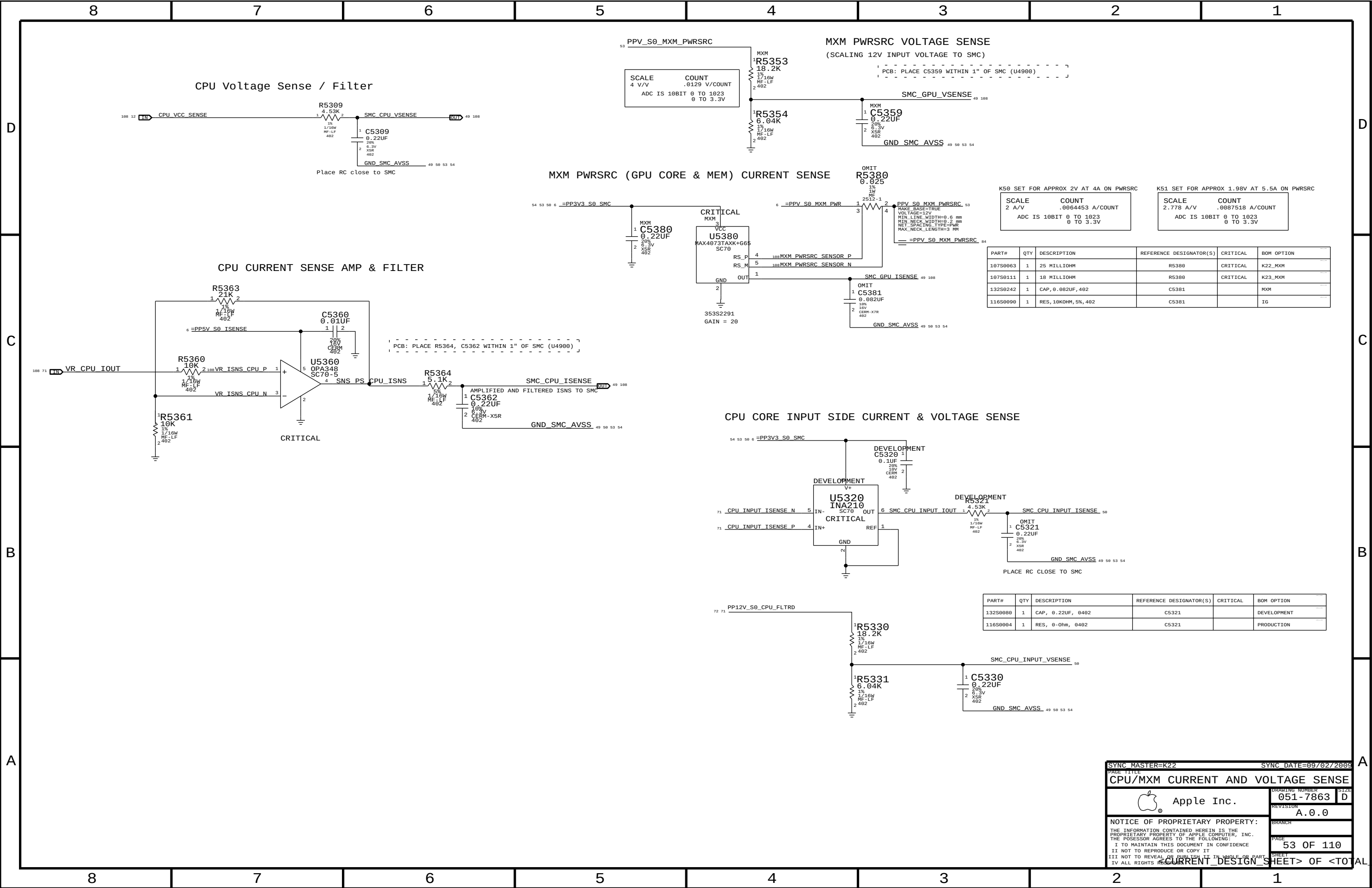
LPCPLUS
R5158
5% 1/16W MF-LF 402

51 51 B1 SPI ALT MISO
51 51 B2 SPI MISO

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LPC+SPI Debug Connector







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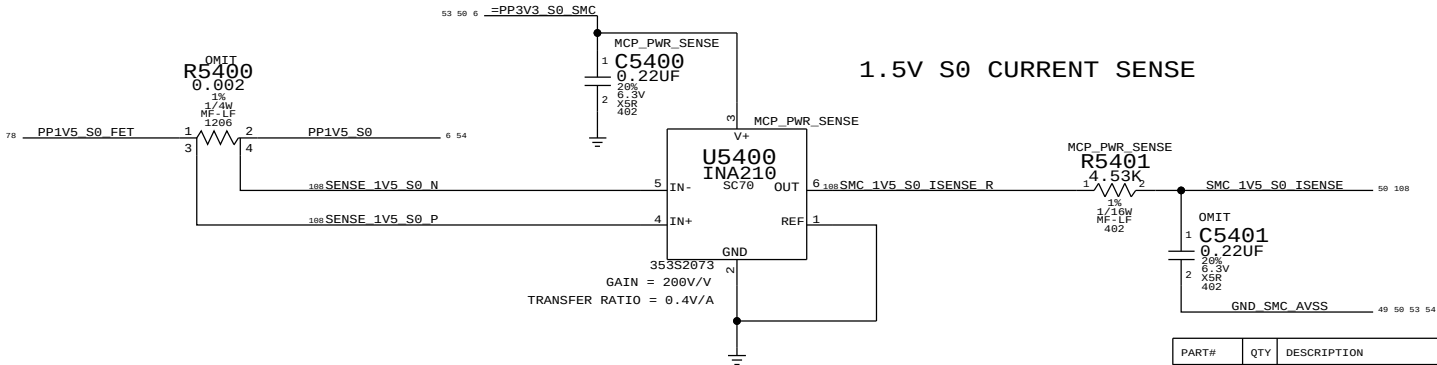
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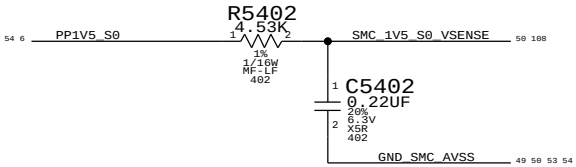
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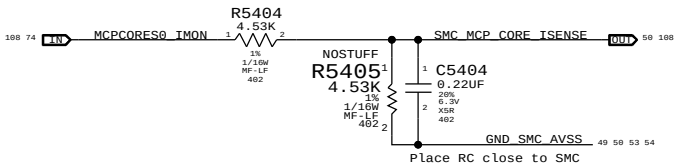
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
104S0018	1	RES, 2 MILLIOHM, 1206	R5400	CRITICAL	MCP_PWR_SENSE
101S0414	1	RES, 0 OHM, 1206, 20MILLIOHM MAX	R5400	CRITICAL	PRODUCTION

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
132S0000	1	CAP, 0.22UF, 0402	C5401		MCP_PWR_SENSE
116S0004	1	RES, 0 OHM, 0402	C5401		PRODUCTION

1.5V S0 VOLTAGE SENSE

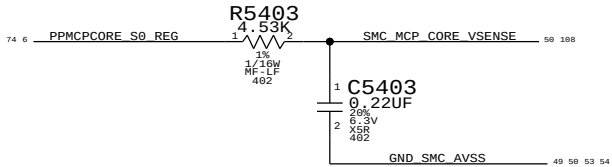


MCP CORE CURRENT SENSE



SCALE IS 0.116 V/A

MCP CORE VOLTAGE SENSE



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MCP CURRENT AND VOLTAGE SENSE

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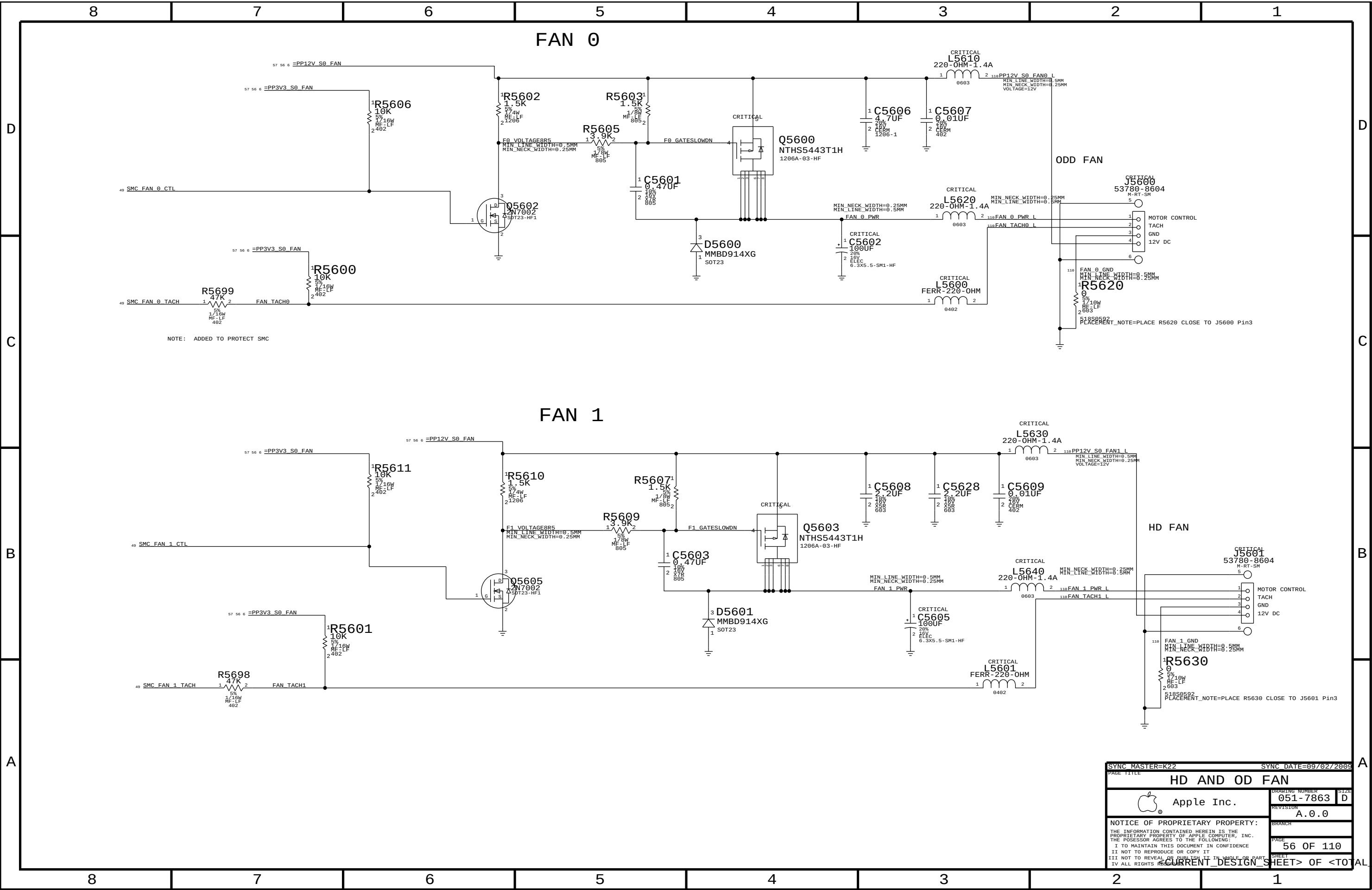
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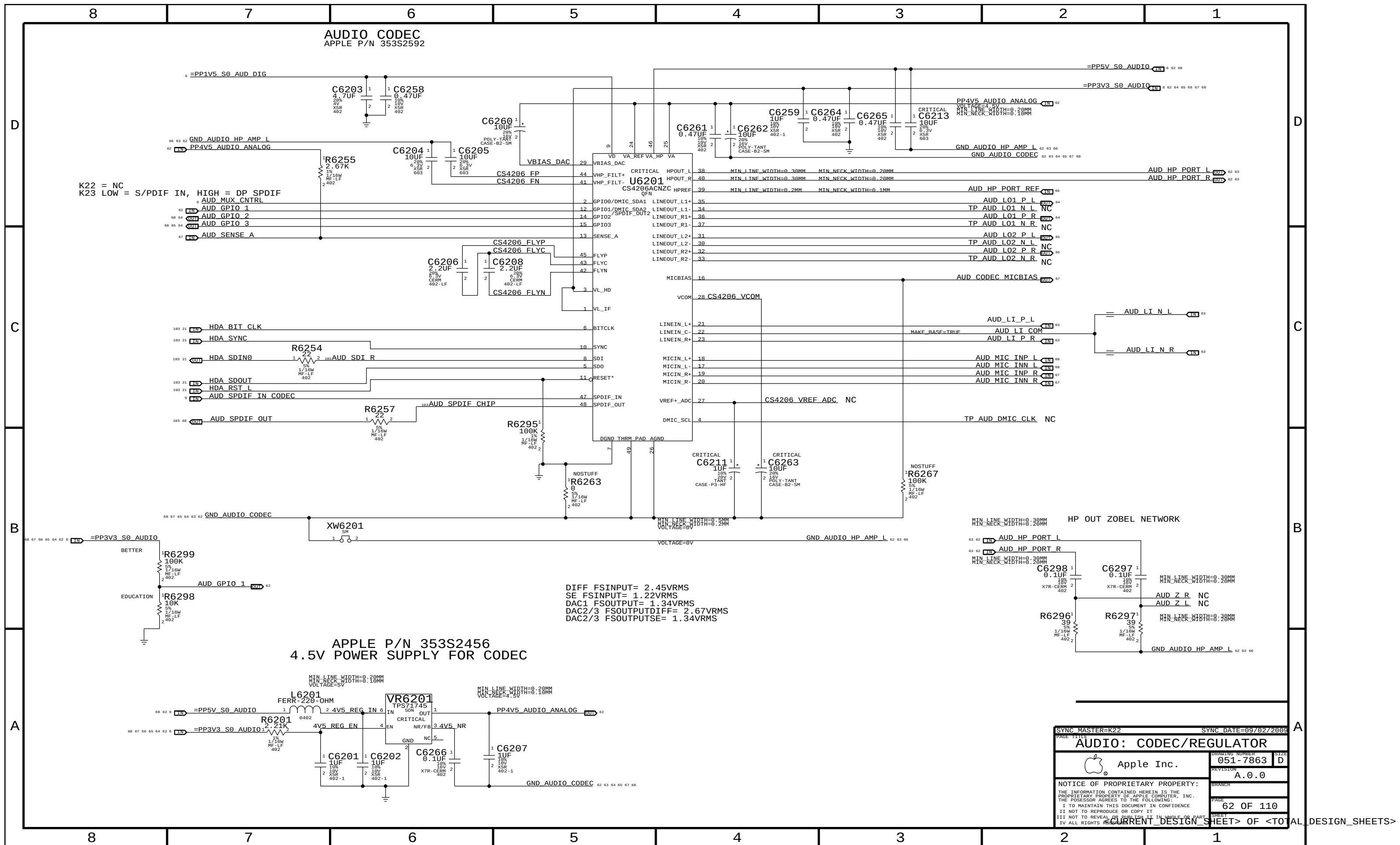
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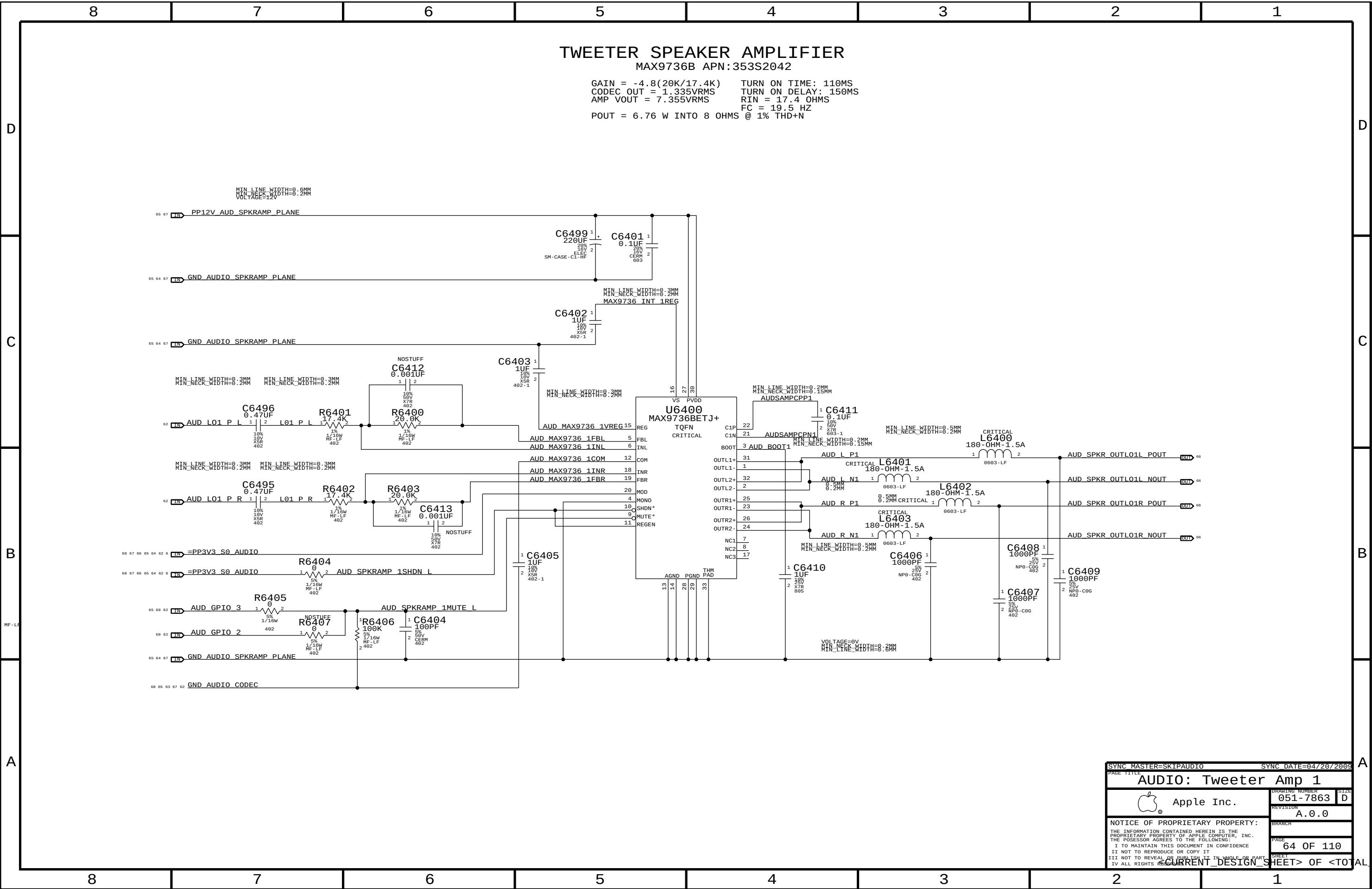
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SYNC MASTER=SKIPAUDIO		SYNC DATE=04/20/2009	
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AUDIO: Tweeter Amp 1			
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WOOFER SPEAKER AMPLIFIER
MAX9736B APN:353S2042

GAIN = -4.8(20K/17.4K) TURN ON TIME: 110MS
CODEC OUT = 1.335VRMS TURN ON DELAY: 150MS
AMP VOUT = 7.355VRMS RIN = 17.4 OHMS
POUT = 6.76 W INTO 8 OHMS @ 1% THD+N FC = 19.5 HZ

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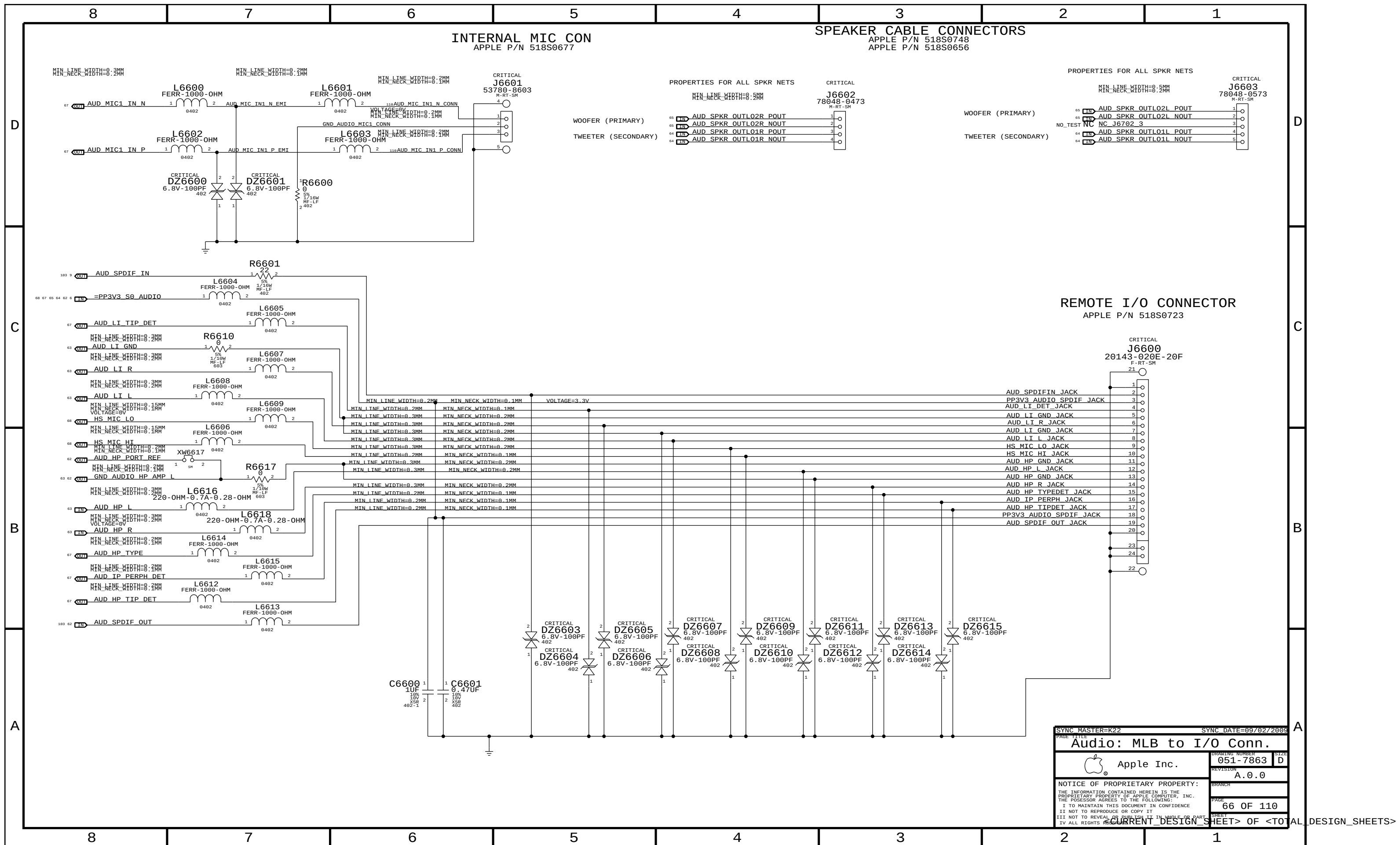
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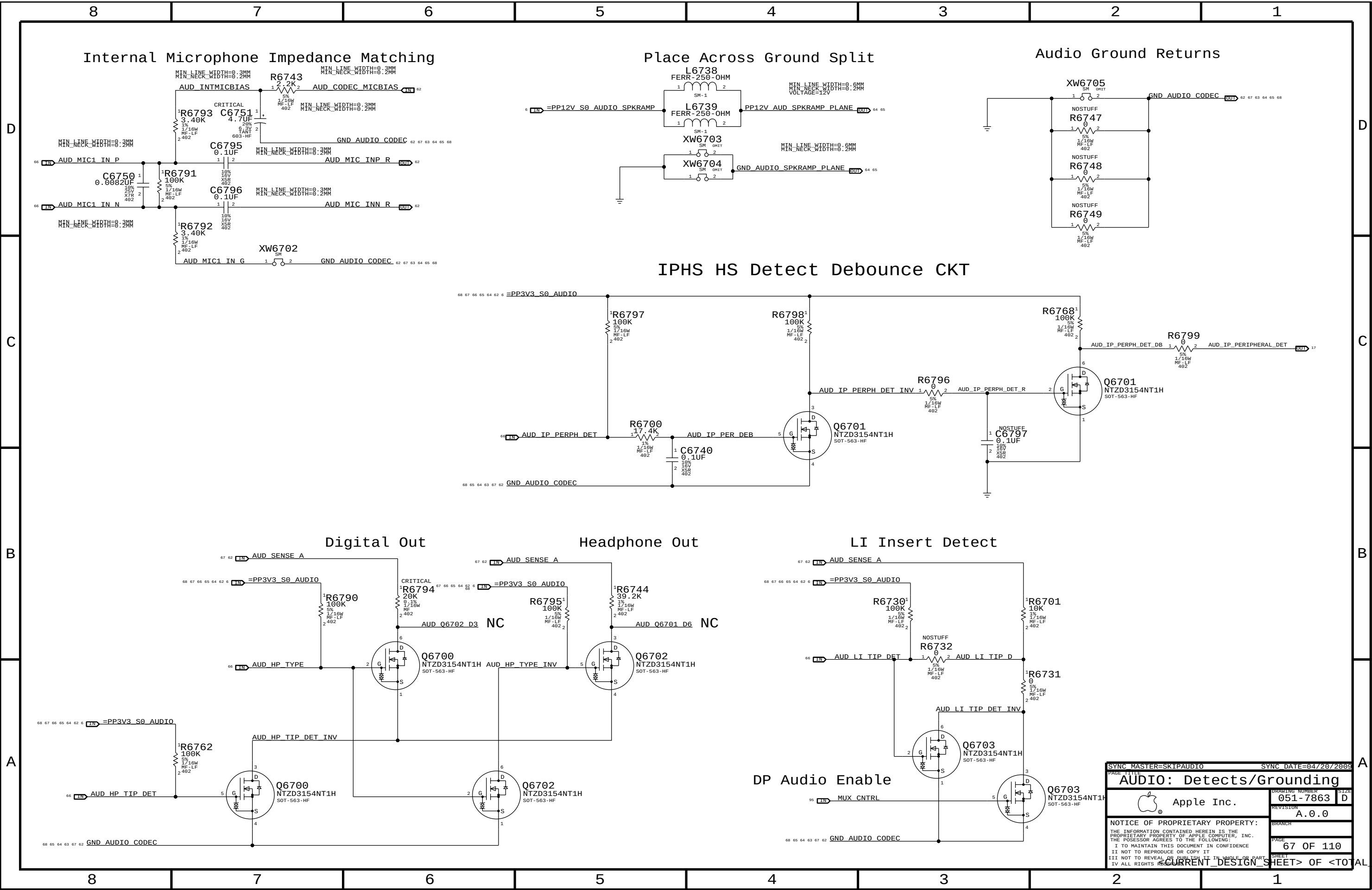
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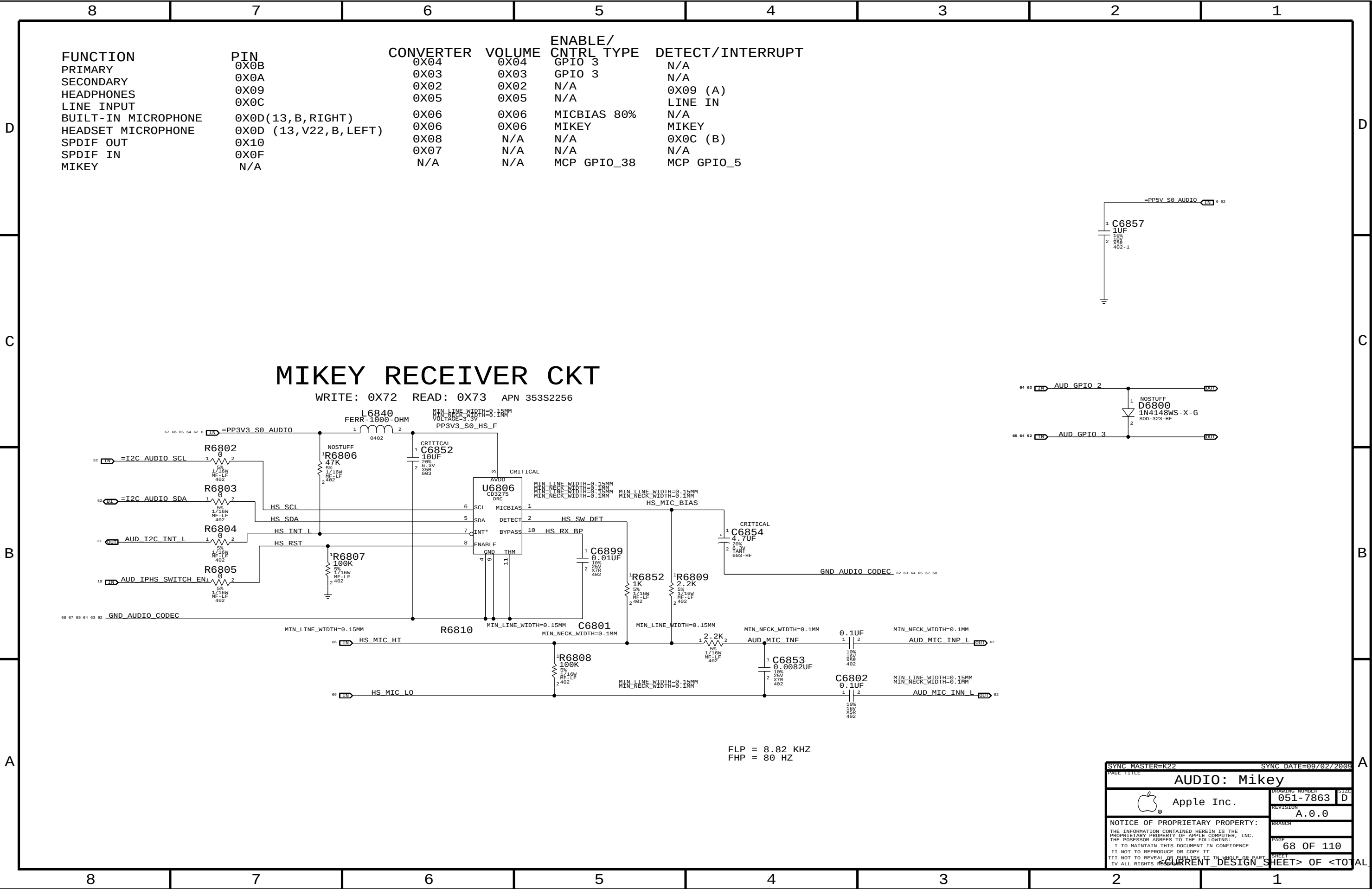
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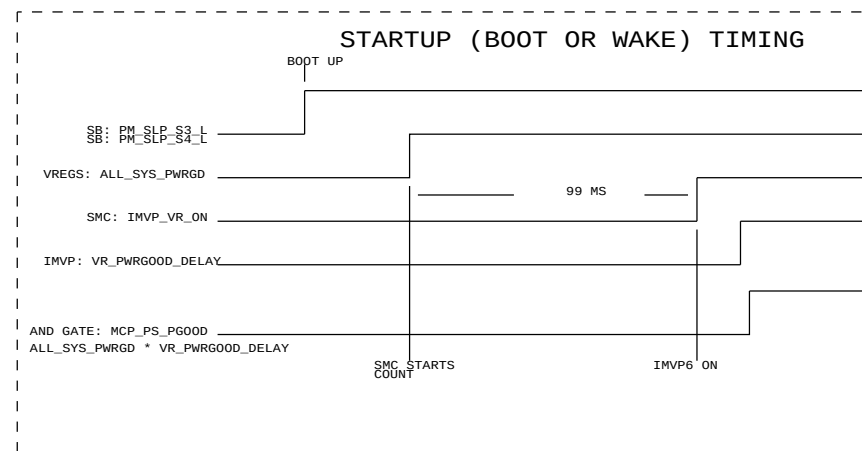
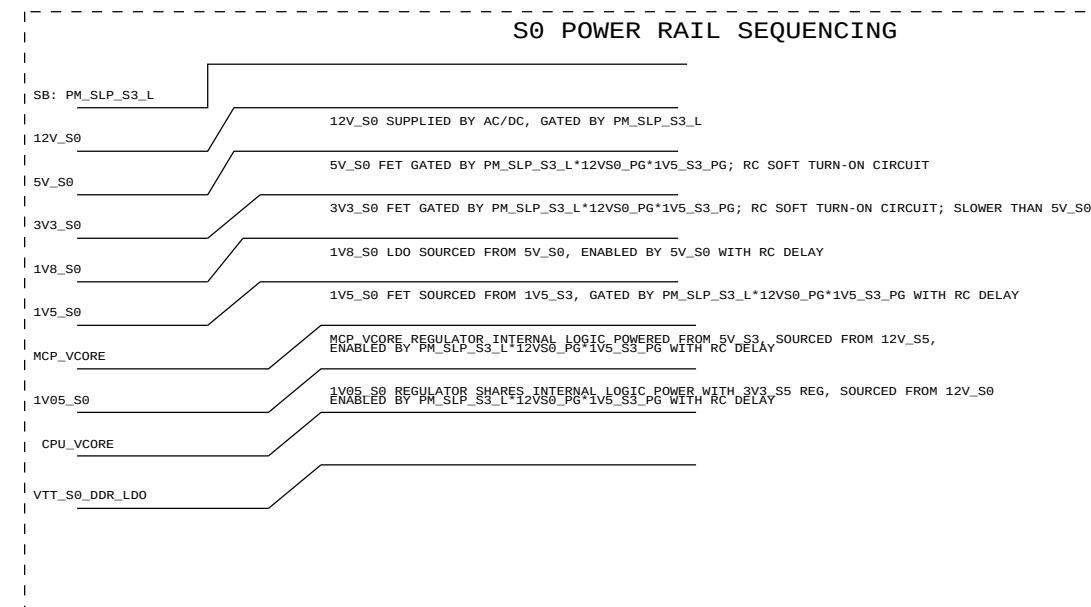
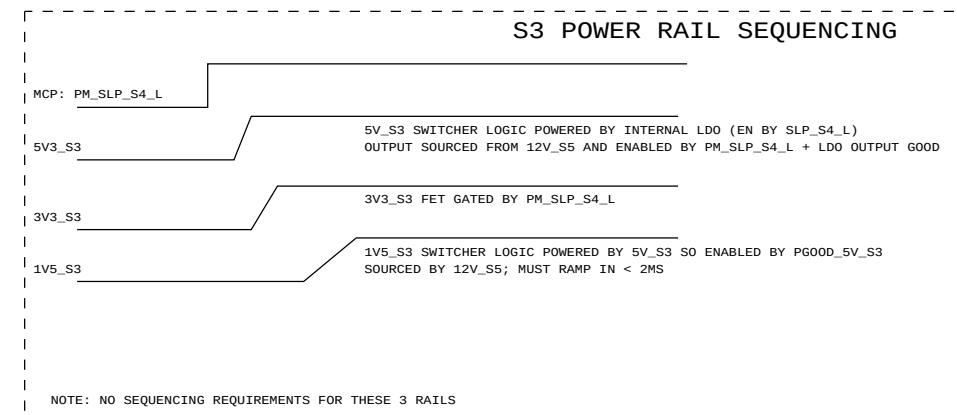
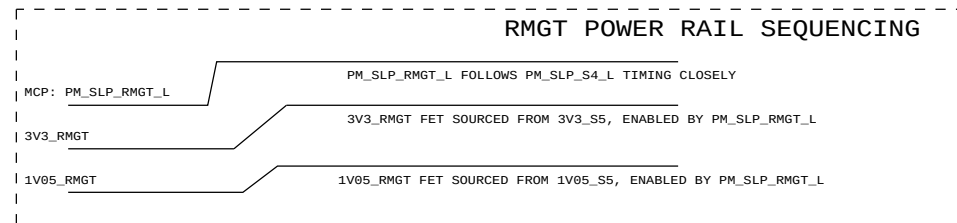
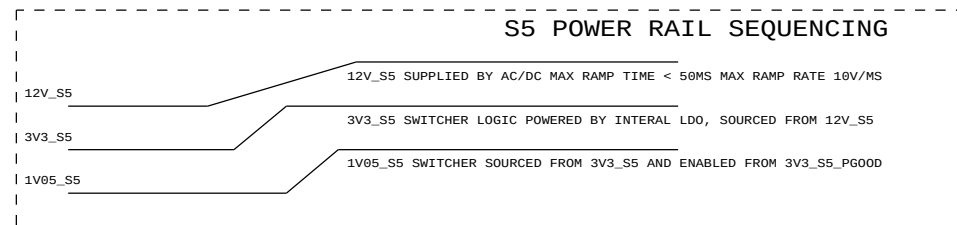
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AUDIO: Woofer Amp			
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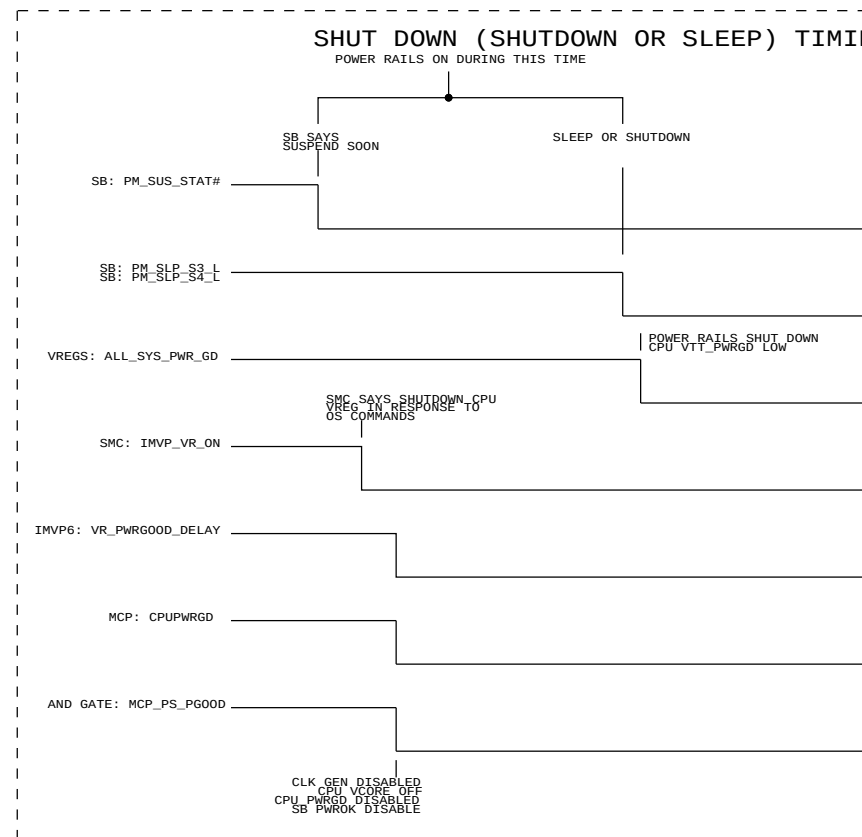


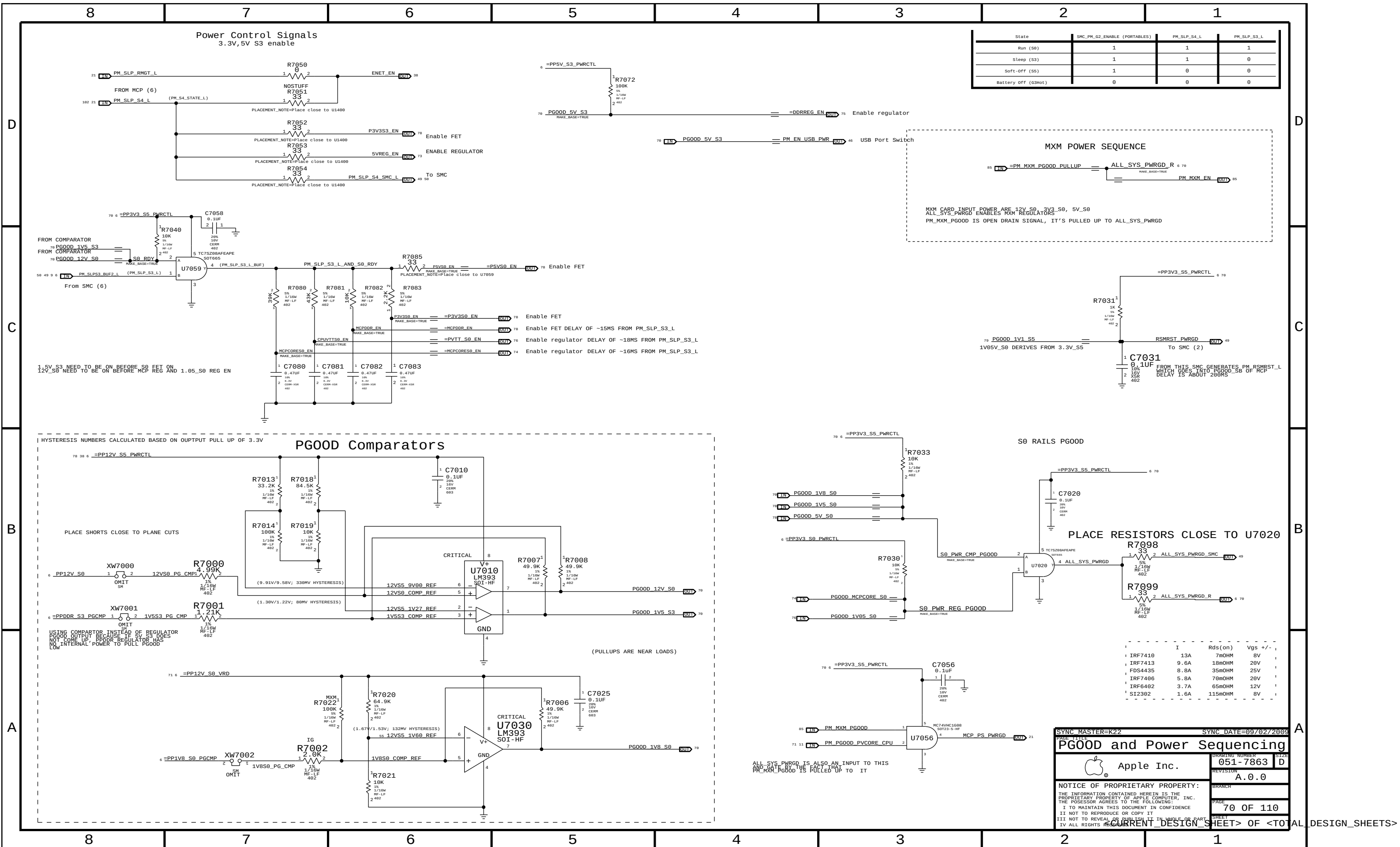


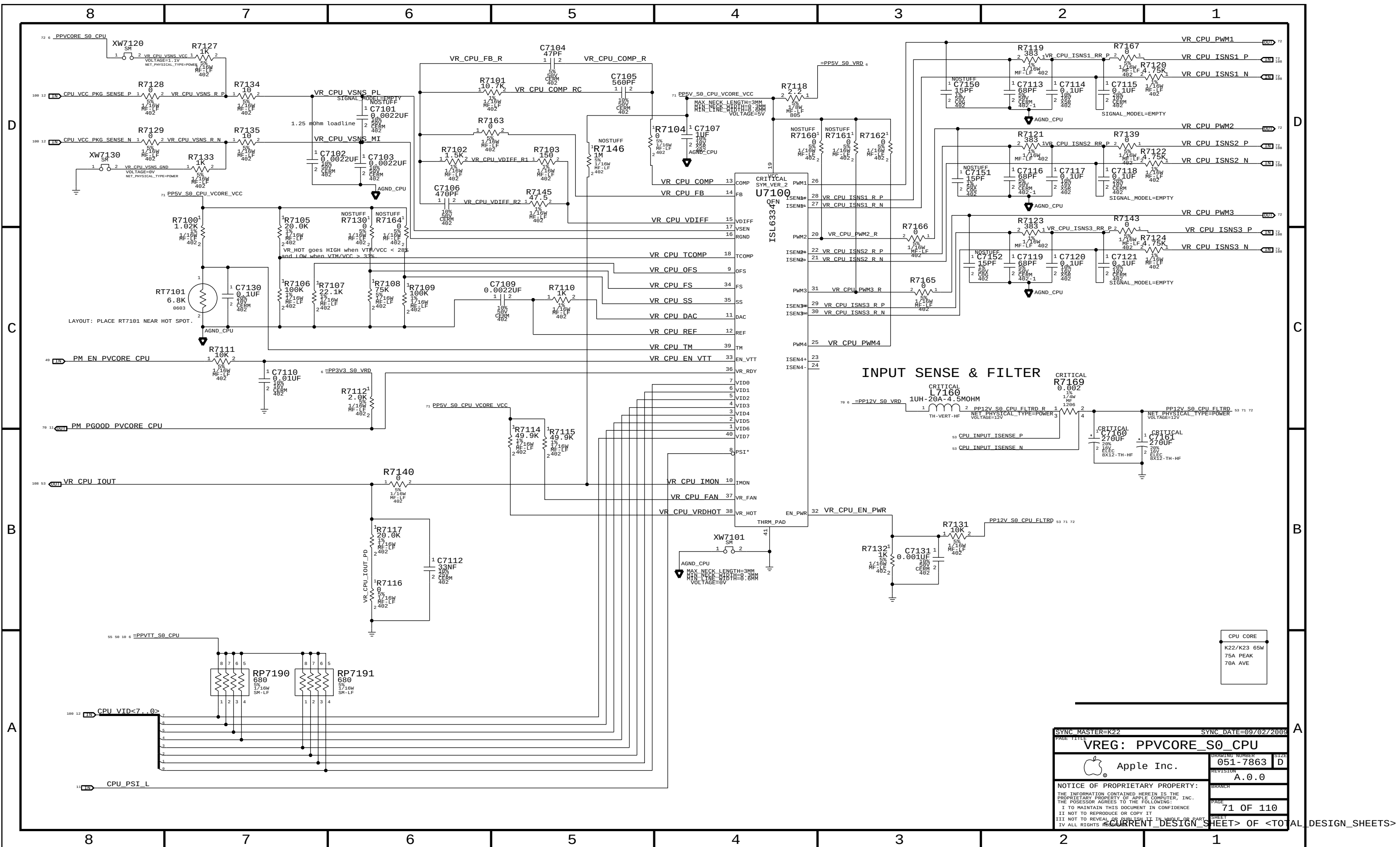


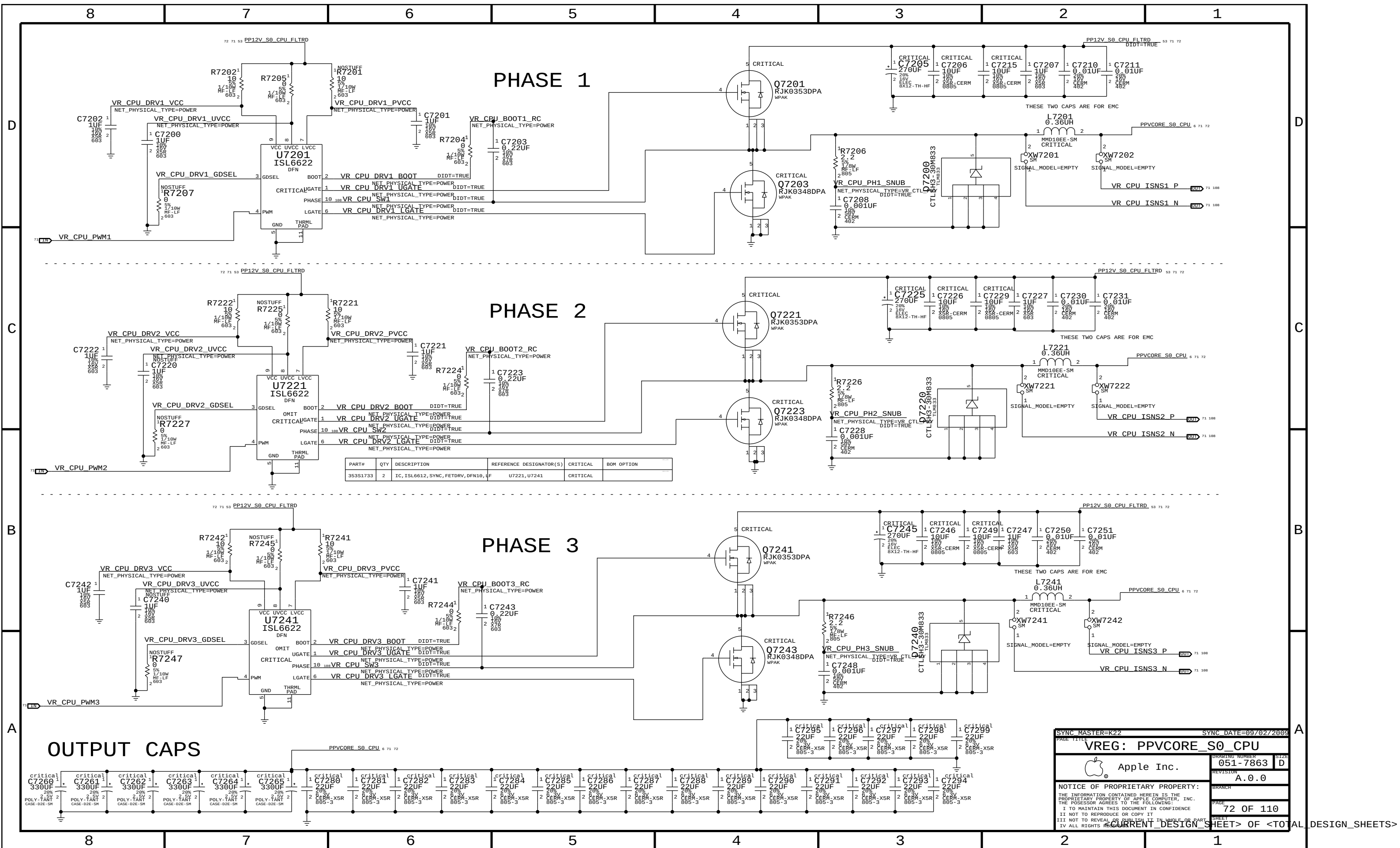


State	Manageability	SMC_PM_G2_ENABLE	PM_S4_STATE_L	PM_SLP_S3_L	PM_SLP_S4_L	PM_SLP_M_L
Run (S0/M0)	N/A	1	1	1	1	1
Sleep (S3/M1)	On	1	1	0	1	1
Soft-Off (S5/M1)	On	1	0	0	1	1
Sleep (S3/M-Off)	Off	1	1	0	1	0
Soft-Off (S5/M-Off)	Off	1	0	0	0	0
Battery Off (G3Hot)	N/A	0	0	0	0	0

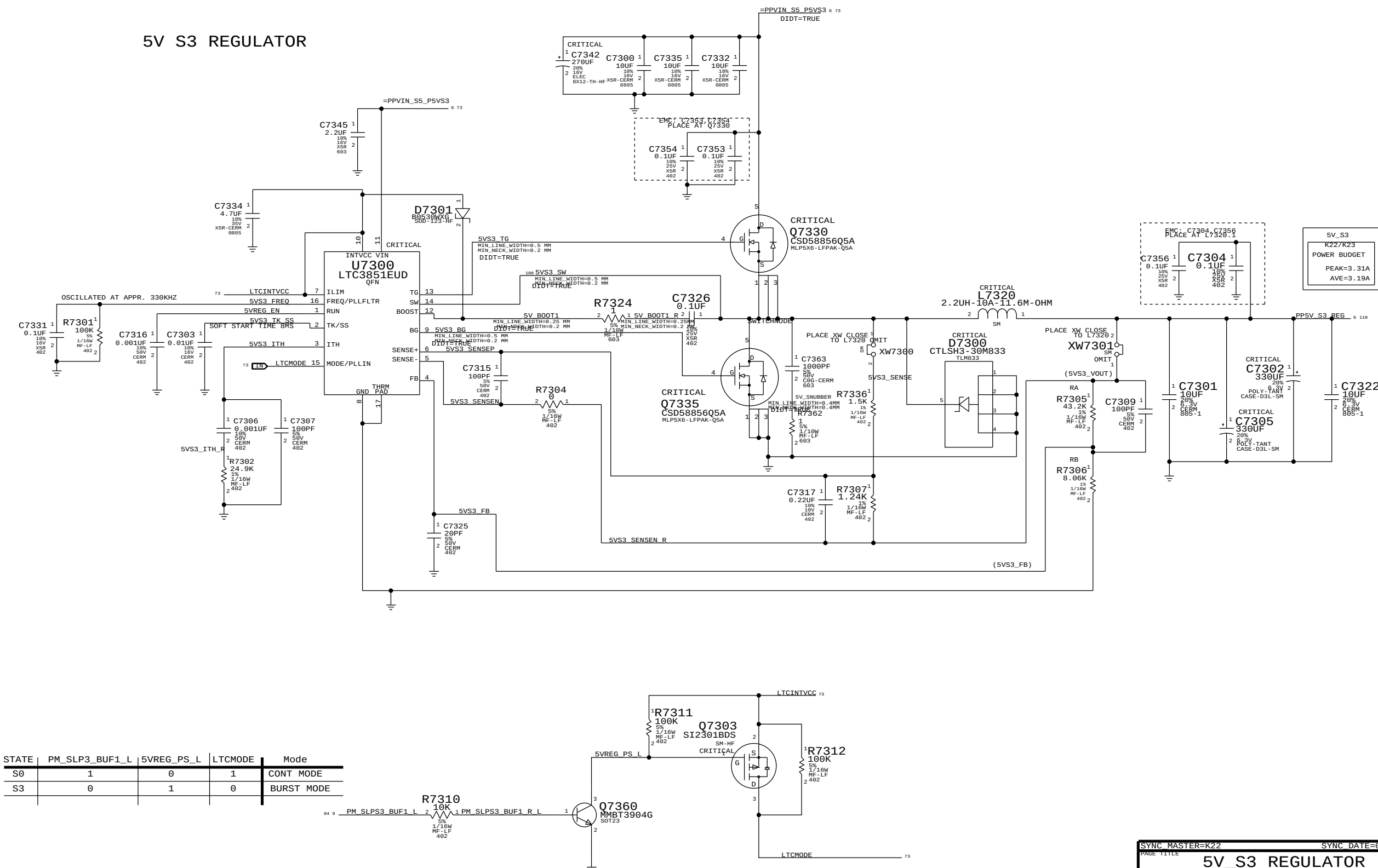








5V S3 REGULATOR



STATE	PM_SLP3_BUF1_L	5VREG_PS_L	LTCMODE	Mode
S0	1	0	1	CONT MODE
S3	0	1	0	BURST MODE

SYNC MASTER=K22

SYNC DATE=09/02/2009

5V_S3 REGULATOR

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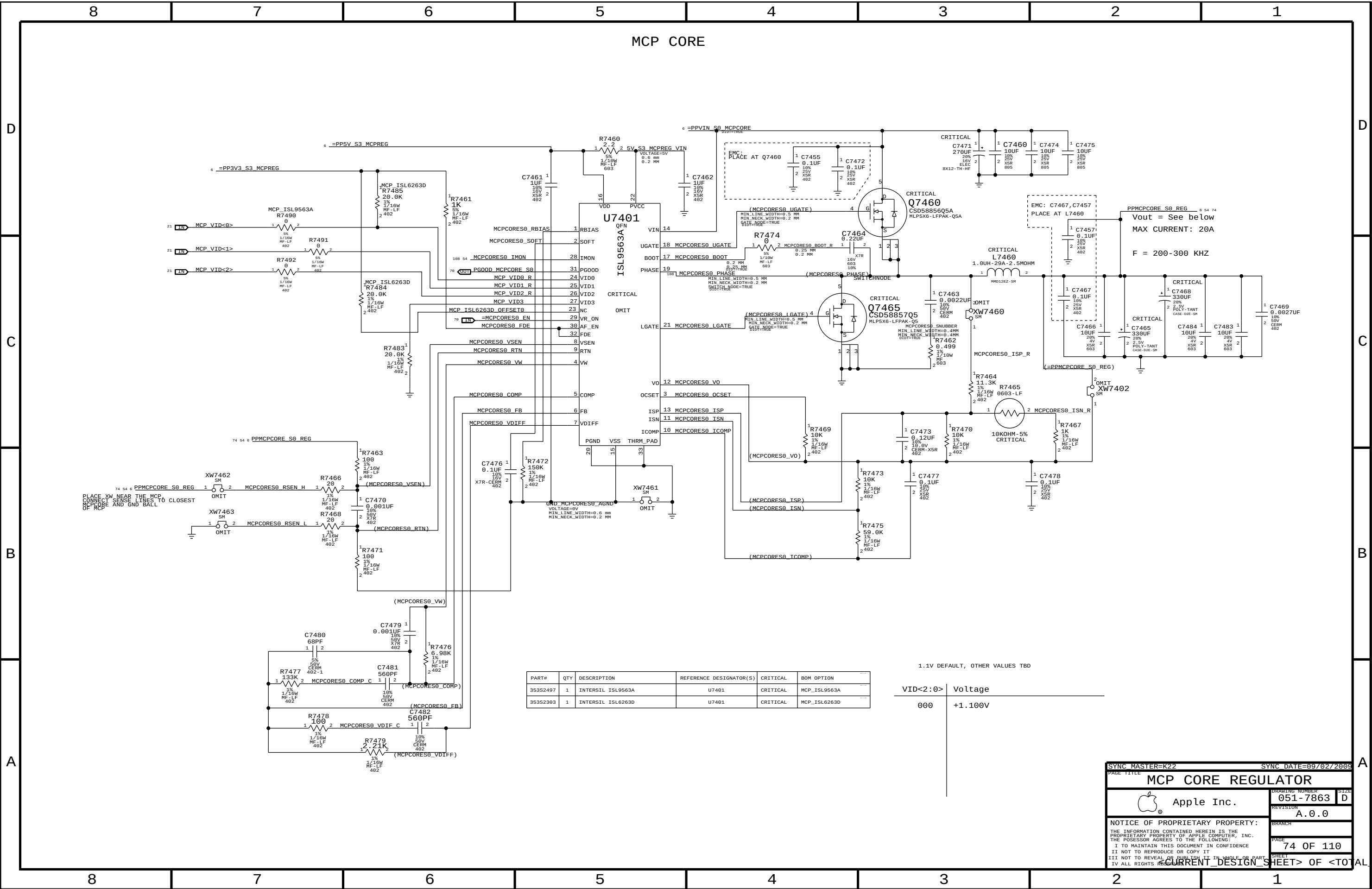
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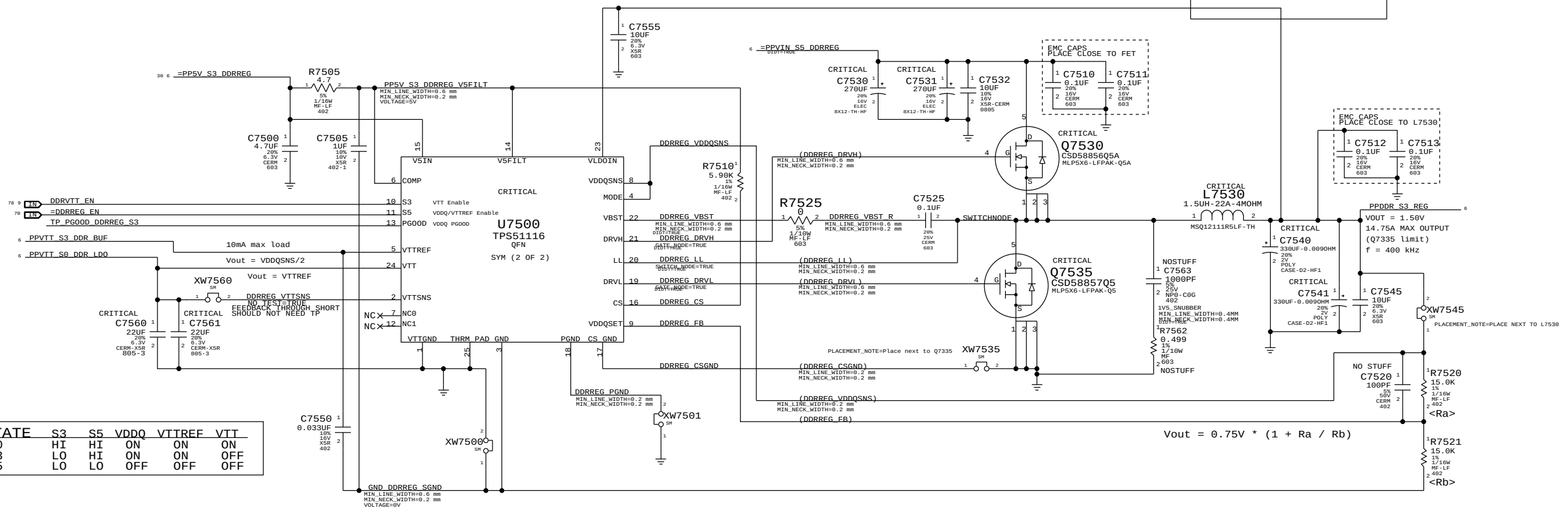
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SHEET

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1.5 V DDR SUPPLY



PPDDR_S3_REG
VOUT = 1.5V
PEAK = 11.28A
AVG = 6.72A

SYNC MASTER=K22

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1.5V DDR SUPPLY

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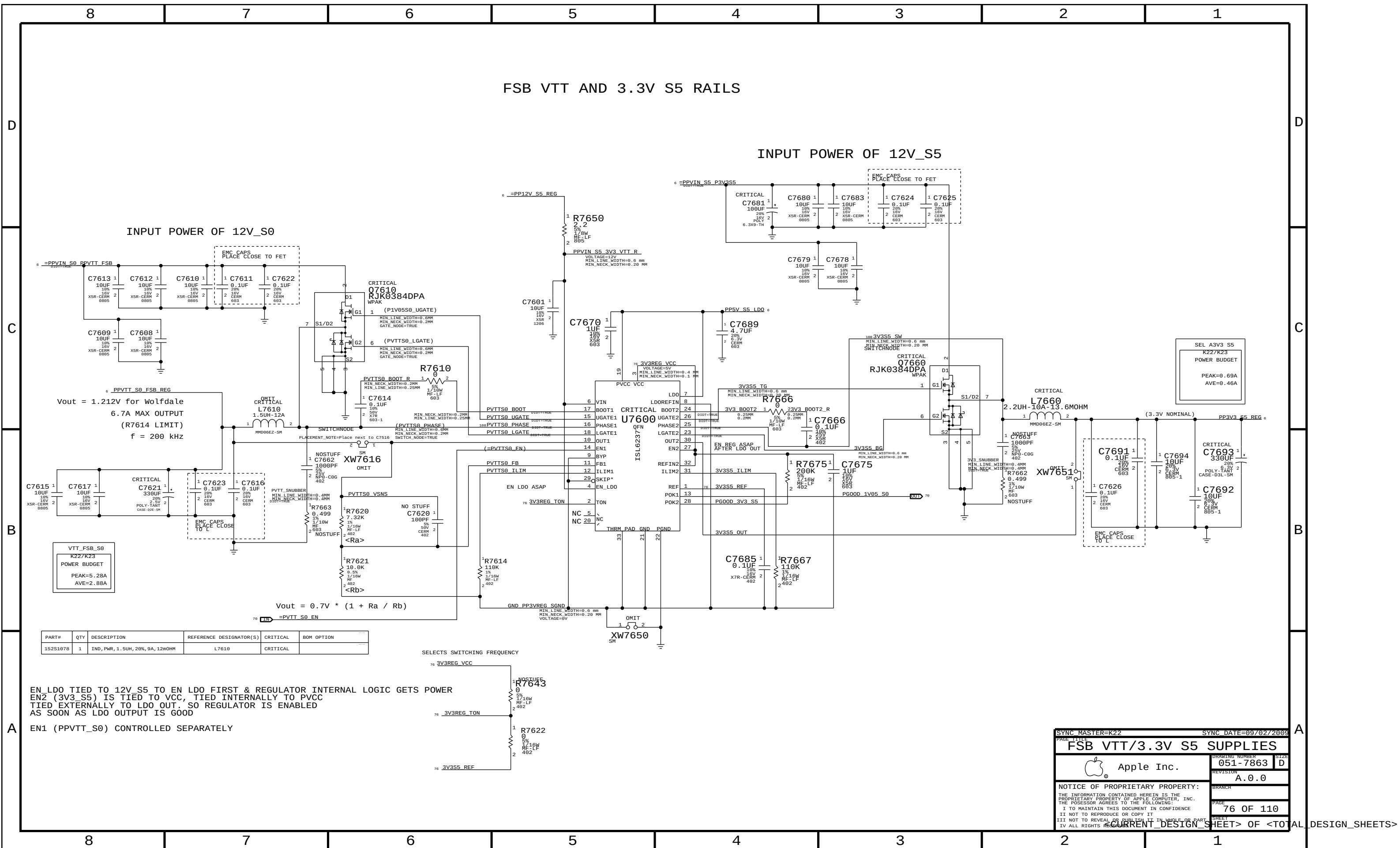
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SHEET

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[illegible]

(ENABLED AT 2,8V MINIMUM)

CRITICAL
L7920
2.2UH-3.25A
IHLP1616BZ-SM

CRITICAL
U7950
TPS62510

CRITICAL
R7980
51.1K

CRITICAL
R7981
60.4K

Vout = 1.1V
MAX Current = 0.55A
FREQ = 1MHz

$V_{OUT} = 0.6V * (1 + R_a / R_b)$

SYNCH MASTER=K22		SYNCH DATE=09/02/2009	
PAGE TITLE			
1V1 S5 POWER SUPPLY			
 Apple Inc.	DRAWING NUMBER		SIZE
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SYNC DATE=12/02/2008

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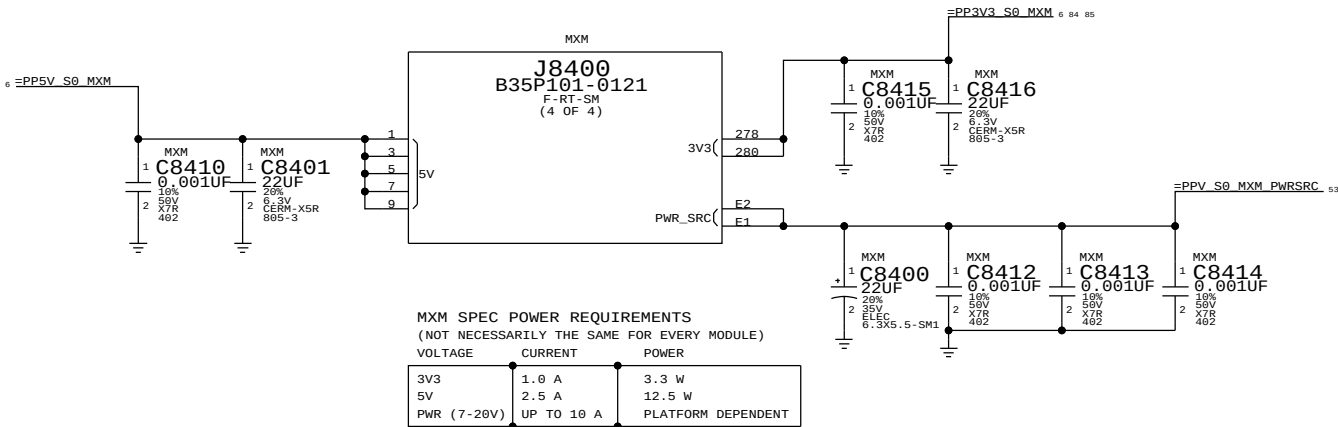
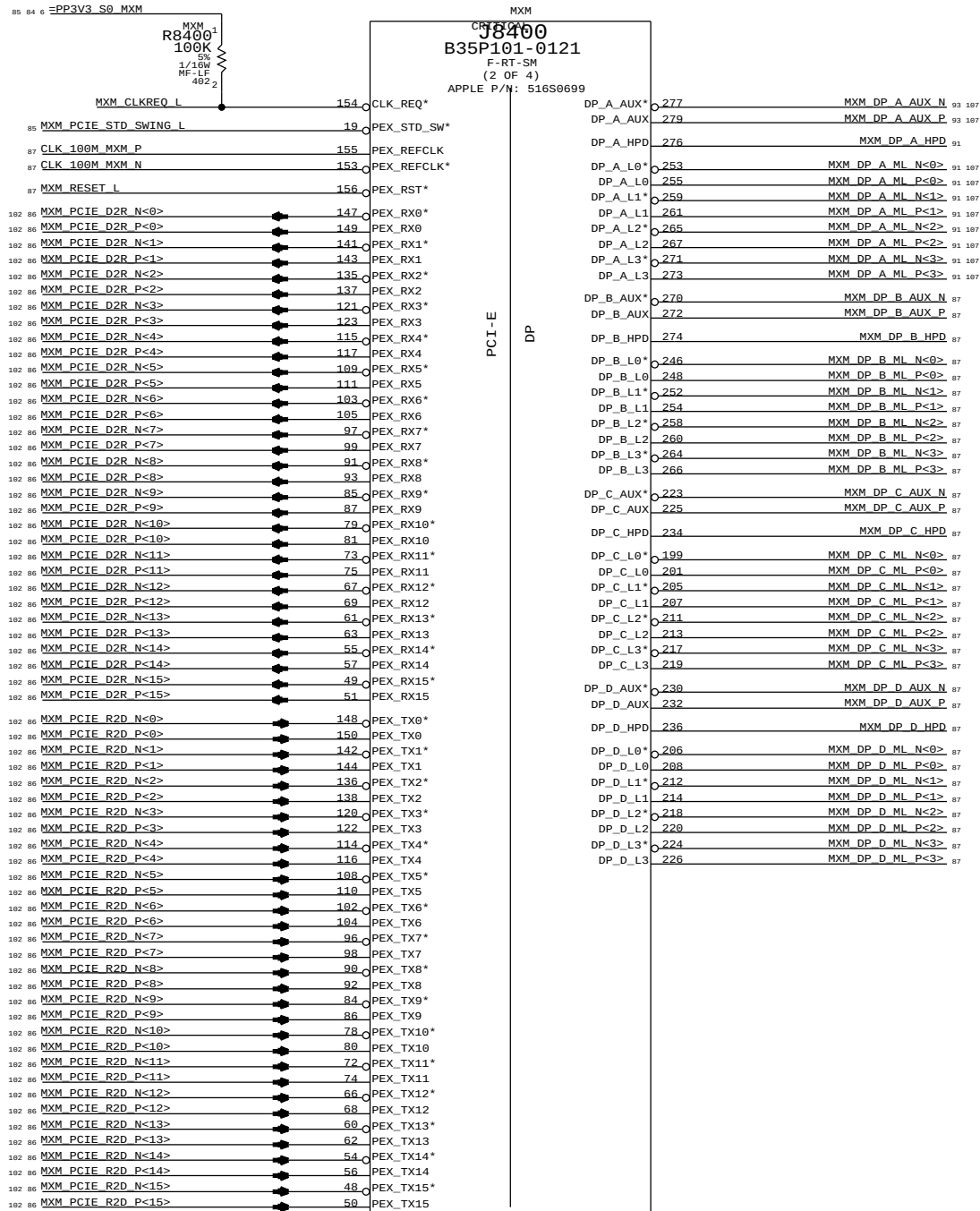
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- =PP3V3_S0_MXM
- =PP5V_S0_MXM
- =PPV_S0_MXM_PWRSRC

Signal aliases required by this page:
(NONE)

BOM options provided by this page:

- MXM



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MXM PCIe, DP & Power			
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MXM PCIE CAPS

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Display: Aliases

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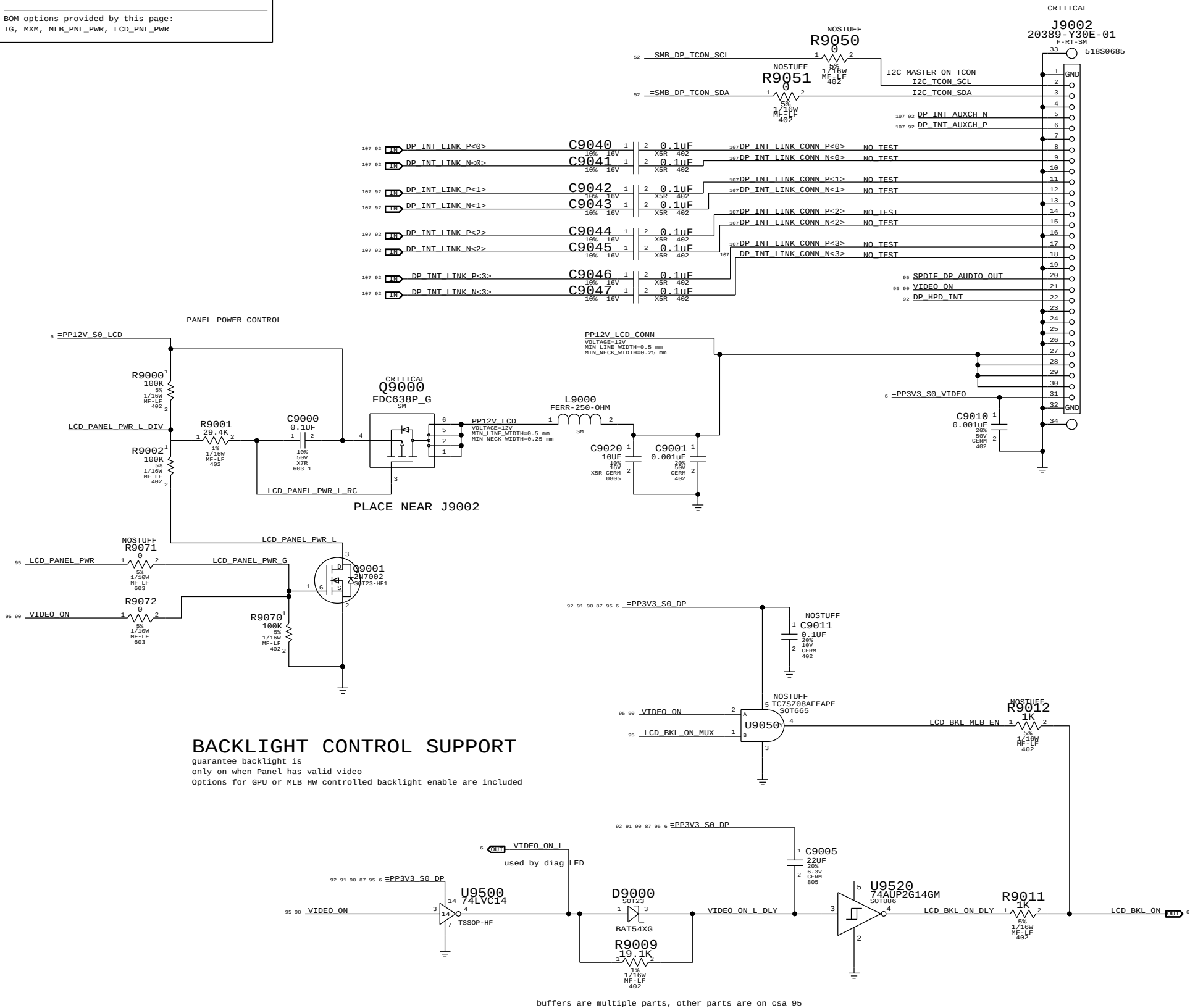
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- =PP12V_S0_LCD
- =PP3V3_S0_VIDEO

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
IG, MXM, MLB_PNL_PWR, LCD_PNL_PWR

INTERNAL DP INTERFACE

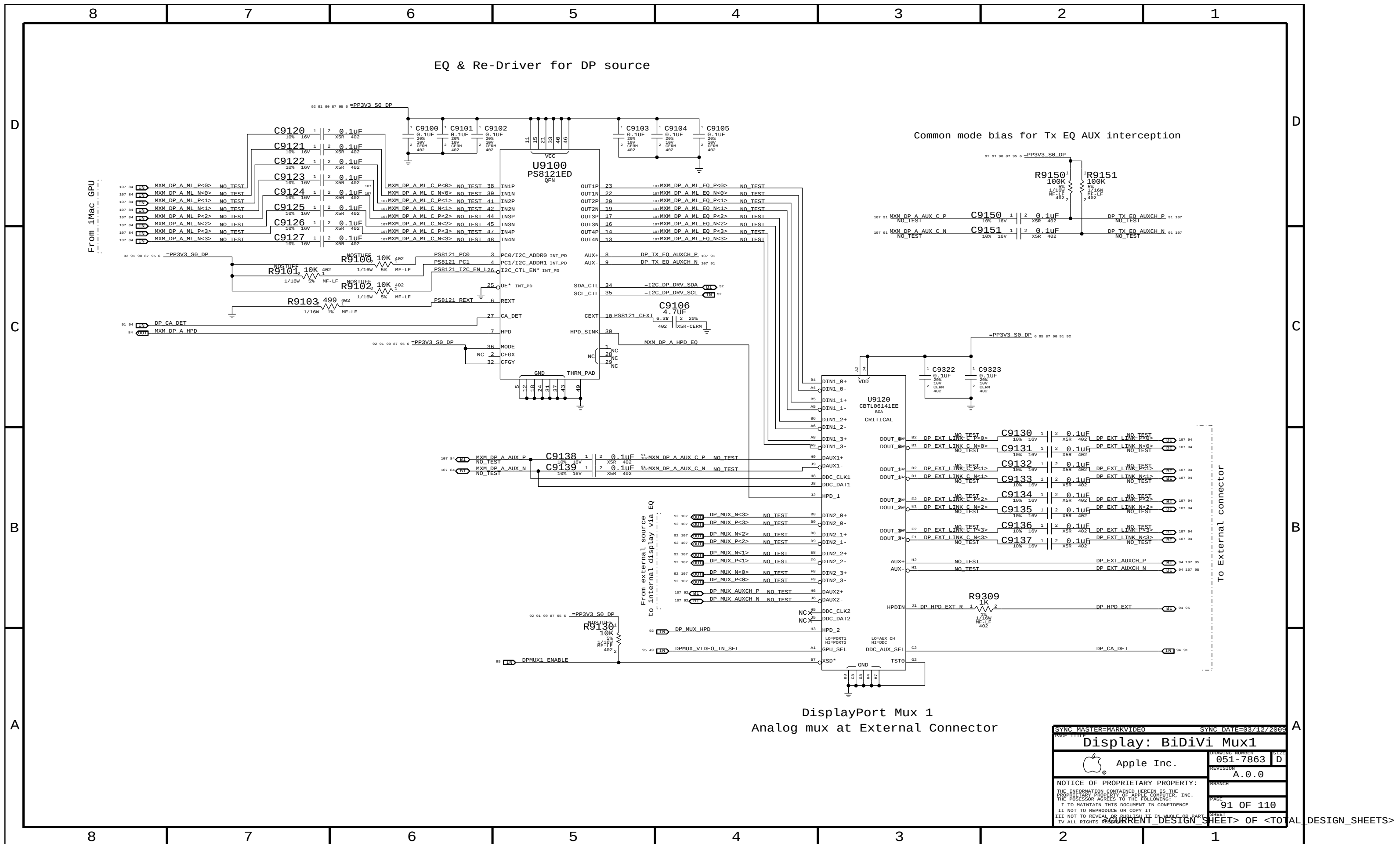


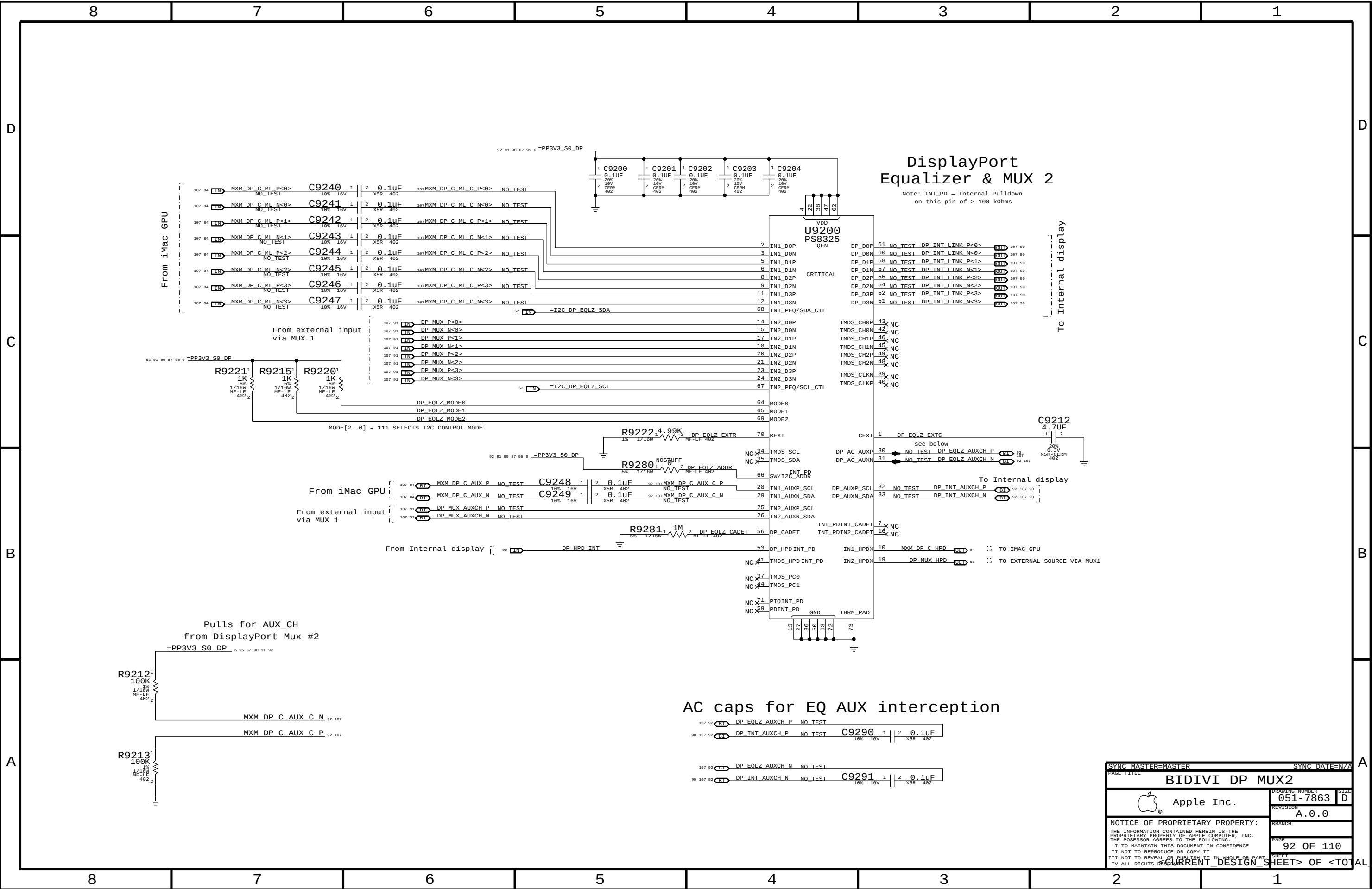
SYNC MASTER=MARKVIDEO SYNC DATE=03/12/2009

Display: Int DP Connector

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DisplayPort Equalizer & MUX 2

Note: INT_PD = Internal Pulldown
on this pin of >=100 kohms

To Internal display

To Internal display

AC caps for EQ AUX interception

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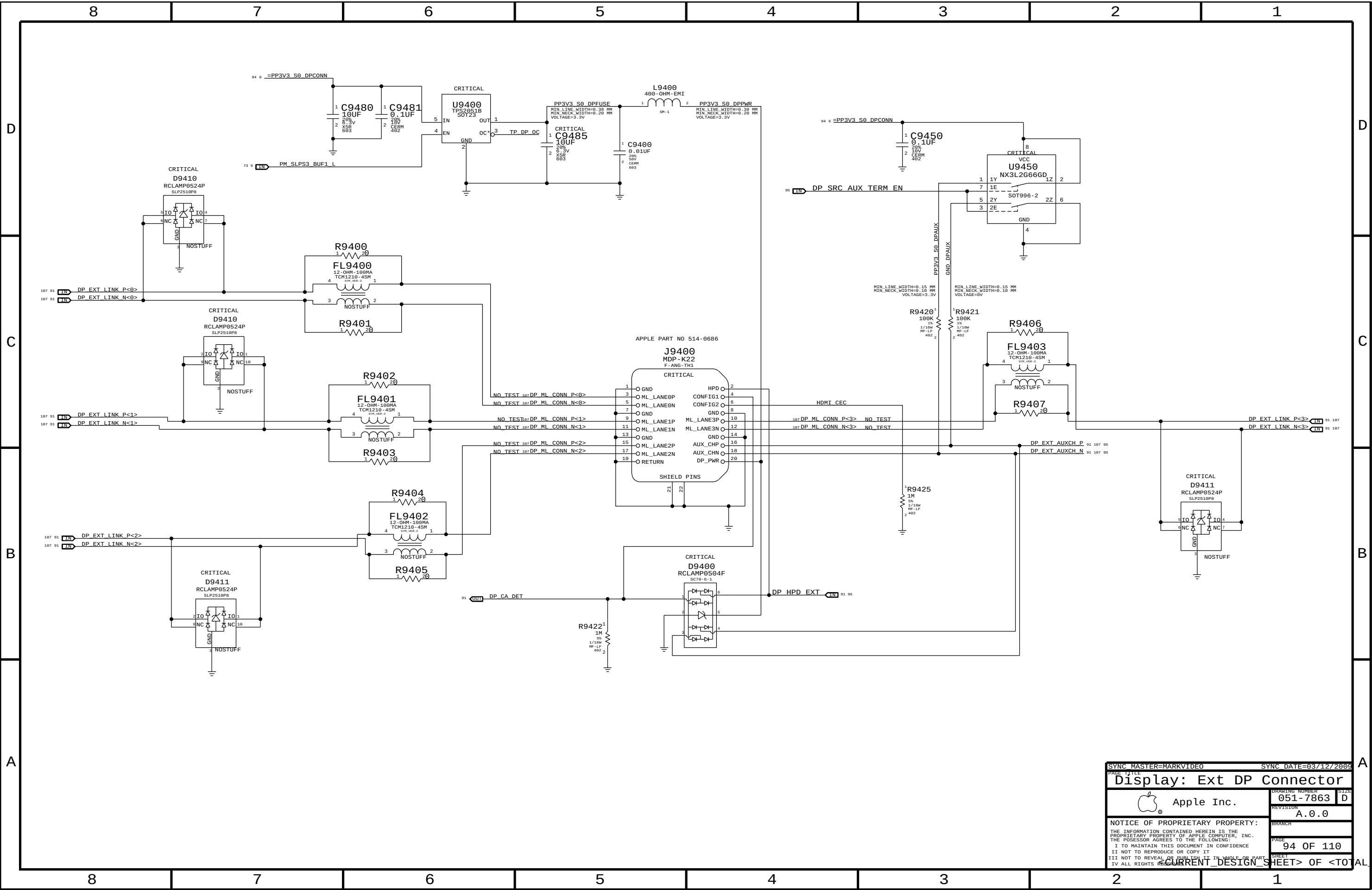
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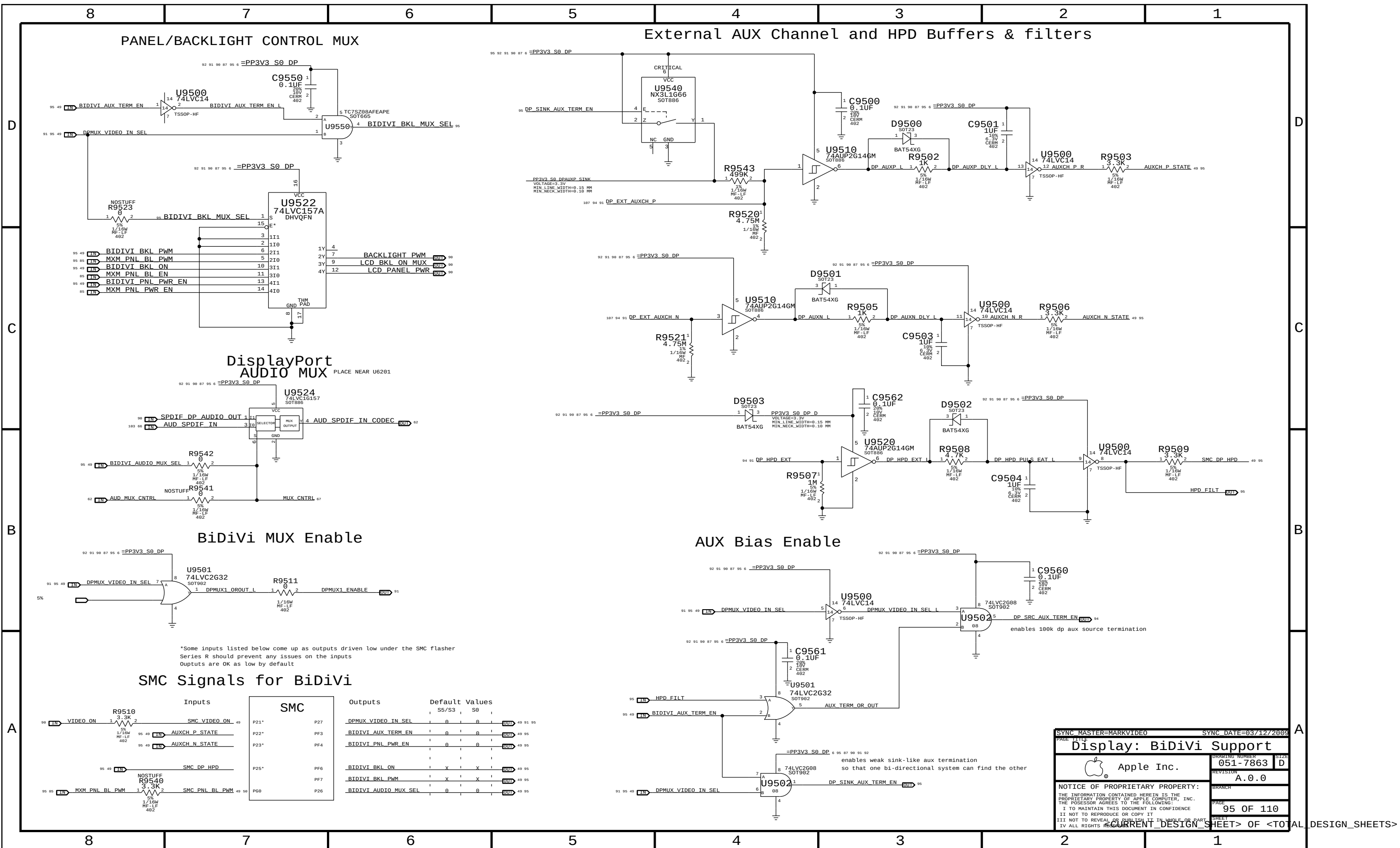
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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_40S_VDD	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
MEM_70D_VDD	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	=3:1_SPACING	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_DATA	*	*	MEM_20THER
MEM_DQS	*	*	MEM_20THER

Need to support MEM_*-style wildcards!

DDR2:

DQ signals should be matched within 20 ps of associated DQS pair.

DQS intra-pair matching should be within 1 ps, no inter-pair matching requirement.

All DQS pairs should be matched within 100 ps of clocks.

CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 140 ps.

A/BA/cmd signals should be matched within 75 ps, no CLK matching requirement.

All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate).

DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.

DDR3:

DQ signals should be matched within 5 ps of associated DQS pair.

DQS intra-pair matching should be within 1 ps, inter-pair matching should be within 180 ps

No DQS to clock matching requirement.

CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps.

A/BA/cmd signals should be matched within 5 ps of CLK pairs.

All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate).

DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.3

SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Section 6.2

MCP MEM COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MCP_MEM_COMP	*	Y	0.175 MM	0.175 MM	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MCP_MEM_COMP	*	0.2 MM	?

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.3.4

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
MEM_70D_VDD	MEM_CLK	MEM A CLK P<1..0>	15 33
MEM_70D_VDD	MEM_CLK	MEM A CLK N<1..0>	15 33
MEM_70D_VDD	MEM_CLK	MEM A CLK P<4..3>	16 33
MEM_70D_VDD	MEM_CLK	MEM A CLK N<4..3>	16 33
MEM_40S_VDD	MEM_CTRL	MEM A CKE<3..0>	15 16 31
MEM_40S_VDD	MEM_CTRL	MEM A CS_L<3..0>	15 16 31
MEM_40S_VDD	MEM_CTRL	MEM A ODT<3..0>	15 16 31
MEM_40S_VDD	MEM_CMD	MEM A A<14..0>	15 31
MEM_40S_VDD	MEM_CMD	MEM A BA<2..0>	15 31
MEM_40S_VDD	MEM_CMD	MEM A RAS_L	15 31
MEM_40S_VDD	MEM_CMD	MEM A CAS_L	15 31
MEM_40S_VDD	MEM_CMD	MEM A WE_L	15 31
MEM_40S	MEM_DATA	MEM A DQ<7..0>	15 33
MEM_40S	MEM_DATA	MEM A DM<0>	15 33
MEM_40S	MEM_DATA	MEM A DQ<15..8>	15 33
MEM_40S	MEM_DATA	MEM A DM<1>	15 33
MEM_40S	MEM_DATA	MEM A DQ<23..16>	15 33
MEM_40S	MEM_DATA	MEM A DM<2>	15 33
MEM_40S	MEM_DATA	MEM A DQ<31..24>	15 33
MEM_40S	MEM_DATA	MEM A DM<3>	15 33
MEM_40S	MEM_DATA	MEM A DQ<39..32>	15 33
MEM_40S	MEM_DATA	MEM A DM<4>	15 33
MEM_40S	MEM_DATA	MEM A DQ<47..40>	15 33
MEM_40S	MEM_DATA	MEM A DM<5>	15 33
MEM_40S	MEM_DATA	MEM A DQ<55..48>	15 33
MEM_40S	MEM_DATA	MEM A DM<6>	15 33
MEM_40S	MEM_DATA	MEM A DQ<63..56>	15 33
MEM_40S	MEM_DATA	MEM A DM<7>	15 33
MEM_70D	MEM_DQS	MEM A DQS P<0>	15 33
MEM_70D	MEM_DQS	MEM A DQS N<0>	15 33
MEM_70D	MEM_DQS	MEM A DQS P<1>	15 33
MEM_70D	MEM_DQS	MEM A DQS N<1>	15 33
MEM_70D	MEM_DQS	MEM A DQS P<2>	15 33
MEM_70D	MEM_DQS	MEM A DQS N<2>	15 33
MEM_70D	MEM_DQS	MEM A DQS P<3>	15 33
MEM_70D	MEM_DQS	MEM A DQS N<3>	15 33
MEM_70D	MEM_DQS	MEM A DQS P<4>	15 33
MEM_70D	MEM_DQS	MEM A DQS N<4>	15 33
MEM_70D	MEM_DQS	MEM A DQS P<5>	15 33
MEM_70D	MEM_DQS	MEM A DQS N<5>	15 33
MEM_70D	MEM_DQS	MEM A DQS P<6>	15 33
MEM_70D	MEM_DQS	MEM A DQS N<6>	15 33
MEM_70D	MEM_DQS	MEM A DQS P<7>	15 33
MEM_70D	MEM_DQS	MEM A DQS N<7>	15 33
MEM_70D_VDD	MEM_CLK	MEM B CLK P<1..0>	15 33
MEM_70D_VDD	MEM_CLK	MEM B CLK N<1..0>	15 33
MEM_70D_VDD	MEM_CLK	MEM B CLK P<4..3>	16 33
MEM_70D_VDD	MEM_CLK	MEM B CLK N<4..3>	16 33
MEM_40S_VDD	MEM_CTRL	MEM B CKE<3..0>	15 16 32
MEM_40S_VDD	MEM_CTRL	MEM B CS_L<3..0>	15 16 32
MEM_40S_VDD	MEM_CTRL	MEM B ODT<3..0>	15 16 32
MEM_40S_VDD	MEM_CMD	MEM B A<14..0>	15 32
MEM_40S_VDD	MEM_CMD	MEM B BA<2..0>	15 32
MEM_40S_VDD	MEM_CMD	MEM B RAS_L	15 32
MEM_40S_VDD	MEM_CMD	MEM B CAS_L	15 32
MEM_40S_VDD	MEM_CMD	MEM B WE_L	15 32
MEM_40S	MEM_DATA	MEM B DQ<7..0>	15 33
MEM_40S	MEM_DATA	MEM B DM<0>	15 33
MEM_40S	MEM_DATA	MEM B DQ<15..8>	15 33
MEM_40S	MEM_DATA	MEM B DM<1>	15 33
MEM_40S	MEM_DATA	MEM B DQ<23..16>	15 33
MEM_40S	MEM_DATA	MEM B DM<2>	15 33
MEM_40S	MEM_DATA	MEM B DQ<31..24>	15 33
MEM_40S	MEM_DATA	MEM B DM<3>	15 33
MEM_40S	MEM_DATA	MEM B DQ<39..32>	15 33
MEM_40S	MEM_DATA	MEM B DM<4>	15 33
MEM_40S	MEM_DATA	MEM B DQ<47..40>	15 33
MEM_40S	MEM_DATA	MEM B DM<5>	15 33
MEM_40S	MEM_DATA	MEM B DQ<55..48>	15 33
MEM_40S	MEM_DATA	MEM B DM<6>	15 33
MEM_40S	MEM_DATA	MEM B DQ<63..56>	15 33
MEM_40S	MEM_DATA	MEM B DM<7>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<0>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<0>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<1>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<1>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<2>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<2>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<3>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<3>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<4>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<4>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<5>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<5>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<6>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<6>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<7>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<7>	15 33
MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM_COMP_VDD	16
MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM_COMP_GND	16

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
MEM_70D	MEM_DQS	MEM B DQS P<0>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<0>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<1>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<1>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<2>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<2>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<3>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<3>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<4>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<4>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<5>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<5>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<6>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<6>	15 33
MEM_70D	MEM_DQS	MEM B DQS P<7>	15 33
MEM_70D	MEM_DQS	MEM B DQS N<7>	15 33
MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM_COMP_VDD	16
MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM_COMP_GND	16

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Memory Constraints

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051-7863

A.0.0

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XTAL Constraints

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT

XTAL	*	=4X_DIELECTRIC	?
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MCP RGMII (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MII_COMP	*	=STANDARD	0.2 MM	0.2 MM	=STANDARD	=STANDARD	=STANDARD
ENET_MII_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_BUF0_CLK	*	=3:1_SPACING	?
ENET_MII	*	0.3 MM	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Sections 2.7.2 & 2.7.4

RTL8211CLGR (ETHERNET PHY) CONSTRAINTS

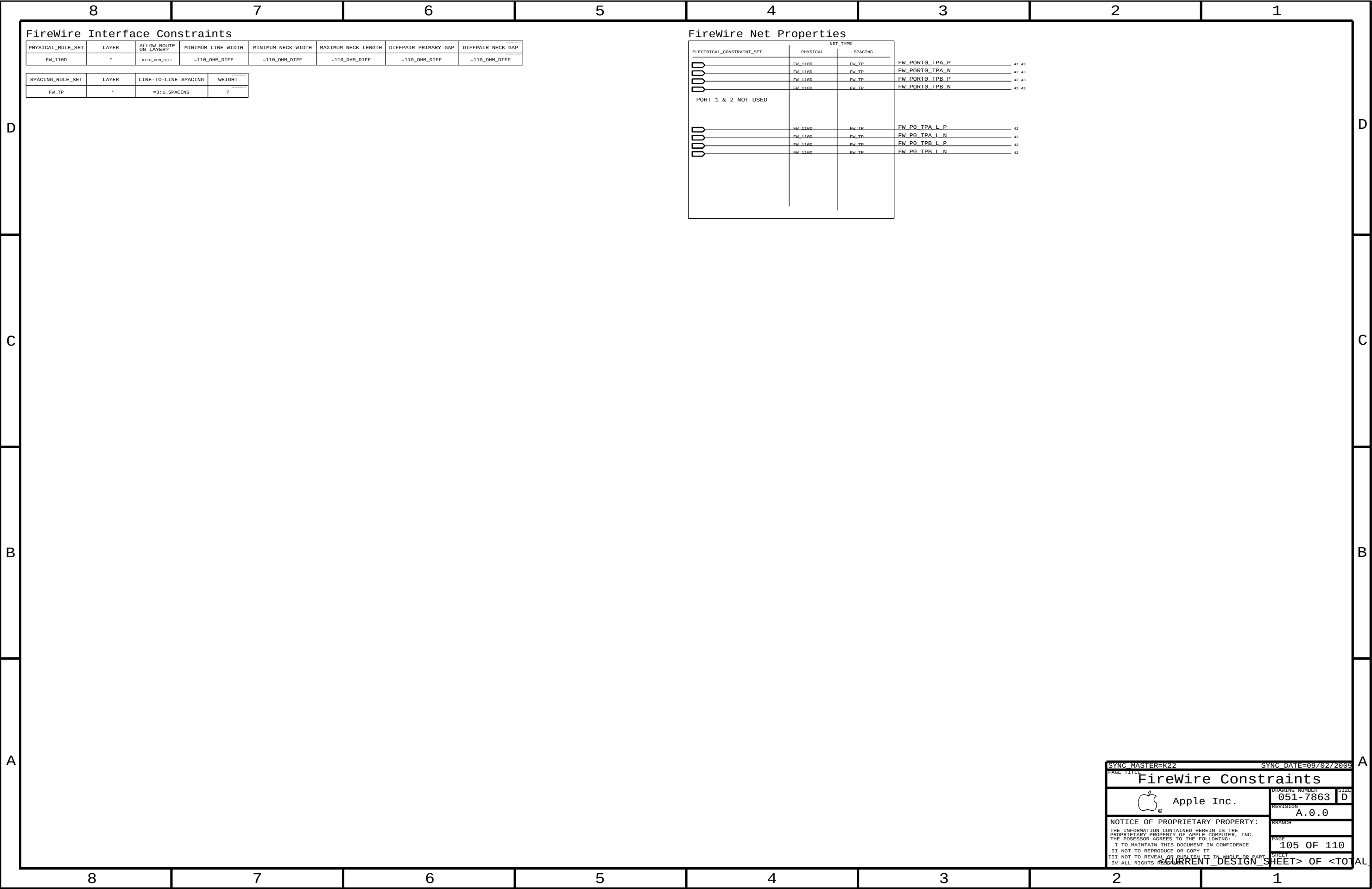
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SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	0.6 MM	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Section 2.7.4

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
		MCP_MII_COMP		MCP_MII_COMP_VDD 18
		MCP_MII_COMP		MCP_MII_COMP_GND 18
		ENET_MII_555	MCP_BUE0_CLK	MCP_CLK25M_BUE0_R 18
		ENET_MII_555	MCP_BUE0_CLK	RTL8211_CLK25M_CKXTAL1 37 38
		ENET_MII_555	ENET_MII	ENET_MDIO 18 37
		ENET_MII_555	ENET_MII	ENET_MDC 18 37
		ENET_MII_555	ENET_MII	ENET_CLK125M_RXCLK 18 37
		ENET_MII_555	ENET_MII	ENET_CLK125M_RXCLK_R 18 37
		ENET_MII_555	ENET_MII	ENET_RXD<0> 37
		ENET_MII_555	ENET_MII	ENET_RXD_R<0> 37
		ENET_MII_555	ENET_MII	ENET_RXD<3...1> 37
		ENET_MII_555	ENET_MII	ENET_RXD_R<3...1> 37
		ENET_MII_555	ENET_MII	ENET_RX_CTRL 37
		ENET_MII_555	ENET_MII	ENET_RXCTL_R 37
		ENET_MII_555	ENET_MII	ENET_CLK125M_TXCLK 37
		ENET_MII_555	ENET_MII	ENET_TXD<0> 18 37
		ENET_MII_555	ENET_MII	ENET_TXD<3...1> 18 37
		ENET_MII_555	ENET_MII	ENET_TX_CTRL 18 37
		ENET_MDI_1680	ENET_MDI	ENET_MDI_P<3...0> 37 39
		ENET_MDI_1680	ENET_MDI	ENET_MDI_N<3...0> 37 39
		ENET_MDI_1680	ENET_MDI	ENET_MDI_T_P<3...0> 39
		ENET_MDI_1680	ENET_MDI	ENET_MDI_T_N<3...0> 39

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Ethernet Constraints															
Apple Inc.	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 60%; padding: 2px;">DRAWING NUMBER</td> <td style="width: 40%; padding: 2px;">SIZE</td> </tr> <tr> <td style="text-align: center; padding: 5px;">051-7863</td> <td style="text-align: center; padding: 5px;">D</td> </tr> <tr> <td colspan="2" style="padding: 2px;">REVISION</td> </tr> <tr> <td colspan="2" style="text-align: center; padding: 5px;">A.0.0</td> </tr> <tr> <td colspan="2" style="padding: 2px;">BRANCH</td> </tr> <tr> <td colspan="2" style="padding: 2px;">PAGE</td> </tr> <tr> <td colspan="2" style="text-align: center; padding: 5px;">104 OF 110</td> </tr> </table>	DRAWING NUMBER	SIZE	051-7863	D	REVISION		A.0.0		BRANCH		PAGE		104 OF 110	
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Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_1080	*	=100_OHM_DIFF	=100_OHM_DIFF	0.08MM	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
MCP_DV_COMP	*	Y	0.5 MM	0.5 MM	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?	DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	*	*	3:1_SPACING
DISPLAYPORT	POWER	*	PWR_P2MM
DISPLAYPORT	GND	*	GND_P2MM

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.
SOURCE: MCP79 Interface D6 (D6-03328-001_v00), Sections 2.5.3 & 2.5.4.

ELECTRICAL_CONSTRAINT_SET ASSIGNED IN CONT. MGR.	NET_TYPE			
	PHYSICAL	SPACING		
	DP_1080	DISPLAYPORT	MXM DP A ML P<3..0>	04 91
	DP_1080	DISPLAYPORT	MXM DP A ML N<3..0>	04 91
	DP_1080	DISPLAYPORT	MXM DP A ML C P<3..0>	91
	DP_1080	DISPLAYPORT	MXM DP A ML C N<3..0>	91
	DP_1080	DISPLAYPORT	MXM DP A ML EQ P<3..0>	91
	DP_1080	DISPLAYPORT	MXM DP A ML EQ N<3..0>	91
	DP_1080	DISPLAYPORT	MXM DP A AUX P	04 91
	DP_1080	DISPLAYPORT	MXM DP A AUX N	04 91
	DP_1080	DISPLAYPORT	MXM DP A AUX C P	91
	DP_1080	DISPLAYPORT	MXM DP A AUX C N	91
	DP_1080	DISPLAYPORT	DP EXT LINK P<3..0>	91 94
	DP_1080	DISPLAYPORT	DP EXT LINK N<3..0>	91 94
	DP_1080	DISPLAYPORT	DP EXT LINK C P<3..0>	91
	DP_1080	DISPLAYPORT	DP EXT LINK C N<3..0>	91
	DP_1080	DISPLAYPORT	DP EXT AUXCH P	91 94 95
	DP_1080	DISPLAYPORT	DP EXT AUXCH N	91 94 95
	DP_1080	DISPLAYPORT	DP ML CONN P<3..0>	94
	DP_1080	DISPLAYPORT	DP ML CONN N<3..0>	94
	DP_1080	DISPLAYPORT	MXM DP C ML P<3..0>	04 92
	DP_1080	DISPLAYPORT	MXM DP C ML N<3..0>	04 92
	DP_1080	DISPLAYPORT	MXM DP C ML C P<3..0>	92
	DP_1080	DISPLAYPORT	MXM DP C ML C N<3..0>	92
	DP_1080	DISPLAYPORT	MXM DP C AUX P	04 92
	DP_1080	DISPLAYPORT	MXM DP C AUX N	04 92
	DP_1080	DISPLAYPORT	MXM DP C AUX C P	92
	DP_1080	DISPLAYPORT	MXM DP C AUX C N	92
	DP_1080	DISPLAYPORT	DP INT LINK P<3..0>	92 98
	DP_1080	DISPLAYPORT	DP INT LINK N<3..0>	92 98
	DP_1080	DISPLAYPORT	DP INT LINK CONN P<3..0>	98
	DP_1080	DISPLAYPORT	DP INT LINK CONN N<3..0>	98
	DP_1080	DISPLAYPORT	DP INT AUXCH P	92 98
	DP_1080	DISPLAYPORT	DP INT AUXCH N	92 98
	DP_1080	DISPLAYPORT	DP MUX P<3..0>	91 92
	DP_1080	DISPLAYPORT	DP MUX N<3..0>	91 92
	DP_1080	DISPLAYPORT	DP MUX AUXCH P	91 92 107
	DP_1080	DISPLAYPORT	DP MUX AUXCH N	91 92 107
	DP_1080	DISPLAYPORT	DP MUX AUXCH P	91 92 107
	DP_1080	DISPLAYPORT	DP MUX AUXCH N	91 92 107
	DP_1080	DISPLAYPORT	DP TX EQ AUXCH P	91
	DP_1080	DISPLAYPORT	DP TX EQ AUXCH N	91
	DP_1080	DISPLAYPORT	DP EQ LZ AUXCH P	92
	DP_1080	DISPLAYPORT	DP EQ LZ AUXCH N	92
	MCP_DV_COMP		MCP HDMI RSET	18 26
	MCP_DV_COMP		MCP HDMI VPROBE	18 26

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
PPDDR_MEM	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM
MEM_CLK	PPDDR_MEM	*	PWR_P2MM
MEM_CMD	PPDDR_MEM	*	PWR_P2MM
MEM_CTRL	PPDDR_MEM	*	PWR_P2MM
MEM_DATA	PPDDR_MEM	*	PWR_P2MM
MEM_DQS	PPDDR_MEM	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_FSB	GND	*	GND_P2MM
CPU_COMP	GND	*	GND_P2MM
CPU_GTLREF	GND	*	GND_P2MM
CPU_VCCSSEF	GND	*	GND_P2MM
FSB_DSTB	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	PWR	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
THERMAL	*	*	4:1_SPACING
SWITCHNODE	*	*	SWITCHNODE
THERMAL	PWR	*	PWR_P2MM
THERMAL	GND	*	GND_P2MM
AUDIO	*	*	AUDIO

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
THERM_DIFF	*	1:1_DIFFPAIR
SNS_DIFF	*	1:1_DIFFPAIR

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	3250 MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S_OVERRIDE	TOP	VERRIDE	VERRIDE	0.1 MM VERRIDE	600 MIL VERRIDE	VERRIDE	VERRIDE
MEM_40S_VDD_OVERRIDE	TOP	VERRIDE	VERRIDE	0.1 MM VERRIDE	600 MIL VERRIDE	VERRIDE	VERRIDE
MEM_78D_OVERRIDE	TOP	VERRIDE	VERRIDE	0.1 MM VERRIDE	600 MIL VERRIDE	VERRIDE	VERRIDE
PCIE_90D_OVERRIDE	TOP	VERRIDE	VERRIDE	VERRIDE	500 MIL VERRIDE	VERRIDE	VERRIDE
USB_90D_OVERRIDE	TOP	VERRIDE	VERRIDE	VERRIDE	500 MIL VERRIDE	VERRIDE	VERRIDE
MCP_DV_COMP_OVERRIDE	TOP	VERRIDE	VERRIDE	0.1 MM VERRIDE	500 MIL VERRIDE	VERRIDE	VERRIDE
MCP_MEM_COMP_OVERRIDE	TOP	VERRIDE	VERRIDE	0.1 MM VERRIDE	500 MIL VERRIDE	VERRIDE	VERRIDE
MCP_MII_COMP_OVERRIDE	TOP	VERRIDE	VERRIDE	0.1 MM VERRIDE	500 MIL VERRIDE	VERRIDE	VERRIDE
MCP_USR_RBIA5_OVERRIDE	TOP	VERRIDE	VERRIDE	0.1 MM VERRIDE	500 MIL VERRIDE	VERRIDE	VERRIDE
MCP_DV_COMP_OVERRIDE	*	VERRIDE	VERRIDE	0.25 MM VERRIDE	250 MIL VERRIDE	VERRIDE	VERRIDE
CPU_27P4S_OVERRIDE	BOTTOM VERRIDE	VERRIDE	VERRIDE	0.23 MM VERRIDE	100 MIL VERRIDE	VERRIDE	VERRIDE

K50/K51 SPECIFIC NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE		
		PHYSICAL	SPACING	
		PPDDR_MEM	=PP1V5_S3_MEM_A	6 30 31
		PPDDR_MEM	=PP1V5_S3_MEM_B	6 30 32
		SWITCHNODE	VR_CPU_SW1	72
		SWITCHNODE	VR_CPU_SW2	72
		SWITCHNODE	VR_CPU_SW3	72
		SWITCHNODE	1V8_SW	80
		SWITCHNODE	1V1S5_SW	79
		SWITCHNODE	PVTTS0_PHASE	76
		SWITCHNODE	3V3S5_SW	76
		SWITCHNODE	5VS3_SW	73
		SWITCHNODE	MPCCORES0_PHASE	74
	THERM_DIFF	THERMAL	SNS_T_DP1_DN6	55
	THERM_DIFF	THERMAL	SNS_T_DN1_DP6	55
	THERM_DIFF	THERMAL	SNS_T_DP2_DN3	55
	THERM_DIFF	THERMAL	SNS_T_DN2_DP3	55
	THERM_DIFF	THERMAL	CPU_THERMD_P	11 55
	THERM_DIFF	THERMAL	CPU_THERMD_N	11 55
	THERM_DIFF	THERMAL	SNS_T_DP4_DN5	55
	THERM_DIFF	THERMAL	SNS_T_DN4_DP5	55
	THERM_DIFF	THERMAL	MCP_THMDIODE_P	21 55
	THERM_DIFF	THERMAL	MCP_THMDIODE_N	21 55
	THERM_DIFF	THERMAL	MXM_PWRSRC_SENSOR_P	53
	THERM_DIFF	THERMAL	MXM_PWRSRC_SENSOR_N	53
	THERM_DIFF	THERMAL	SENSE_1V5_S0_P	54
	THERM_DIFF	THERMAL	SENSE_1V5_S0_N	54
	THERM_DIFF	THERMAL	SNS_LCD_P	55 110
	THERM_DIFF	THERMAL	SNS_LCD_N	55 110
	THERM_DIFF	THERMAL	SNS_ODD_P	55 110
	THERM_DIFF	THERMAL	SNS_ODD_N	55 110
	THERM_DIFF	THERMAL	SNS_CPU_H_P	55 110
	THERM_DIFF	THERMAL	SNS_CPU_H_N	55
	THERM_DIFF	THERMAL	SNS_MCP_P	55
	THERM_DIFF	THERMAL	SNS_MCP_N	55
	THERM_DIFF	THERMAL	SNS_AMB_P	55 110
	THERM_DIFF	THERMAL	SNS_AMB_N	55 110
	THERM_DIFF	THERMAL	SNS_MXM_P	55
	THERM_DIFF	THERMAL	SNS_MXM_N	55
	SNS_DIFF	THERMAL	VR_CPU_ISNS1_P	71 72
	SNS_DIFF	THERMAL	VR_CPU_ISNS1_N	71 72
	SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_P	71
	SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_N	71
	SNS_DIFF	THERMAL	VR_CPU_ISNS2_P	71 72
	SNS_DIFF	THERMAL	VR_CPU_ISNS2_N	71 72
	SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_P	71
	SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_N	71
	SNS_DIFF	THERMAL	VR_CPU_ISNS3_P	71 72
	SNS_DIFF	THERMAL	VR_CPU_ISNS3_N	71 72
	SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_P	71
	SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_N	71
1230		THERMAL	SMC_CPU_ISENSE	49 53
1231		THERMAL	VR_CPU_IOUT	53 71
1232		THERMAL	VR_ISNS_CPU_P	53
1233		THERMAL	VR_ISNS_CPU_N	53
1234		THERMAL	SNS_PS_CPU_ISNS	53
1235		THERMAL	SMC_CPU_VSENSE	49 53
1236		THERMAL	CPU_VCC_SENSE	12 53
1237		THERMAL	SMC_GPU_VSENSE	49 53
1238		THERMAL	SMC_GPU_ISENSE	49 53
1239		THERMAL	SMC_1V5_S0_ISENSE	50 54
1240		THERMAL	SMC_1V5_S0_ISENSE_R	54
1241		THERMAL	SMC_1V5_S0_VSENSE	50 54
1242		THERMAL	SMC_MCP_CORE_ISENSE	50 54
1243		THERMAL	SMC_MCP_CORE_VSENSE	50 54
1244		THERMAL	MPCCORES0_IMON	54 74
1245		THERMAL	CPU_PECI_L	11 55
1246		THERMAL	SMB_PECI_L	55
1247		THERMAL	CPU_PECI_MCP	14 55
1248		THERMAL	HDD_OOB_TEMP_FILT	55
1249		THERMAL	HDD_OOB_TEMP	55
1250		THERMAL	HDD_OOB_TEMP_R	55
1251		THERMAL	SMC_HDD_OOB_TEMP	55

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SPACING RULE SET

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SPACING_0.5MM	*	0.5 MM	?
CLK_SPACING_0.6MM	*	0.6 MM	?
GND_P2MM	*	0.2 MM	1000
PWR_P2MM	*	0.2 MM	1000
SWITCHNODE	*	0.6 MM	1000

CONSTRAINTS FOR BGA AREA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	0.2 MM	?
BGA_P3MM	*	0.3 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
MEM_CLK	*	BGA_P1MM	BGA_P2MM
CLK_FSB	*	BGA_P1MM	BGA_P2MM
CLK_PCIE	*	BGA_P1MM	BGA_P1MM
FSB_DSTB	FSB_DSTB	BGA_P1MM	BGA_P1MM
CLK_LPC	*	BGA_P1MM	BGA_P1MM
CLK_PCI	*	BGA_P1MM	BGA_P1MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MCP_FSB_COMP	*	BGA_P1MM	BGA_P2MM
MCP_MEM_COMP	*	BGA_P1MM	BGA_P2MM
MCP_PEX_COMP	*	BGA_P1MM	BGA_P2MM

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D	FUNCTIONAL TESTPOINTS FOR MAC-1 & ICT								D
	J4700 USB CAMERA 103 47 USB_CAMERA_L_P FUNC_TEST=TRUE 103 47 USB_CAMERA_L_N FUNC_TEST=TRUE 1 PP5V_S3_REG Testpoint near J4700 2 Ground Testpoints near J4700	J5520 ANALOG LCD TEMP SENSOR 108 55 SNS_LCD_P FUNC_TEST=TRUE 108 55 SNS_LCD_N FUNC_TEST=TRUE	J6601 AUDIO MICROPHONE 66 AUD_MIC_IN1_N_CONN FUNC_TEST=TRUE 66 GND_AUDIO_MIC1_CONN FUNC_TEST=TRUE 66 AUD_MIC_IN1_P_CONN FUNC_TEST=TRUE 1 Ground Testpoint near J6601	 GND 16 TP'S FUNC_TEST=TRUE RIN_ALLOWED_TPS=16 78 6 PP3V3_S3 2 TP'S FUNC_TEST=TRUE RIN_ALLOWED_TPS=2 73 6 PP5V_S3_REG 2 TP'S FUNC_TEST=TRUE RIN_ALLOWED_TPS=2 78 6 PP5V_S0 FUNC_TEST=TRUE RIN_ALLOWED_TPS=1					
	J4750 USB CARD READER 103 47 USB_SDCARD_L_P FUNC_TEST=TRUE 103 47 USB_SDCARD_L_N FUNC_TEST=TRUE 1 PP3V3_S3 Testpoint near J4750 2 Ground Testpoints near J4750	J5521 AMBIENT TEMP SENSOR 108 55 SNS_AMB_P FUNC_TEST=TRUE 108 55 SNS_AMB_N FUNC_TEST=TRUE	J6602 AUDIO RIGHT SPEAKER 66 AUD_SPKR_OUTL02R_P FUNC_TEST=TRUE 66 AUD_SPKR_OUTL02R_N FUNC_TEST=TRUE 66 AUD_SPKR_OUTL01R_P FUNC_TEST=TRUE 66 AUD_SPKR_OUTL01R_N FUNC_TEST=TRUE						
	J4720 USB BLUETOOTH 103 47 USB_BT_L_P FUNC_TEST=TRUE 103 47 USB_BT_L_N FUNC_TEST=TRUE 1 PP3V3_S3 Testpoint near J4720 2 Ground Testpoints near J4720	J5551 ODD TEMP SENSOR 108 55 SNS_ODD_P FUNC_TEST=TRUE 108 55 SNS_ODD_N FUNC_TEST=TRUE	J6603 AUDIO LEFT SPEAKER 66 AUD_SPKR_OUTL02L_P FUNC_TEST=TRUE 66 AUD_SPKR_OUTL02L_N FUNC_TEST=TRUE 66 AUD_SPKR_OUTL01L_P FUNC_TEST=TRUE 66 AUD_SPKR_OUTL01L_N FUNC_TEST=TRUE						
C	J4780 IR BOARD 103 47 USB_IR_L_P FUNC_TEST=TRUE 103 47 USB_IR_L_N FUNC_TEST=TRUE 1 PP5V_S3_REG Testpoint near J4780 2 Ground Testpoints near J4780	J5600 ODD FAN 56 FAN_0_PWR_L FUNC_TEST=TRUE 56 FAN_TACH0_L FUNC_TEST=TRUE 56 PP12V_S0_FAN0_L FUNC_TEST=TRUE 56 FAN_0_GND FUNC_TEST=TRUE							C
	J4520 SATA ODD (HIGH SPEED) 102 45 SATA_ODD_R2D_P FUNC_TEST=TRUE 102 45 SATA_ODD_R2D_N FUNC_TEST=TRUE 102 45 SATA_ODD_D2R_C_N FUNC_TEST=TRUE 102 45 SATA_ODD_D2R_C_P FUNC_TEST=TRUE 49 45 SMC_ODD_DETECT FUNC_TEST=TRUE 1 PP5V_S0 Testpoint near J4520 5 Ground Testpoints near J4520	J5700 CPU FAN 57 FAN_2_PWR_L FUNC_TEST=TRUE 57 FAN_TACH2_L FUNC_TEST=TRUE 57 PP12V_S0_FAN2_L FUNC_TEST=TRUE 57 FAN_2_GND FUNC_TEST=TRUE	J5601 HD FAN 56 FAN_1_PWR_L FUNC_TEST=TRUE 56 FAN_TACH1_L FUNC_TEST=TRUE 56 PP12V_S0_FAN1_L FUNC_TEST=TRUE 56 FAN_1_GND FUNC_TEST=TRUE						
	J4510 SATA HDD (HIGH SPEED) 102 45 SATA_HDD_R2D_P FUNC_TEST=TRUE 102 45 SATA_HDD_R2D_N FUNC_TEST=TRUE 102 45 SATA_HDD_D2R_C_N FUNC_TEST=TRUE 102 45 SATA_HDD_D2R_C_P FUNC_TEST=TRUE 3 Ground Testpoints near J4510								
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